INCH-POUND

MIL-M-28787/357
29 March 1991

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

CONTROLLER, DMA MODULE, KEY CODE FUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic controller Direct Memory Access (DMA) module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.4873 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be FUM.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 120 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A23	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B6	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B7-B12	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	---------Module shall pass all remaining tests---------						
D1-D21	Ibd	V	S	0.0	100.0	0.0	100.0	0.0	100.0	uA
D22-D26	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D27,D28	Ibd	V	S	0.0	105.2	0.0	105.2	0.0	105.2	uA
D29	Ibd	V	S	0.0	1.005	0.0	1.005	0.0	1.005	mA
D30	Ibd	V	S	0.0	200.0	0.0	200.0	0.0	200.0	uA
D31	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
E1-E24	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E25-E30	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E31	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F16	Ios	VII	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
F17-F33	Ios	VII	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
F34	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
F35	Ios	VII	S	-100.4	-40.0	-100.4	-40.0	-100.4	-40.0	mA
G1-G21	Iih	VIII	X	0	20	0	20	0	20	uA
G22-G25	Iih	VIII	X	0	50	0	50	0	50	uA
G26,G27	Iih	VIII	X	-294	-248	-294	-248	-294	-248	uA
G28	Iih	VIII	X	-294	-216	-294	-216	-294	-216	uA
G29-G31	Iih	VIII	X	0	40	0	40	0	40	uA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H1-H21	Iil	IX	X	-200.0	0.0	-200.0	0.0	-200.0	0.0	uA
H22-H25	Iil	IX	X	-2.0	-1.0	-2.0	-1.0	-2.0	-1.0	mA
H26	Iil	IX	X	-1.6	0.0	-1.6	0.0	-1.6	0.0	mA
H27,H28	Iil	IX	X	-736.0	-485.0	-736.0	-485.0	-736.0	-485.0	uA
H29	Iil	IX	X	-2.53	-1.48	-2.53	-1.48	-2.53	-1.48	mA
H30	Iil	IX	X	-400.0	0.0	-400.0	0.0	-400.0	0.0	uA
H31	Iil	IX	X	-4.8	-2.0	-4.8	-2.0	-4.8	-2.0	mA
J1-J47	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J48,J49	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J50	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J51,J52	Voh	X	X	2.4	2.6	2.4	2.6	2.4	2.6	V
K1-K33	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K34-K50	Vol	XI	X	0.0	0.45	0.0	0.45	0.0	0.45	V
K51,K52	Vol	XI	X	0.2	0.5	0.2	0.5	0.2	0.5	V
K53	Vol	XI	X	0.0	0.45	0.0	0.45	0.0	0.45	V
L1-L32	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L33-L46	Iozh	XII	X	-10	10	-10	10	-10	10	uA
M1-M16	Iozl	XIII	X	-200	0	-200	0	-200	0	uA
M17-M32	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M33-M43	Iozl	XIII	X	-10	10	-10	10	-10	10	uA
N	Icc	XIV	X	82.3	823.0	82.3	823.0	82.3	823.0	mA
P1,P2	Functional	XV	X	----------		----------	See table XV	----------		
Q1,Q2	Dynamic	XVI	S	----------		----------	See table XVI	----------		
Q3,Q4	Dynamic	XVI	Q	----------		----------	See table XVI	----------		
R	PSFC	XVIII	S	3.92	7.28	3.92	7.28	3.92	7.28	uF
S	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
T1-T51	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
T52,T53	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
U1-U33	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U34	Output crosstalk (low level)	XXI	Q	0	564	0	564	0	564	mV
U35	Output crosstalk (low level)	XXI	Q	0	540	0	540	0	540	mV
U36-U39	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U40	Output crosstalk (low level)	XXI	Q	0	576	0	576	0	576	mV
U41	Output crosstalk (low level)	XXI	Q	0	564	0	564	0	564	mV
U42	Output crosstalk (low level)	XXI	Q	0	672	0	672	0	672	mV
U43-U51	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U52,U53	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
U54	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V1-V3	Crosstalk duration (low level)	XXII	Q	0	50	0	50	0	50	nS
V4	Crosstalk duration (low level)	XXII	Q	0	60	0	60	0	60	nS
V5	Crosstalk duration (low level)	XXII	Q	0	50	0	50	0	50	nS
W1,W2	Crosstalk pulse width (low level)	XXIII	Q	0	10	0	10	0	10	nS
X1-X6	Ci/o	XXIV	Q	2	30	2	30	2	30	pF
X7	Ci/o	XXIV	Q	2	55	2	55	2	55	pF
X8	Ci/o	XXIV	Q	2	30	2	30	2	30	pF
X9-X16	Ci/o	XXIV	Q	2	25	2	25	2	25	pF
X17-X34	Ci/o	XXIV	Q	2	35	2	35	2	35	pF
X35-X64	Ci/o	XXIV	Q	2	20	2	20	2	20	pF
X65	Ci/o	XXIV	Q	6	75	6	75	6	75	pF
X66,X67	Ci/o	XXIV	Q	4	45	4	45	4	45	pF
Y	Threshold	XXV	X	See table XXV						
Z1-Z32	Disable time	XXVI	Q	4	53	4	58	4	58	nS
Z33-Z48	Disable time	XXVI	Q	2	56	2	56	2	56	nS
Z49-Z64	Disable time	XXVI	Q	2	39	2	39	2	39	nS
Z65-Z67	Disable time	XXVI	Q	4	206	4	206	4	206	nS
Z68-Z70	Disable time	XXVI	Q	4	203	4	203	4	203	nS
Z71-Z78	Disable time	XXVI	Q	4	296	4	296	4	296	nS
Z79-Z86	Disable time	XXVI	Q	4	163	4	163	4	163	nS

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/357-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

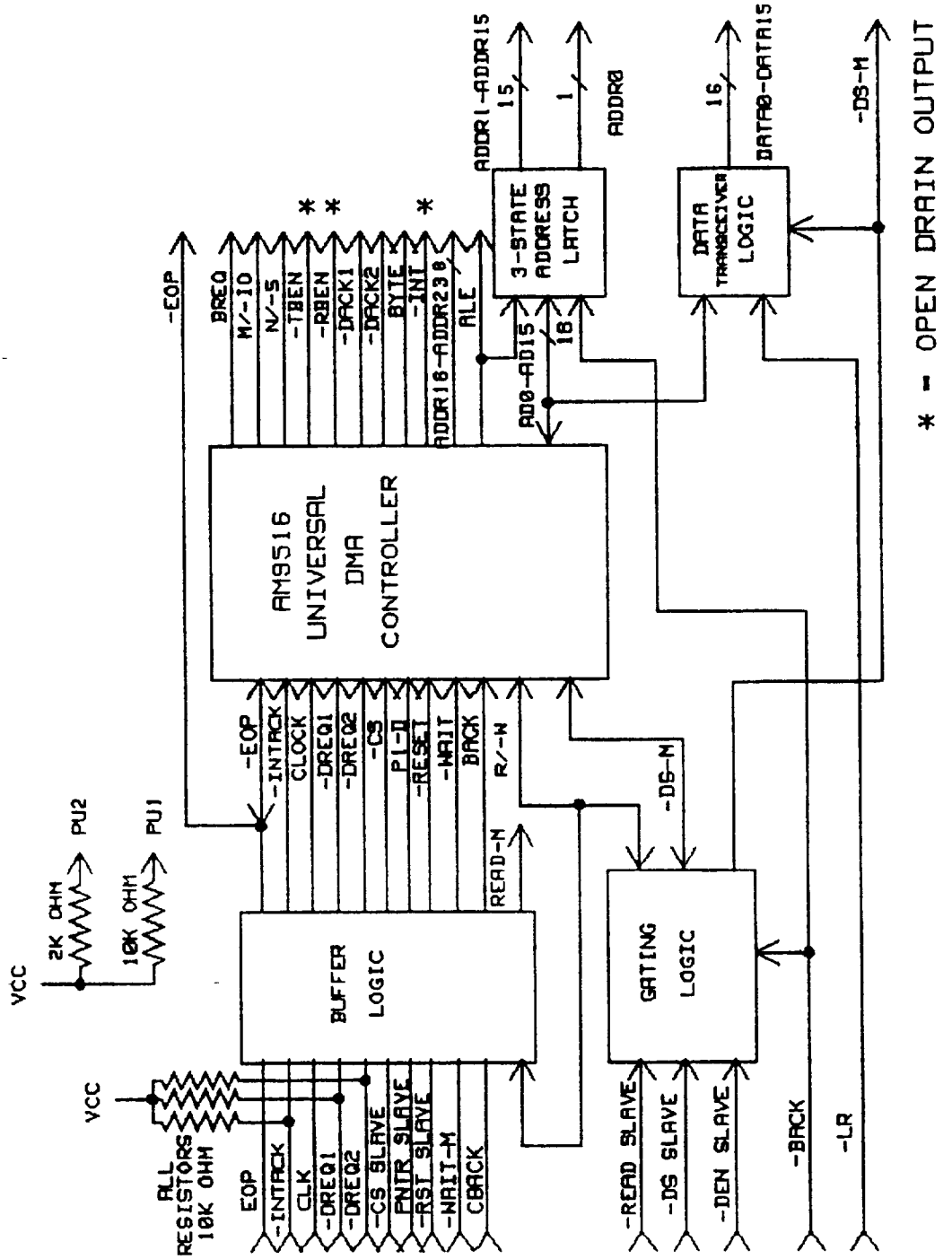


FIGURE 1. FUM functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

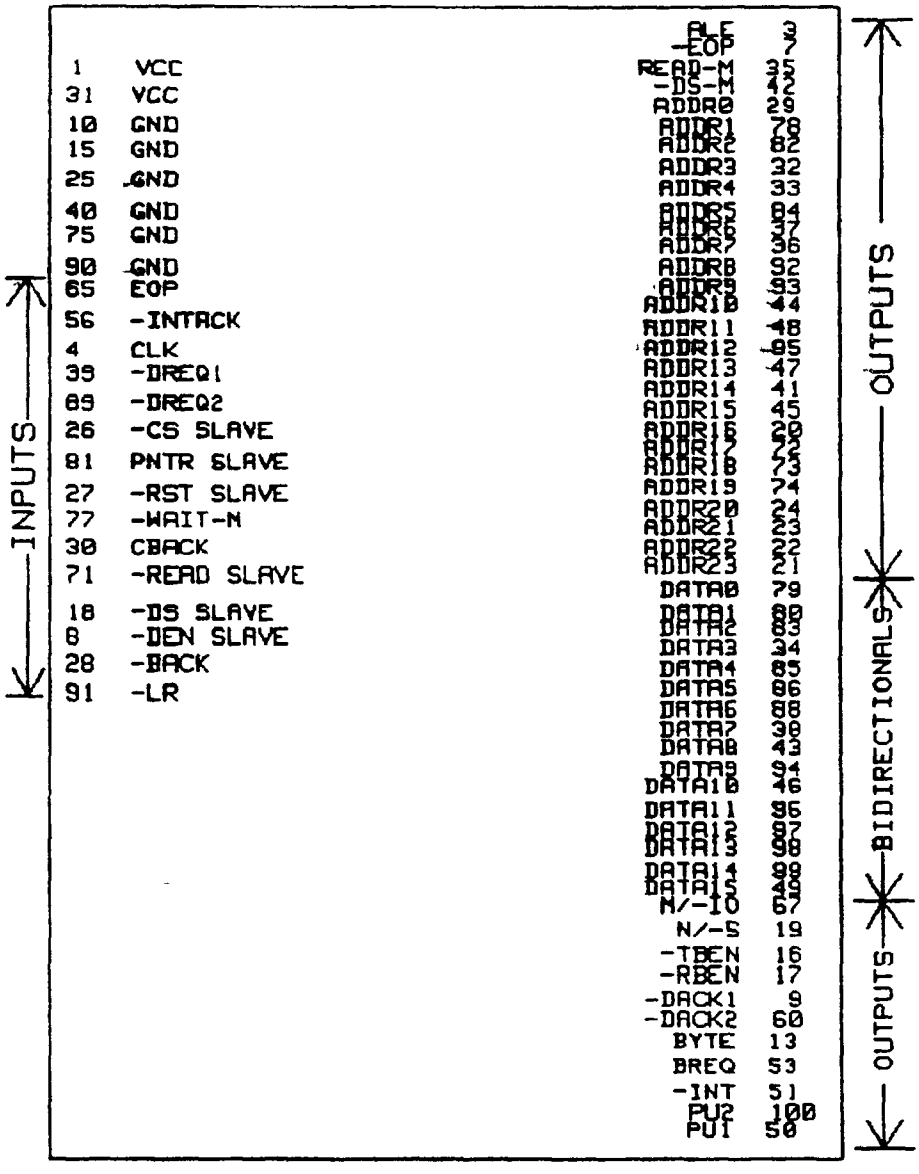


FIGURE 2. FUM pin assignment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

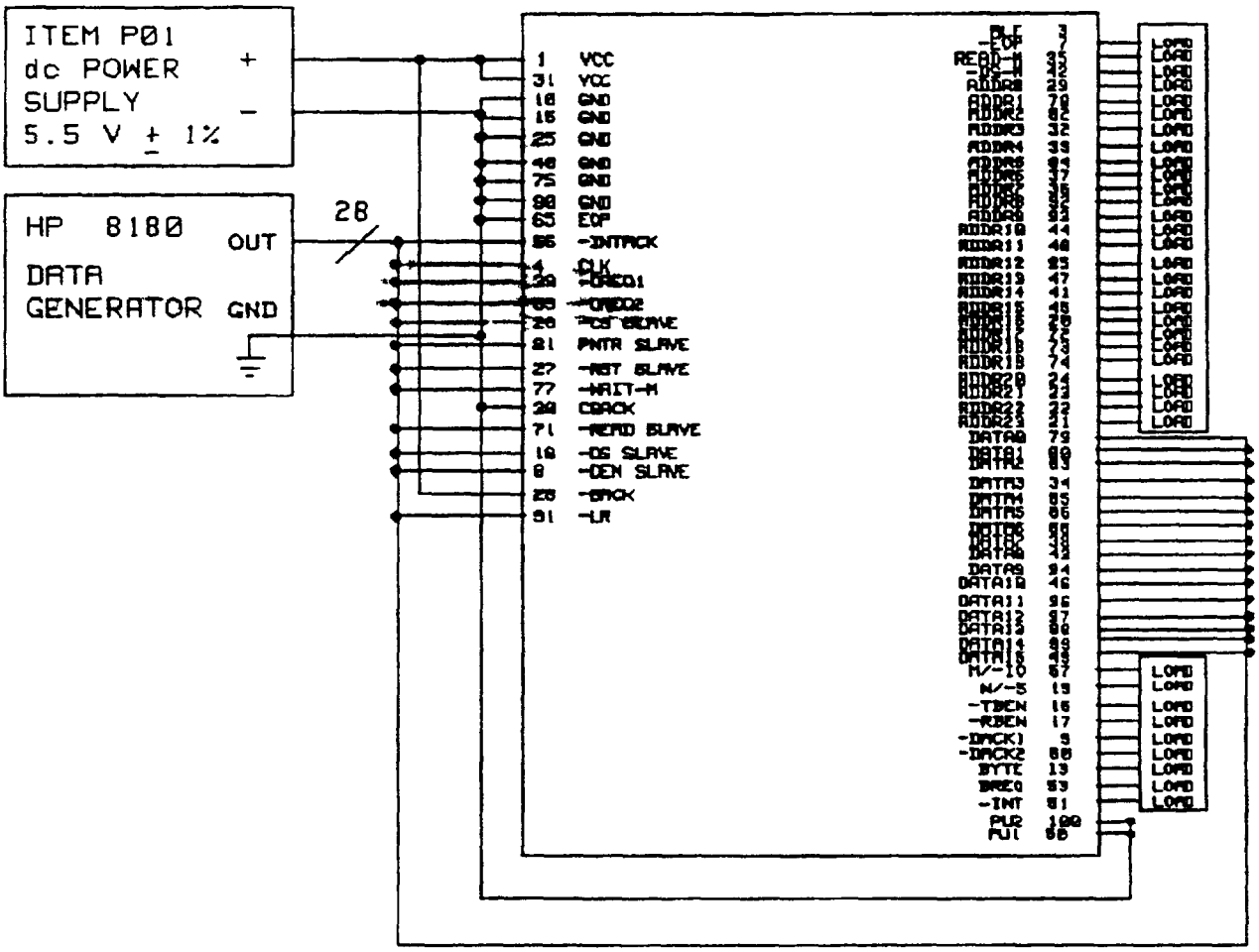
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through W and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. The load for each output is as specified in the output voltage test, except for output pins 79, 80, 83, 34, 85, 86, 88, 38, 43, 94, 46, 96, 97, 98, 99 and 49.

FIGURE 3. FUM operating temperature and life test circuit.

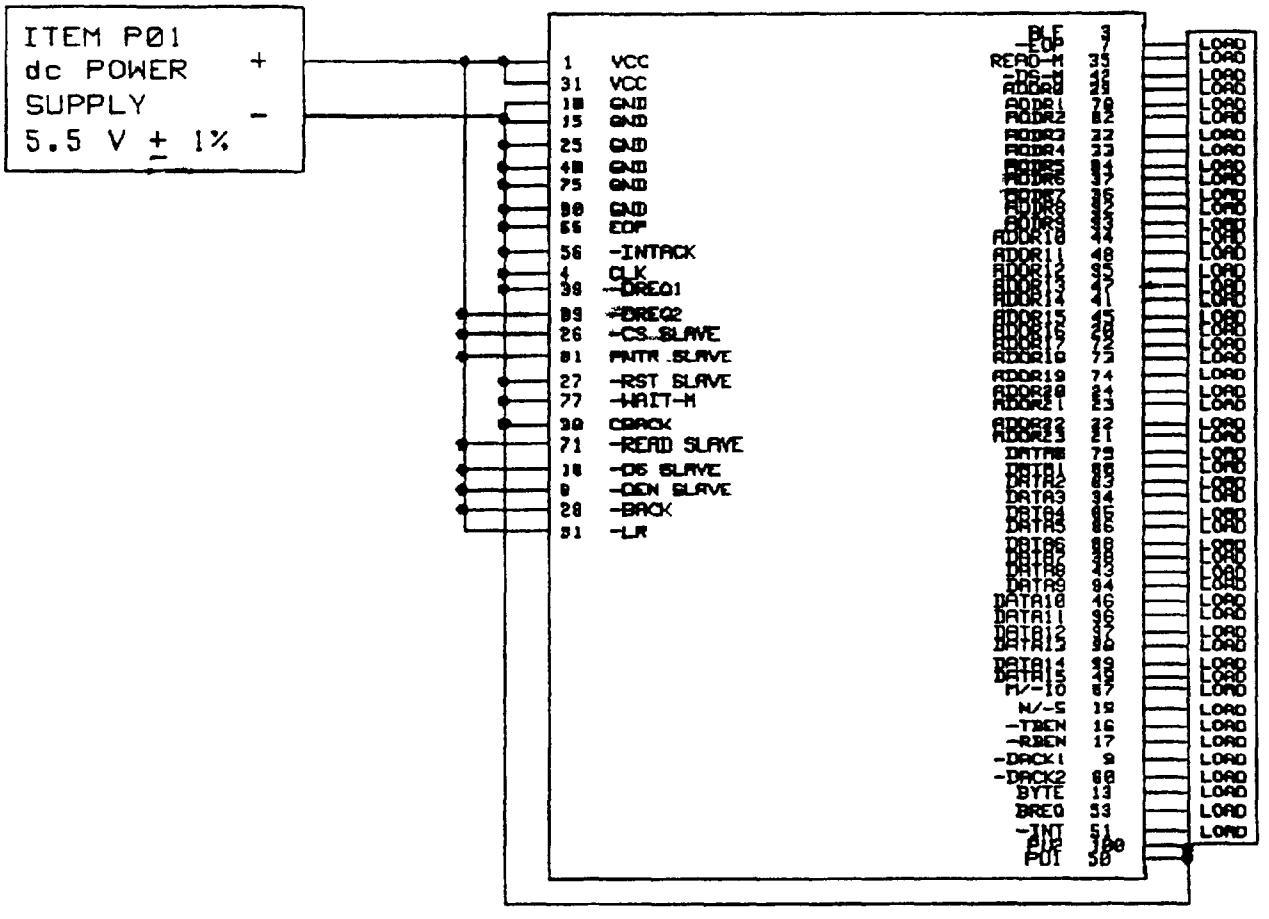
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. Test vector for operating temperature and life test:

Pin number	7883888349499994 9034568834667899	950067138228273 164851899861770	
	1111111111111111	011101011111010	
	1111111111111111	010000111111010	
	1111111111111111	011000111111010	
	1111111111111111	010000111111010	
	1111111111111111	011000111111010	
	1111111111111111	010000111111010	
	1111111111111111	011000111111010	RESET
	1111111100100000	010000111111110	
	1111111100100000	011001111101110	
	1111111100100000	010001011101110	
	1111111100100000	011001111101110	
	1111111100100000	010001111111110	
	1111111111111111	011001111110110	
	1111111111111111	010001111100110	
	1111111111111111	011001011100110	
	1111111111111111	010001111100110	
	1111111111111111	011001111111110	
	1111111111111111	010001111111110	
	1111111110100001	010000111111110	
	1111111110100001	011001111101110	
	1111111110100001	010001011101110	
	1111111110100001	011001111101110	
	1111111110100001	010001111111110	
	0000000000000000	011001111110110	
	0000000000000000	010001111100110	
	0000000000000000	011001011100110	
	0000000000000000	010001111100110	
	0000000000000000	011001111111110	
	0000000000000000	010001111111110	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTE:

1. The load for each output is as specified in the output voltage test.

FIGURE 4. FUM component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

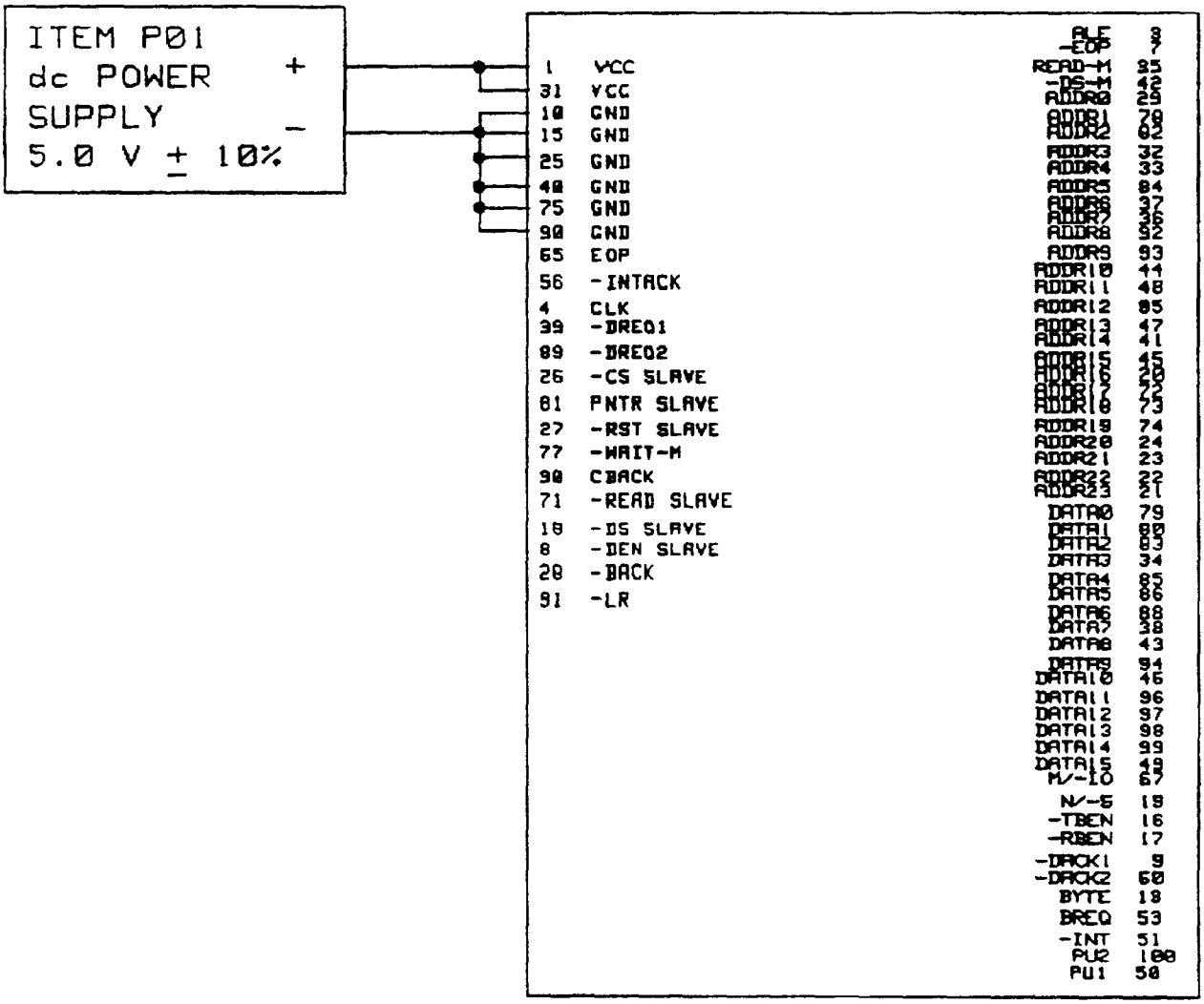


FIGURE 5. FUM vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through W and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through W and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ci/o = input/output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.954 W (max); 2.677 W (typical).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
-CS SLAVE	-1.5	0.7	2.0	-0.20	20	0.1	30
PNTR SLAVE	-1.5	0.7	2.0	-0.20	20	0.1	30
-RST SLAVE	-1.5	0.7	2.0	-0.20	20	0.1	30
-WAIT-M	-1.5	0.7	2.0	-0.20	20	0.1	25
CBACK	-1.5	0.7	2.0	-0.20	20	0.1	25
DATA0	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA1	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA2	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA3	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA4	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA5	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA6	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA7	-1.5	0.7	2.0	-0.20	20	0.1	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
DATA8	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA9	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA10	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA11	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA12	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA13	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA14	-1.5	0.7	2.0	-0.20	20	0.1	35
DATA15	-1.5	0.7	2.0	-0.20	20	0.1	35
-LR	-1.5	0.7	2.0	-0.40	40	0.2	45
-DREQ1	-1.5	0.7	2.0	-0.736	-246	0.11	35
-DREQ2	-1.5	0.7	2.0	-0.736	-246	0.11	30
-DS SLAVE	-1.2	0.7	2.0	-2.0	50	1.0	30
-READ SLAVE	-1.2	0.7	2.0	-2.0	50	1.0	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
EOP	-1.2	0.7	2.0	-2.0	50	1.0	35
-DEN SLAVE	-1.2	0.7	2.0	-2.0	50	1.0	30
-INTACK	-1.2	0.7	2.0	-2.53	-216	1.005	25
-BACK	-1.2	0.7	2.7	-4.8	140	2.2	55
CLK	-1.5	0.7	2.4	-1.6	40	1.0	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
ADDR0	2.4	0.4	-1.0	12.0	20	-20	-130	25
ADDR1	2.4	0.4	-1.0	12.0	20	-20	-130	25
ADDR2	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR3	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR4	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR5	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR6	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR7	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR8	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR9	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR10	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR11	2.4	0.4	-1.0	12.0	20	-20	-130	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
ADDR12	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR13	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR14	2.4	0.4	-1.0	12.0	20	-20	-130	20
ADDR15	2.4	0.4	-1.0	12.0	20	-20	-130	20
DATA0	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA1	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA2	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA3	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA4	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA5	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA6	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA7	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA8	2.4	0.4	-3.0	12.0	20	-200	-225	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
DATA9	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA10	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA11	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA12	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA13	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA14	2.4	0.4	-3.0	12.0	20	-200	-225	35
DATA15	2.4	0.4	-3.0	12.0	20	-200	-225	35
READ-M	2.4	0.4	-3.0	12.0	NA	NA	-225	30
ADDR16	2.4	0.45	-0.25	3.2	10	-10	NA	20
ADDR17	2.4	0.45	-0.25	3.2	10	-10	NA	20
ADDR18	2.4	0.45	-0.25	3.2	10	-10	NA	20
ADDR19	2.4	0.45	-0.25	3.2	10	-10	NA	25
ADDR20	2.4	0.45	-0.25	3.2	10	-10	NA	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
ADDR21	2.4	0.45	-0.25	3.2	10	-10	NA	20
ADDR22	2.4	0.45	-0.25	3.2	10	-10	NA	20
ADDR23	2.4	0.45	-0.25	3.2	10	-10	NA	20
M/-IO	2.4	0.45	-0.25	3.2	10	-10	NA	20
N/-S	2.4	0.45	-0.25	3.2	10	-10	NA	20
-TBEN	NA	0.45	NA	3.2	10	NA	NA	20
-RBEN	NA	NA	NA	3.2	10	NA	NA	20
-DACK1	2.4	0.45	-0.25	3.2	NA	NA	NA	20
-DACK2	2.4	0.45	-0.25	3.2	NA	NA	NA	20
BYTE	2.4	0.45	-0.25	3.2	10	-10	NA	20
BREQ	2.4	0.45	-0.25	3.2	NA	NA	NA	20
-INT	2.4	0.45	NA	3.2	10	NA	NA	25
-DS-M	2.5	0.5	-0.96	19.6	NA	NA	-100.4	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
-EOP	2.5	0.5	-1.0	20.0	NA	NA	-100	35
ALE	2.4	0.45	-0.23	2.4	NA	NA	NA	75
PU1	2.4	NA	0.21	NA	NA	NA	NA	NA
PU2	2.4	NA	-1.05	NA	NA	NA	NA	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in the appendix.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
PNTR SLAVE	81	I	This input signal identifies the information on the data bus (DATA0 - DATA15) when the FUM module is a bus slave. A high on PNTR SLAVE indicates that the information on the data bus is an address of the internal register to be accessed. The data on the bus is loaded into the pointer register of the AM9516. A low on PNTR SLAVE indicates that a data transfer is taking place between the bus and the internal register designated by the AM9516 pointer register. Note that if a transaction is carried out with the -READ SLAVE input low and the PNTR SLAVE input high, the contents of the AM9516 pointer register will be read.
-RST SLAVE	27	I	This active low input signal disables the AM9516 and clears its master mode register.
-WAIT-M	77	I	This active low input extends the DMA Master Data Strobe (-DS-M) output and the Buffer Enable outputs (-TBEN, -RBEN) during a DMA operation to accommodate slow memories and peripherals. The -WAIT-M function may be disabled using a control bit in the master mode register (bit 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CBACK	30	I	This active high input indicates that the current bus master has relinquished control of the bus and that DMA operations can begin. In a typical System, the CBACK signal is generated by the DMA support module (FUN) as the logical AND of the bus acknowledge (-BACK) output of the system bus arbiter (microprocessor communication support module AHH) and the DMA bus request (BREQ) output of the FUM module.
-READ SLAVE	71	I	This active low input signal is typically the read output signal from a microprocessor. -READ SLAVE is used in conjunction with the PNTR SLAVE input to define the type and direction of the current data bus transaction.
-BACK	28	I	This active low input indicates that the current bus master has relinquished the bus and that no higher priority device has assumed bus control. -BACK is used on the FUM module to enable the three-state address latch outputs (ADDR0 - ADDR15) and to condition the DMA Slave Read (-READ SLAVE) and DMA Slave Data Strobe (-DS SLAVE) inputs. In a typical system, -BACK is sourced by the microprocessor communication support module AHH in response to a DMA bus request (BREQ) from the FUM module.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

+326.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DS SLAVE	18	I	This active low input is used in DMA slave mode as a timing input to the AM9516 to move data to or from the data bus (DATA0 - DATA15). Data is written into the AM9516 by the external system on the rising edge of -DS SLAVE. Data is read from the AM9516 by the external system while -DS SLAVE is low. -DS SLAVE is conditioned in the FUM by -BACK to ensure that -DS SLAVE is only recognized in DMA slave mode. There are no timing requirements between -DS slave and the CLK input; this allows use of the FUM module with a system bus which does not have a bussed clock. -DS SLAVE is derived from the Lower Data Strobe (-LDS) output of the 68000 microprocessor module (NAH).
READ-M	35	O	This active high output signal indicates that a read of the DMA module is being conducted in either the master or slave modes.
-DEN SLAVE	8	I	This active low input enables the FUM module data transceivers output. -DEN SLAVE should only be active during DMA slave operations. -DEN SLAVE is generated by the DMA support module (FUN).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DS-M	42	O	This active low output is generated by the AM9516 in the master mode and is used by the external system to move data to or from the FUM data bus (DATA0 - DATA15). -DS-M is used in the FUM module to enable the data bus transceivers. During DMA write operations, the AM9516 places output data on the internal FUM data bus before the falling edge of -DS-M and holds the data stable until after the rising edge of -DS-M. In a DMA read operation, the AM9516 strobes input data from the internal FUM data bus on the rising edge of -DS-M. -DS-M is supplied to the DMA support module (FUN) where it is used to condition the generation of bus control signals (-AS, -UDS, -LDS).
-LR	91	I	This input signal controls the direction of the data bus in the FUM module if in both the DMA master and slave modes. Data flow is into the FUM module when -LR is low and out of the module when -LR is high.
ALE	3	O	This active high output signal is provided by the AM9516 to latch its address output signals (AD0 - AD15) into the three-state address latch in the FUM module. This output is never placed in the high impedance state.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
ADDR0	29	O	These three-state output signals are the lower order address lines activated only when the FUM module is bus master. In the FUM module, these signals are the outputs of a three-state latch which is controlled by the ALE output of the AM9516 and the -BACK input signal.
ADDR1	78	O	
ADDR2	82	O	
ADDR3	32	O	
ADDR4	33	O	
ADDR5	84	O	
ADDR6	37	O	
ADDR7	36	O	
ADDR8	92	O	
ADDR9	93	O	
ADDR10	44	O	
ADDR11	48	O	
ADDR12	95	O	
ADDR13	47	O	
ADDR14	41	O	
ADDR15	45	O	
ADDR16	20	O	These three-state output signals are the higher order address lines activated only when the FUM module is bus master. In the FUM module, these signals are the direct A16 - A23 outputs of the AM9516.
ADDR17	72	O	
ADDR18	73	O	
ADDR19	74	O	
ADDR20	24	O	
ADDR21	23	O	
ADDR22	22	O	
ADDR23	21	O	
DATA0	79	I/O	This 16-bit, bidirectional, three-state bus is the general purpose data path into and out of the module. In the FUM module, the data bus is provided by a set of data transceivers whose direction is determined by the -LR input and which are enabled by the -DS-M output signal.
DATA1	80	I/O	
DATA2	83	I/O	
DATA3	34	I/O	
DATA4	85	I/O	
DATA5	86	I/O	
DATA6	88	I/O	
DATA7	38	I/O	
DATA8	43	I/O	
DATA9	94	I/O	
DATA10	46	I/O	
DATA11	96	I/O	
DATA12	97	I/O	
DATA13	98	I/O	
DATA14	99	I/O	
DATA15	49	I/O	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
BREQ	53	O	This active high output signal is used by the FUM module to request control of the system bus for DMA master operations.
BYTE	13	O	This output signal is used in DMA master operations to select the size of the data transferred. When BYTE is high, a byte (8-bit) transfer is selected. A word (16-bit) transfer is selected when BYTE is low.
-DACK1 -DACK2	9 60	O O	These signals are active low outputs, one per channel, which indicate that the channel is performing a DMA operation. -DACKX is pulsed, held active and programmed in the channel mode register. -DACKX is not output during chaining operations.
-INT	51	O	-INT is +45an active low, open drain output used to interrupt the system microprocessor. It is driven low whenever the IP and CIE bits of the AM9516 status register are set. It is cleared by the AM9516 after receiving a clear IP command.
M/-IO	67	O	This three-state output signal specifies the type of DMA transaction. A high on the M/-IO output indicates a memory transaction. A low on the M/-IO indicates an Input/Output (I/O) transaction. M/-IO is placed in the high impedance state when the FUM module is not bus master.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
N/-S	19	O	This three-state output signal is activated only when the FUM module is bus master. Normal is indicated when N/-S is high and system is indicated when N/-S is low. This signal supplements the M/-IO output and is used to indicate which memory or I/O space is being accessed.
-TBEN	16	O	-TBEN is an active low, open drain output used when the FUM is bus master. A low on -TBEN indicates that output data is being transferred from the AM9516 to the FUM data bus (DATA0 - DATA15). The purpose of this signal is to allow external logic to be implemented to eliminate bus contention. When the FUM module is not bus master, -TBEN is placed in the high impedance mode.
-RBEN	17	O	-RBEN is an active low, open drain output used when the FUM is bus master. A low on -RBEN indicates that input data is being transferred from the FUM data bus (DATA0 - DATA15) to the AM9516. The purpose of this signal is to allow external logic to be implemented to eliminate bus contention. When the FUM module is not bus master, -RBEN is placed in the high impedance mode.
PU1 PU2	50 100	O O	These outputs are connections to internal pullup resistors: PU1 is connected to a 10 KOhm resistor and PU2 to to a 2 KOhm resistor. These signals are available for general purpose use in tying module inputs to a logic high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
EOP	65	I	This active high input allows an external source to terminate a DMA operation in progress. Since there is a single EOP input for both channels, EOP should be driven high only by a channel while that channel is being serviced.
-EOP	7	O	This active low output is the logical -OR of the external module EOP input and internal AM9516 generated EOP output. The AM9516 generates an EOP output under two conditions: when a channel's current operation count goes to 0 (Terminal Count (TC)), and when the data being transferred and searched, or searched, meets the match condition programmed in channel mode register bits CM17 - CM16 (Match Condition (MC)).
-INTACK	56	I	This active low input indicates that an FUM request for interrupt has been granted.
CLK	4	I	The CLK input signal controls the module internal operations and the rates of data transfers. The input CLK frequency can range from 0.5 to 4.0 MHz.
-DREQ1 -DREQ2	39 89	I I	These active low input signals are DMA request lines, one per channel. They may make transitions independent of the CLK input.
-CS SLAVE	26	I	This is an active low input used by a microprocessor or other external device to activate the module for reading or writing its internal register. The -CS SLAVE input is ignored when the FUM module is bus master.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 6.

(j) Vol test - table XI, figure 6.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 7.

(o) Dynamic test - tables XVI and XVII, figures 6, 8 and 9.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 10.

(r) Output crosstalk tests - tables XX and XXI, figures 6 and 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

(s) Output crosstalk duration test - table XXII, figures 6 and 12.

(t) Output crosstalk, pulse width, test - table XXIII, figures 6 and 13.

(u) Input/Output capacitance test - table XXIV.

(v) Threshold test - table XXV.

(w) Disable time test - table XXVI, figures 14, 15, 16 and 17.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0056)

Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused	2	5,6,11,12, 14,52,54, 55,57,58, 59,61,62, 63,64,66, 68,69,70, 76,87				
A4	Isolation, unused to unused	5	6,11,12, 14,52,54, 55,57,58, 59,61,62, 63,64,66, 68,69,70, 76,87				
A5	Isolation, unused to unused	6	11,12,14, 52,54,55, 57,58,59, 61,62,63, 64,66,68, 69,70,76, 87				
A6	Isolation, unused to unused	11	12,14,52, 54,55,57, 58,59,61, 62,63,64, 66,68,69, 70,76,87	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A7	Isolation, unused to unused	12	14,52,54, 55,57,58, 59,61,62, 63,64,66, 68,69,70, 76,87	10	INF	MOhm	1-7
A8	Isolation, unused to unused	14	52,54,55, 57,58,59, 61,62,63, 64,66,68, 69,70,76, 87				
A9	Isolation, unused to unused	52	54,55,57 58,59,61, 62,63,64, 66,68,69, 70,76,87				
A10	Isolation, unused to unused	54	55,57,58, 59,61,62, 63,64,66, 68,69,70, 76,87				
A11	Isolation, unused to unused	55	57,58,59, 61,62,63, 64,66,68, 69,70,76, 87				
A12	Isolation, unused to unused	57	58,59,61, 62,63,64, 66,68,69, 70,76,87				
A13	Isolation, unused to unused	58	59,61,62, 63,64,66, 68,69,70, 76,87	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A14	Isolation, unused to unused	59	61,62,63, 64,66,68, 69,70,76, 87	10	INF	MOhm	1-7
A15	Isolation, unused to unused	61	62,63,64, 66,68,69, 70,76,87				
A16	Isolation, unused to unused	62	63,64,66, 68,69,70, 76,87				
A17	Isolation, unused to unused	63	64,66,68, 69,70,76, 87				
A18	Isolation, unused to unused	64	66,68,69, 70,76,87				
A19	Isolation, unused to unused	66	68,69,70, 76,87				
A20	Isolation, unused to unused	68	69,70,76, 87				
A21	Isolation, unused to unused	69	70,76,87				
A22	Isolation, unused to unused	70	76,87				
A23	Isolation, unused to unused	76	87	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A23
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A23 A1-A23

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40				
B5	Continuity, GND	40	75				
B6	Continuity, GND	75	90	0.0	1.0	Ohm	1-4
B7	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B8	Continuity, GND	10	15				
B9	Continuity, GND	15	25				
B10	Continuity, GND	25	40				
B11	Continuity, GND	40	75				
B12	Continuity, GND	75	90	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B6 B7-B12

2. The ohmmeter used shall be item R10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40 75,90	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.235 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	-CS SLAVE	26	5.5	0.0	100.0	uA	1-3
D2	PNTR SLAVE	81					
D3	-RST SLAVE	27					
D4	-WAIT-M	77					
D5	CBACK	30					
D6	DATA0	79					
D7	DATA1	80					
D8	DATA2	83					
D9	DATA3	34					
D10	DATA4	85					
D11	DATA5	86					
D12	DATA6	88					
D13	DATA7	38					
D14	DATA8	43					
D15	DATA9	94					
D16	DATA10	46					
D17	DATA11	96					
D18	DATA12	97					
D19	DATA13	98					
D20	DATA14	99	5.5	0.0	100.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	DATA15	49	5.5	0.0	100.0	uA	1-3
D22	-DS SLAVE	18	5.5	0.0	1.0	mA	1-3
D23	-READ SLAVE	71					
D24	EOP	65					
D25	-DEN SLAVE	8					
D26	CLK	4	5.5	0.0	1.0	mA	1-3
D27	-DREQ1	39	5.5	0.0	105.2	uA	1-3
D28	-DREQ2	89	5.5	0.0	105.2	uA	1-3
D29	-INTACK	56	5.5	0.0	1.005	mA	1-3
D30	-LR	91	5.5	0.0	200.0	uA	1-3
D31	-BACK	28	5.5	0.0	2.2	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V 5.5 V	± 55 mV +0, -55mV	D1-D26, D30,D31 D27-D29
Power supply current limit	1.23 A	NA	D1-D31

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D26, D30,D31
	5.5 V	+0, -55 mV	D27-D29
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D31
low, (VL)	0.0 V	± 100 mV	D1-D31
Input current measurement	NA	± 5.0 uA	D1-D21
	NA	± 50.0 uA	D22-D26
	NA	± 5.26 uA	D27,D28
	NA	± 50.2 uA	D29
	NA	± 10.0 uA	D30
	NA	± 110.0 uA	D31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	-CS SLAVE	26	-18.0	-1.5	-0.4	V	1-3
E2	PNTR SLAVE	81					
E3	-RST SLAVE	27					
E4	-WAIT-M	77					
E5	CBACK	30					
E6	DATA0	79					
E7	DATA1	80					
E8	DATA2	83					
E9	DATA3	34					
E10	DATA4	85					
E11	DATA5	86					
E12	DATA6	88					
E13	DATA7	38					
E14	DATA8	43					
E15	DATA9	94					
E16	DATA10	46					
E17	DATA11	96					
E18	DATA12	97					
E19	DATA13	98					
E20	DATA14	99	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	DATA15	49	-18.0	-1.5	-0.4	V	1-3
E22	-LR	91					
E23	-DREQ1	39					
E24	-DREQ2	89	-18.0	-1.5	-0.4	V	1-3
E25	-DS SLAVE	18	-18.0	-1.2	-0.4	V	1-3
E26	-READ SLAVE	71					
E27	EOP	65					
E28	-DEN SLAVE	8					
E29	-INTACK	56					
E30	-BACK	28	-18.0	-1.2	-0.4	V	1-3
E31	CLK	4	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E31
Power supply current limit	1.23 A	NA	E1-E31

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iin)	-18 mA -12 mA	± 1.8 mA ± 1.2 mA	E1-E30 E31
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E31 E1-E31
Input voltage measurement	NA NA NA	± 55 mV ± 40 mV ± 55 mV	E1-E24 E25-E30 E31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	ADDR0	29	0.0	-130.0	-15.0	mA	1-4
F2	ADDR1	78					
F3	ADDR2	82					
F4	ADDR3	32					
F5	ADDR4	33					
F6	ADDR5	84					
F7	ADDR6	37					
F8	ADDR7	36					
F9	ADDR8	92					
F10	ADDR9	93					
F11	ADDR10	44					
F12	ADDR11	48					
F13	ADDR12	95					
F14	ADDR13	47					
F15	ADDR14	41					
F16	ADDR15	45	0.0	-130.0	-15.0	mA	1-4
F17	DATA0	79	0.0	-225.0	-40.0	mA	1-4
F18	DATA1	80					
F19	DATA2	83					
F20	DATA3	34	0.0	-225.0	-40.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	DATA4	85	0.0	-225.0	-40.0	mA	1-4
F22	DATA5	86					
F23	DATA6	88					
F24	DATA7	38					
F25	DATA8	43					
F26	DATA9	94					
F27	DATA10	46					
F28	DATA11	96					
F29	DATA12	97					
F30	DATA13	98					
F31	DATA14	99					
F32	DATA15	49					
F33	READ-M	35	0.0	-225.0	-40.0	mA	1-4
F34	-EOP	7	0.0	-100.0	-40.0	mA	1-4
F35	-DS-M	42	0.0	-100.4	-40.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F35
Power supply current limit	1.23 A	NA	F1-F35
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F35
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F35 F1-F35
Output current measurement	 NA NA NA NA	 ± 5.75 mA ± 9.25 mA ± 3.0 mA ± 3.02 mA	 F1-F16 F17-F33 F34 F35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	-CS SLAVE	26	2.7	0	20	uA	1-3
G2	PNTR SLAVE	81					
G3	-RST SLAVE	27					
G4	-WAIT-M	77					
G5	CBACK	30					
G6	DATA0	79					
G7	DATA1	80					
G8	DATA2	83					
G9	DATA3	34					
G10	DATA4	85					
G11	DATA5	86					
G12	DATA6	88					
G13	DATA7	38					
G14	DATA8	43					
G15	DATA9	94					
G16	DATA10	46					
G17	DATA11	96					
G18	DATA12	97					
G19	DATA13	98					
G20	DATA14	99	2.7	0	20	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	DATA15	49	2.7	0	20	uA	1-3
G22	-DS SLAVE	18	2.7	0	50	uA	1-3
G23	-READ SLAVE	71					
G24	EOP	65					
G25	-DEN SLAVE	8	2.7	0	50	uA	1-3
G26	-DREQ1	39	2.7	-294	-246	uA	1-3
G27	-DREQ2	89	2.7	-294	-246	uA	1-3
G28	-INTACK	56	2.7	-294	-216	uA	1-3
G29	-LR	91	2.7	0	40	uA	1-3
G30	-BACK	28	2.7	0	40	uA	1-3
G31	CLK	4	2.4	0	40	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G25, G29-G31
	5.5 V	+0, -55mV	G26-G28
Power supply current limit	1.23 A	NA	G1-G31

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vih)	2.7 V 2.4 V	± 135 mV ± 120 mV	G1-G30 G31
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G31 G1-G31
Input current measurement	NA NA NA NA NA	± 1.0 uA ± 2.5 uA ± 2.4 uA ± 3.9 uA ± 2.0 uA	G1-G21 G22-G25 G26,G27 G28 G29-G31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	-CS SLAVE	26	0.4	-200.0	0.0	uA	1-3
H2	PNTR SLAVE	81					
H3	-RST SLAVE	27					
H4	-WAIT-M	77					
H5	CBACK	30					
H6	DATA0	79					
H7	DATA1	80					
H8	DATA2	83					
H9	DATA3	34					
H10	DATA4	85					
H11	DATA5	86					
H12	DATA6	88					
H13	DATA7	38					
H14	DATA8	43					
H15	DATA9	94					
H16	DATA10	46					
H17	DATA11	96					
H18	DATA12	97					
H19	DATA13	98					
H20	DATA14	99	0.4	-200.0	0.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	DATA15	49	0.4	-200.0	0.0	uA	1-3
H22	-DS SLAVE	18	0.5	-2.0	-1.0	mA	1-3
H23	-READ SLAVE	71					
H24	EOP	65					
H25	-DEN SLAVE	8	0.5	-2.0	-1.0	mA	1-3
H26	CLK	4	0.4	-1.6	0.0	mA	1-3
H27	-DREQ1	39	0.4	-736.0	-485.0	uA	1-3
H28	-DREQ2	89	0.4	-736.0	-485.0	uA	1-3
H29	-INTACK	56	0.5	-2.53	-1.48	mA	1-3
H30	-LR	91	0.5	-400.0	0.0	uA	1-3
H31	-BACK	28	0.4	-4.8	-2.0	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H31
Power supply current limit	1.23 A	NA	H1-H31

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vil)	0.4 V 0.5 V 0.4 V 0.5 V 0.4 V	± 20 mV ± 25 mV ± 20 mV ± 25 mV ± 20 mV	H1-H21 H22-H25 H26-H28 H29,H30 H31
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H31 H1-H31
Input current measurement	NA NA NA NA NA NA NA	± 10.0 uA ± 50.0 uA ± 80.0 uA ± 12.6 uA ± 52.5 uA ± 20.0 uA ± 0.14 mA	H1-H21 H22-H25 H26 H27,H28 H29 H30 H31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ADDR0	29	-1.0	2.4	4.5	V	1-4
J2	ADDR1	78					
J3	ADDR2	82					
J4	ADDR3	32					
J5	ADDR4	33					
J6	ADDR5	84					
J7	ADDR6	37					
J8	ADDR7	36					
J9	ADDR8	92					
J10	ADDR9	93					
J11	ADDR10	44					
J12	ADDR11	48					
J13	ADDR12	95					
J14	ADDR13	47					
J15	ADDR14	41					
J16	ADDR15	45	-1.0	2.4	4.5	V	1-4
J17	DATA0	79	-3.0	2.4	4.5	V	1-4
J18	DATA1	80					
J19	DATA2	83					
J20	DATA3	34	-3.0	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	DATA4	85	-3.0	2.4	4.5	V	1-4
J22	DATA5	86					
J23	DATA6	88					
J24	DATA7	38					
J25	DATA8	43					
J26	DATA9	94					
J27	DATA10	46					
J28	DATA11	96					
J29	DATA12	97					
J30	DATA13	98					
J31	DATA14	99					
J32	DATA15	49					
J33	READ-M	35	-3.0	2.4	4.5	V	1-4
J34	ADDR16	20	-0.25	2.4	4.5	V	1-4
J35	ADDR17	72					
J36	ADDR18	73					
J37	ADDR19	74					
J38	ADDR20	24					
J39	ADDR21	23					
J40	ADDR22	22	-0.25	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	ADDR23	21	-0.25	2.4	4.5	V	1-4
J42	M/-IO	67					
J43	N/-S	19					
J44	-DACK1	9					
J45	-DACK2	60					
J46	BYTE	13					
J47	BREQ	53	-0.25	2.4	4.5	V	1-4
J48	-DS-M	42	-0.96	2.5	4.5	V	1-4
J49	-EOP	7	-1.0	2.5	4.5	V	1-4
J50	ALE	3	-0.21	2.4	4.5	V	1-4
J51	PU2	100	-1.00	2.4	2.6	V	1-4
J52	PU1	50	-0.20	2.4	2.6	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J52
Power supply current limit	1.23 A	NA	J1-J52

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-1.0 mA	± 10.0 uA	J1-J16
	-3.0 mA	± 30.0 uA	J17-J33
	-0.25 mA	± 2.5 uA	J34-J47
	-0.96 mA	± 9.6 uA	J48
	-1.0 mA	± 10.0 uA	J49
	-0.21 mA	± 2.1 uA	J50
	-1.00 mA	± 10.0 uA	J51
	-0.20 mA	± 2.0 uA	J52
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J52
low, (VL)	0.0 V	± 100 mV	J1-J52
Output voltage measurement	NA	± 105 mV	J1-J47
	NA	± 100 mV	J48,J49
	NA	± 105 mV	J50
	NA	± 10 mV	J51,J52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

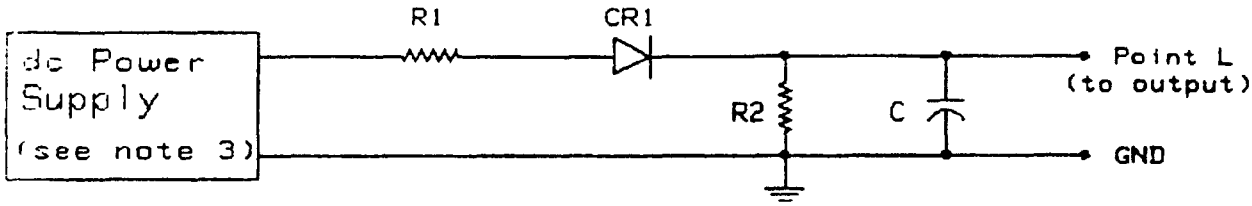
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. The load shall be as indicated on figure 6.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 pF load (including probe and parasitic capacitance) attached to each output.

(e) C is not required for dc electrical test conditions J and K.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

4. Pins 16, 17 and 51 are to be loaded with 1437.5 ohm resistors connected to Vcc and 50 pF capacitors connected to ground.

FIGURE 6. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

5. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

6. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (Ohms)	Iol (mA)	Vol (Volts)
9,13,19,20,21,22, 23,24,53,60,67,72, 73,74	9600	3.2	0.4
34,35,38,43,46,49, 79,80,83,85,86,88, 94,96,97,98,99	800	12.0	0.4
29,32,33,36,37,41, 44,45,47,48,78,82, 84,92,93,95	2400	12.0	0.4
42	2604	19.6	0.5
50	12000	NA	NA
7	2500	20.0	0.5
100	24000	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	DATA0	79	12.0	0.0	0.4	V	1-5
K2	DATA1	80					
K3	DATA2	83					
K4	DATA3	34					
K5	DATA4	85					
K6	DATA5	86					
K7	DATA6	88					
K8	DATA7	38					
K9	DATA8	43					
K10	DATA9	94					
K11	DATA10	46					
K12	DATA11	96					
K13	DATA12	97					
K14	DATA13	98					
K15	DATA14	99					
K16	DATA15	49					
K17	READ-M	35					
K18	ADDR0	29					
K19	ADDR1	78					
K20	ADDR2	82	12.0	0.0	0.4	V	1-5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K21	ADDR3	32	12.0	0.0	0.4	V	1-5
K22	ADDR4	33					
K23	ADDR5	84					
K24	ADDR6	37					
K25	ADDR7	36					
K26	ADDR8	92					
K27	ADDR9	93					
K28	ADDR10	44					
K29	ADDR11	48					
K30	ADDR12	95					
K31	ADDR13	47					
K32	ADDR14	41					
K33	ADDR15	45	12.0	0.0	0.4	V	1-5
K34	ADDR16	20	3.2	0.0	0.45	V	1-5
K35	ADDR17	72					
K36	ADDR18	73					
K37	ADDR19	74					
K38	ADDR20	24					
K39	ADDR21	23					
K40	ADDR22	22	3.2	0.0	0.45	V	1-5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K41	ADDR23	21	3.2	0.0	0.45	V	1-5
K42	M/-IO	67					
K43	N/-S	19					
K44	-TBEN	16					
K45	-RBEN	17					
K46	-DACK1	9					
K47	-DACK2	60					
K48	BYTE	13					
K49	BREQ	53					
K50	-INT	51	3.2	0.0	0.45	V	1-5
K51	-EOP	7	20.0	0.2	0.5	V	1-5
K52	-DS-M	42	19.6	0.2	0.5	V	1-5
K53	ALE	3	2.4	0.0	0.45	V	1-5

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K53
Power supply current limit	1.23 A	NA	K1-K53

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iol)	12.0 mA	± 120 uA	K1-K33
	3.2 mA	± 32 uA	K34-K50
	20.0 mA	± 200 uA	K51
	19.6 mA	± 196 uA	K52
	2.4 mA	± 24 uA	K53
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K53
low, (VL)	0.0 V	± 100 mV	K1-K53
Output voltage measurement	NA	± 20.0 mV	K1-K33
	NA	± 22.5 mV	K34-K50
	NA	± 15.0 mV	K51,K52
	NA	± 22.5 mV	K53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 6.

4. To perform test conditions K44, K45 and K50, the respective outputs are to be loaded with 1437.5 ohm resistors connected to Vcc.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	DATA0	79	2.7	-20	20	uA	1-3
L2	DATA1	80					
L3	DATA2	83					
L4	DATA3	34					
L5	DATA4	85					
L6	DATA5	86					
L7	DATA6	88					
L8	DATA7	38					
L9	DATA8	43					
L10	DATA9	94					
L11	DATA10	46					
L12	DATA11	96					
L13	DATA12	97					
L14	DATA13	98					
L15	DATA14	99					
L16	DATA15	49					
L17	ADDR0	29					
L18	ADDR1	78					
L19	ADDR2	82					
L20	ADDR3	32	2.7	-20	20	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L21	ADDR4	33	2.7	-20	20	uA	1-3
L22	ADDR5	84					
L23	ADDR6	37					
L24	ADDR7	36					
L25	ADDR8	92					
L26	ADDR9	93					
L27	ADDR10	44					
L28	ADDR11	48					
L29	ADDR12	95					
L30	ADDR13	47					
L31	ADDR14	41					
L32	ADDR15	45	2.7	-20	20	uA	1-3
L33	ADDR16	20	2.7	-10	10	uA	1-3
L34	ADDR17	72					
L35	ADDR18	73					
L36	ADDR19	74					
L37	ADDR20	24					
L38	ADDR21	23					
L39	ADDR22	22					
L40	ADDR23	21	2.7	-10	10	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	M/-IO	67	2.7	-10	10	uA	1-3
L42	N/-S	19					
L43	BYTE	13					
L44	-TBEN	16					
L45	-RBEN	17					
L46	-INT	51	2.7	-10	10	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L46
Power supply current limit	1.23 A	NA	L1-L46
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L46
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L46 L1-L46
Output current measurement	NA NA	± 2 uA ± 1 uA	L1-L32 L33-L46

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
M1	DATA0	79	0.4	-200	0	uA	1-3
M2	DATA1	80					
M3	DATA2	83					
M4	DATA3	34					
M5	DATA4	85					
M6	DATA5	86					
M7	DATA6	88					
M8	DATA7	38					
M9	DATA8	43					
M10	DATA9	94					
M11	DATA10	46					
M12	DATA11	96					
M13	DATA12	97					
M14	DATA13	98					
M15	DATA14	99					
M16	DATA15	49	0.4	-200	0	uA	1-3
M17	ADDR0	29	0.4	-20	20	uA	1-3
M18	ADDR1	78					
M19	ADDR2	82					
M20	ADDR3	32	0.4	-20	20	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M21	ADDR4	33	0.4	-20	20	uA	1-3
M22	ADDR5	84					
M23	ADDR6	37					
M24	ADDR7	36					
M25	ADDR8	92					
M26	ADDR9	93					
M27	ADDR10	44					
M28	ADDR11	48					
M29	ADDR12	95					
M30	ADDR13	47					
M31	ADDR14	41					
M32	ADDR15	45	0.4	-20	20	uA	1-3
M33	ADDR16	20	0.4	-10	10	uA	1-3
M34	ADDR17	72					
M35	ADDR18	73					
M36	ADDR19	74					
M37	ADDR20	24					
M38	ADDR21	23					
M39	ADDR22	22					
M40	ADDR23,	21	0.4	-10	10	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M41	M/-IO	67	0.4	-10	10	uA	1-3
M42	N/-S	19					
M43	BYTE	13	0.4	-10	10	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M43
Power supply current limit	1.23 A	NA	M1-M43
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M43
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M43
low, (VL)	0.0 V	± 100 mV	M1-M43
Output current measurement	NA	± 10 uA	M1-M16
	NA	± 2 uA	M17-M32
	NA	± 1 uA	M33-M43

2. The following procedure is required to perform this test:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TES: - Continued.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	82.3	823.0	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.23 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 8.15 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-8
P2	Functional test maximum power supply voltage maximum pattern rate	1-8

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.23 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.5 V 0.4 V	 ± 125 mV ± 25 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) high, (Voh) low, (Vol) low, (Vol) low, (Vol)	 NA NA NA NA NA NA	 ± 10 mV ± 105 mV ± 100 mV ± 20 mV ± 22.5 mV ± 15 mV	 P1,P2 P1,P2 P1,P2 P1,P2 P1,P2 P1,P2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 6). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

3. Input parameters for figures 7a-7f: PRR1=250 nS, PWA=132 nS, PWB=132 nS, PWC=170 nS, PWD=170 nS (see note 4), PWE=41 nS, PWF=30 nS, PWG=30 nS, ts1=42 nS, ts2=42 nS, ts3=42 nS, ts4=4 nS, ts5=58 nS, ts6=28 nS, ts7=75 nS, ts8=28 nS, ts9=79 nS, ts10=49 nS, ts11=83 nS, ts12=22 nS, ts13=0 nS, ts14=58 nS, ts15=58 nS, ts16=73 nS, ts17=47 nS, ts18=58 nS, ts19=72 nS, ts20=97 nS, th1=42 nS, th2=42 nS, th3=60 nS, th4=4 nS, th5=192 nS, th6=154 nS, th7=47 nS, th8=184 nS, th9=45 nS, th10=0 nS, th11=0 nS.

4. 2000 nS for slow readable registers (worst case).

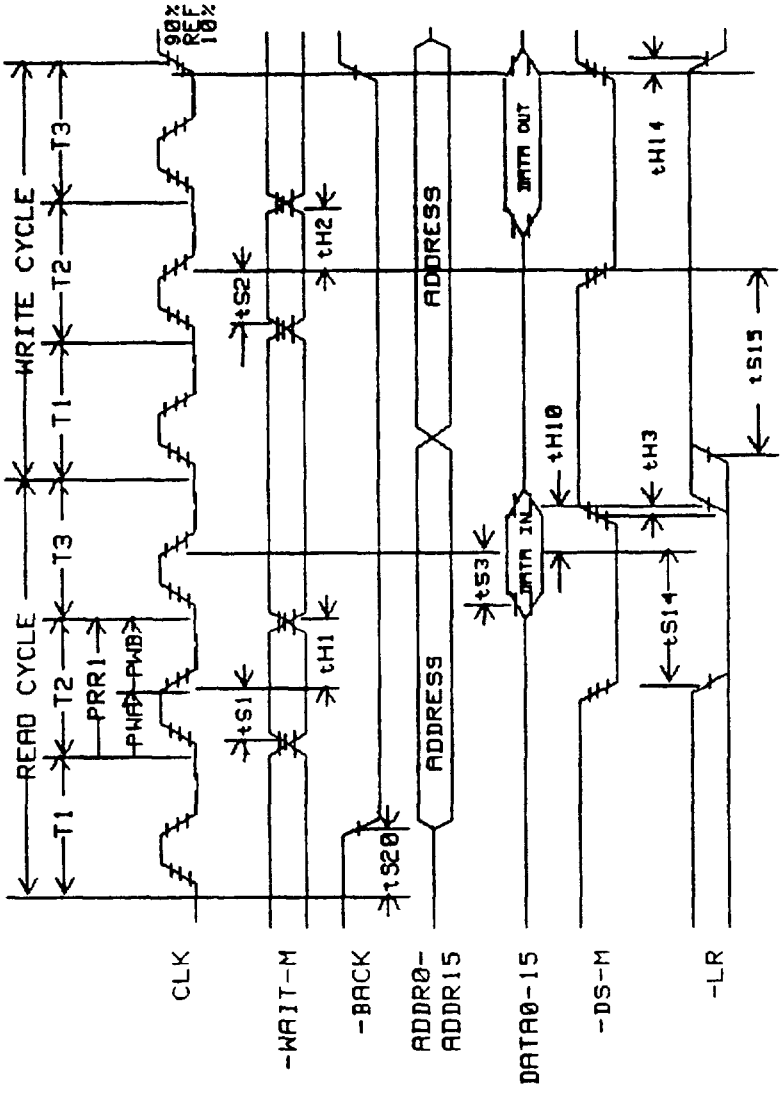
5. VREF = 1.3 V + 0.1 V.

6. Rise times (as measured between the 10 percent and 90 percent points) shall be 15 nS maximum.

7. Fall times (as measured between the 90 percent and 10 percent points) shall be 6 nS maximum.

8. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

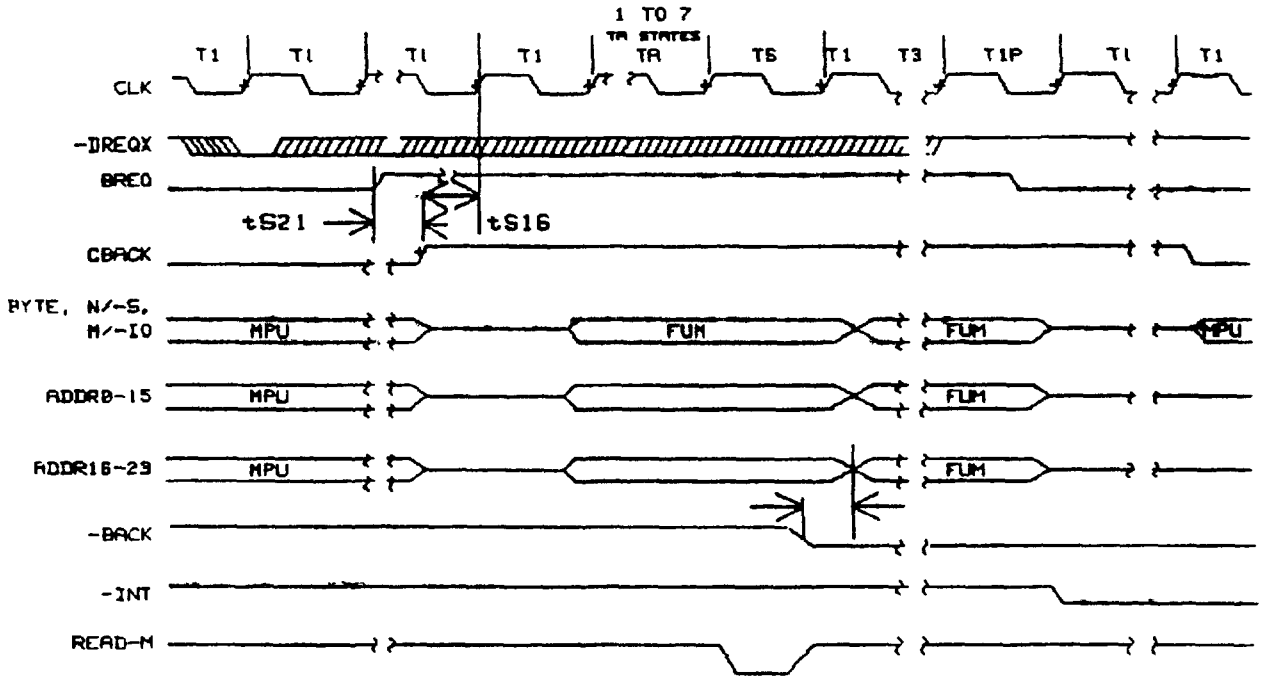


NOTE:

1. Above waveforms are ac timing waveforms when FUM module is a bus master.

FIGURE 7a. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

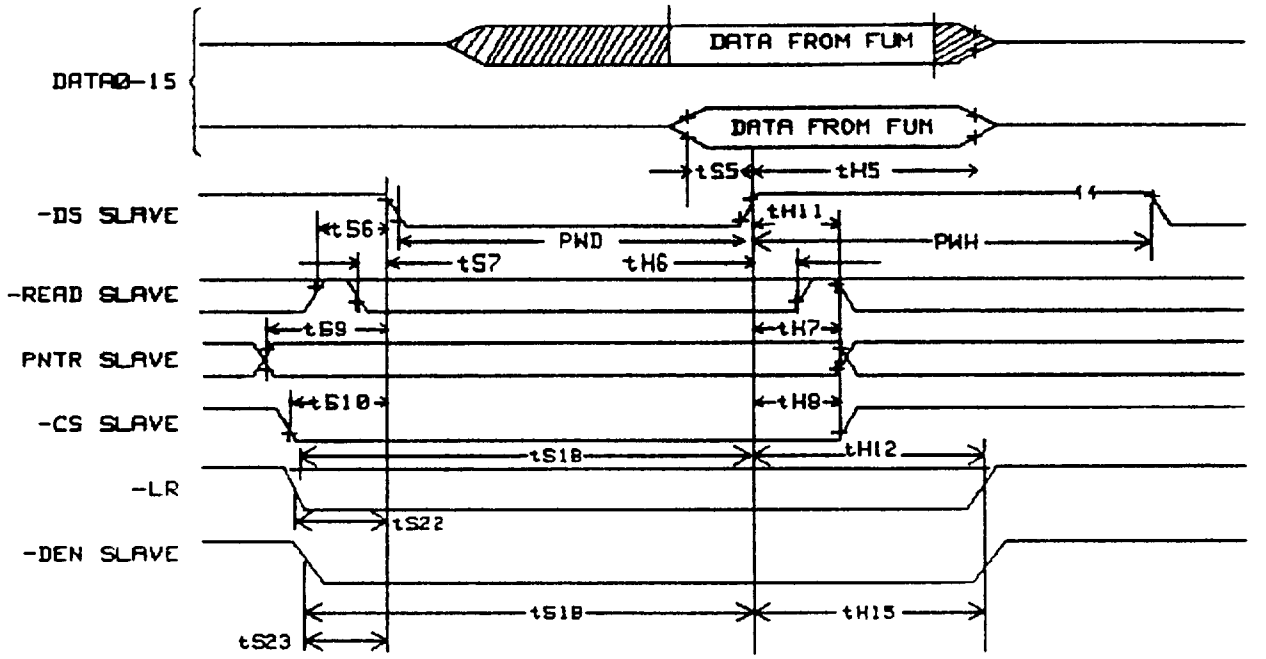


NOTES:

1. Under no circumstances can CBACK be removed prior to BREQ.
2. Above waveforms are bus exchange timing waveforms.

FIGURE 7b. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

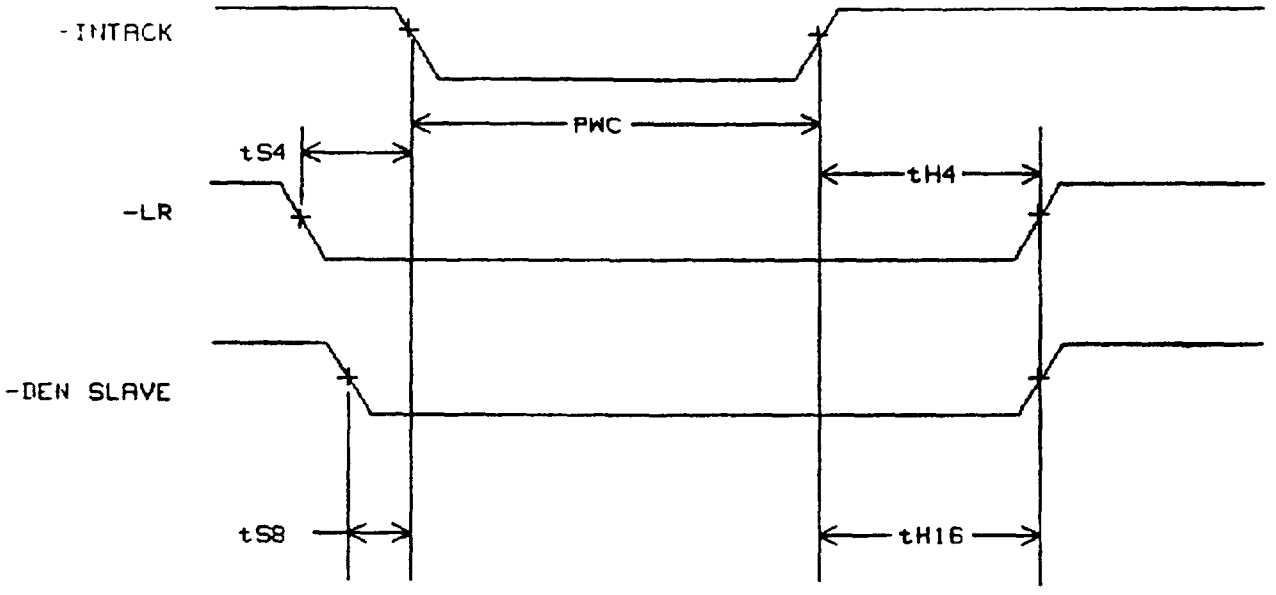


NOTE:

1. Above waveforms are ac timing when FUM module is a bus slave.

FIGURE 7c. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

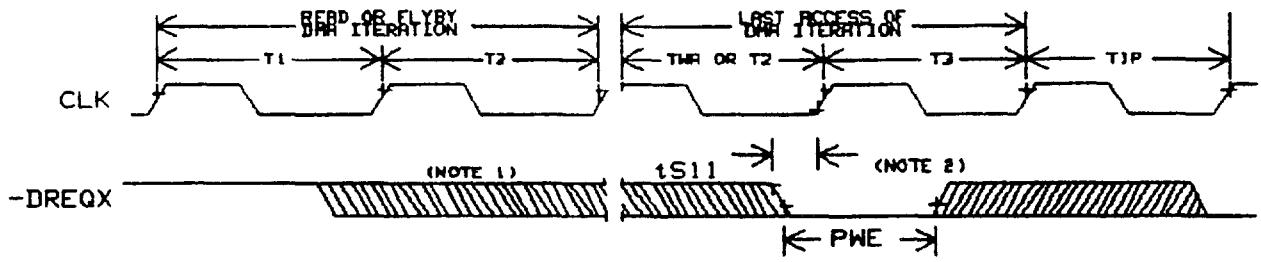


NOTE:

1. Above waveforms are ac timing for FUM interrupt acknowledge cycle.

FIGURE 7d. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. High-to-low -DREQX transitions will only be recognized after the high-to-low transition of the clock during T1 of a read or flyby DMA iteration.

2. A high-to-low -DREQX transition must meet the conditions in note 1 and must occur tS11 before state T3 of the last access of the DMA iteration if the channel is to retain bus control and immediately start the next iteration. -DREQX may go high before tS11 if it has met the PWE parameter.

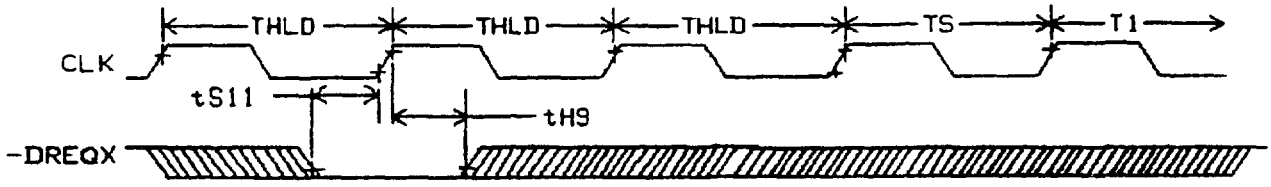
3. Flyby and search transactions have only a single access; parameter tS11 should be referenced to the start of T3 of the access. All other operations will always have two or three accesses per iteration.

4. The above waveform is a sampling -DREQX waveform during single transfer DMA operations.

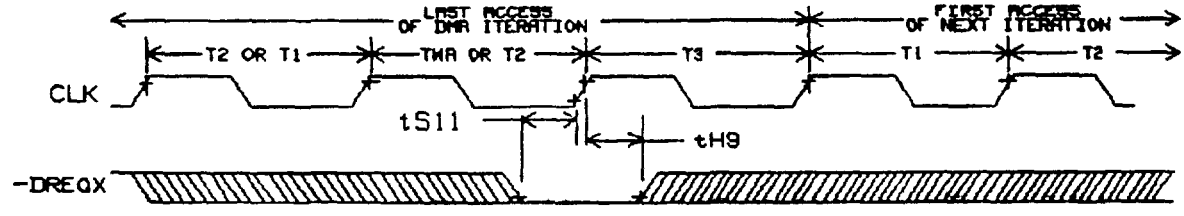
FIGURE 7e. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

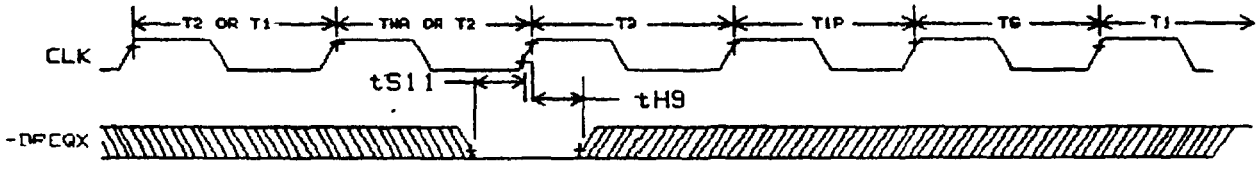
a. Sampling of -DREQX while in bus hold mode.



b. Sampling of -DREQX in demand mode during DMA operations.



c. Sampling of -DREQX at the end of chaining.



d. Sampling of -DREQX at the end of base-to-current reloading.

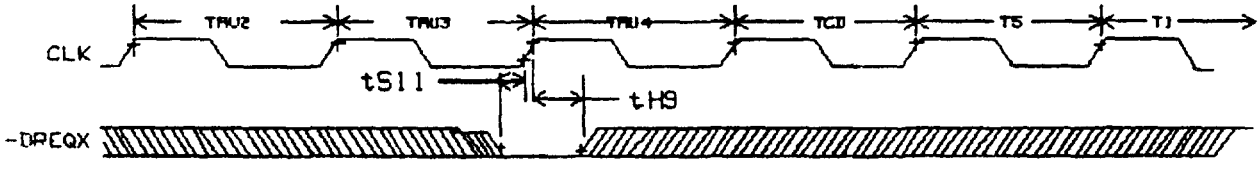


FIGURE 7f. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES:

1. -DREQX must be low from the start of tS11 to the end of tH9 to ensure that the request is recognized.

2. Failure to meet this setup time will result in the channel releasing the bus.

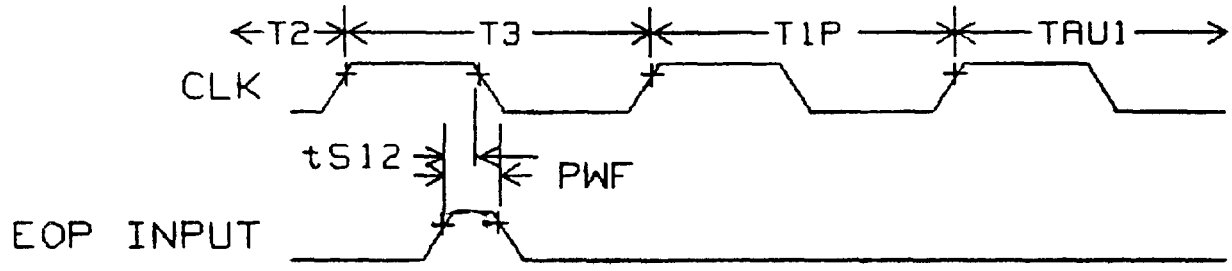
3. Ts is a setup state, generated before entering DMA operation cycle.

4. TAU2, TAU3 and TAU4 are auto-reload states, followed by TCD (chain decision) state.

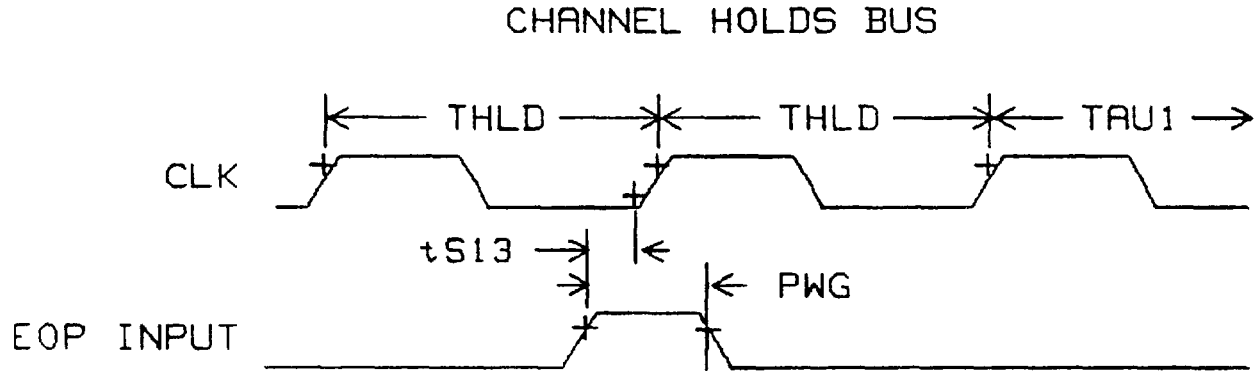
5. Above waveforms are sampling of -DREQX in demand mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

a. Sampling and generation of EOP during DMA operations.



b. Sampling of EOP during bus hold.



NOTES:

1. The diagram lists state names for both I/O and memory accesses. Sampling of EOP will occur on the falling edge of state T3.

2. State T1P is a pseudo-T1 state, generated following termination of any DMA operation.

3. TAU1 is an auto-initialization state, generated following the TC, MC or EOP termination.

4. Above waveforms are end-of-process timing waveforms.

FIGURE 7g. Minimum input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.23 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 6.

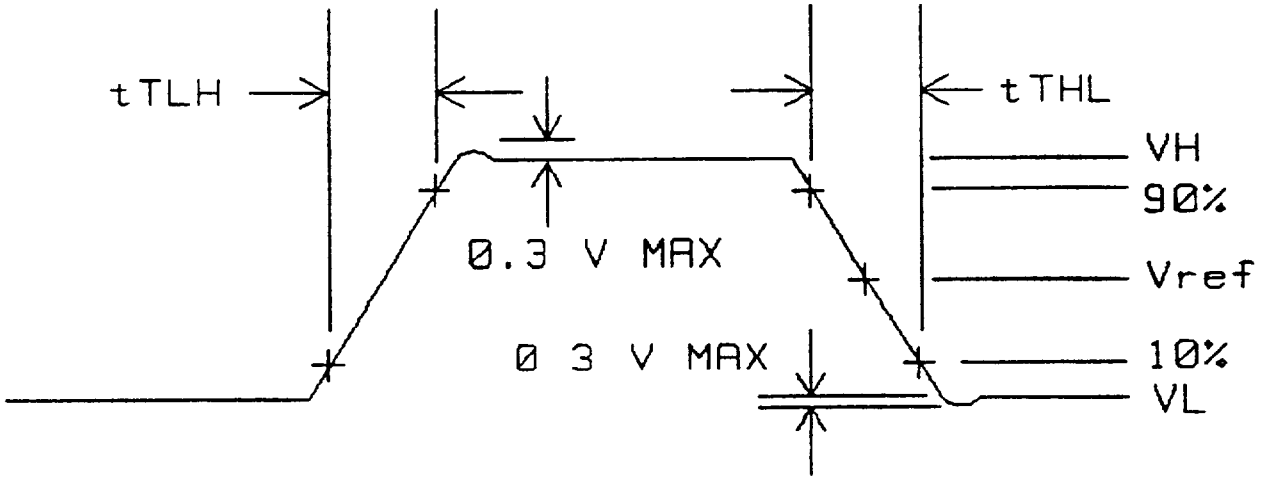
4. All of the input waveforms shall meet the requirements shown on figure 8.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. I/O waveforms are as referenced on figure 9.

7. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
4, 8, 18, 28, 56, 65, 71, 91	2.5 ± 1.5 nS	2.5 ± 1.5 nS
26, 27, 30, 34, 38, 39, 43, 46, 49, 77, 79, 80, 81, 83, 85, 86, 88, 89, 94, 96, 97, 98, 99	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 8. Sample input waveform for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
1	1	CLK cycle time	250	2000	250	2000	nS	9a
2	2	CLK PWH	118	-	118	-	nS	9a
3	3	CLK PWL	118	-	118	-	nS	9a
6	4-11	CLK to ADDR16-23 valid	-	125	-	125	nS	9b
8	12	CLK to BYTE valid	-	145	-	145	nS	9b
9	13-28	CLK to ADDR0-15 valid	5	149	5	149	nS	9b
11	29	CLK to ALE; tPLH	-	103	-	103	nS	9b
12	30	CLK to ALE; tPHL	-	105	-	105	nS	9b
13, 14	31	CLK to -DS-M; tPHL	2	109	2	109	nS nS	9d
19	48-63	ADDR0-15 to ALE(H)	26	-	26	-	nS	9b
15	64	CLK to -DS-M; tPLH	2	109	2	109	nS	9b
17	65-80	DATA0-15 in to CLK (setup time)	42	-	42	-	nS	9b
20	81	ALE PWH	80	-	80	-	nS	9b
18	82-89	ADDR16-23 valid to ALE delay; tPHL	50	-	50	-	nS	9b
22	90	ALE to -DS-M; tPHL	89	-	89	-	nS	9b
23	91-106	ALE to DATA0-15 IN required valid delay	-	322	-	322	nS	9b
24	107-122	ADDR0-15 to DATA0-15 IN required valid delay	-	471	-	471	nS	9b
25	123-138	-DS-M to ADDR0-15 active	71	-	71	-	nS	9b
26	139	-DS-M to ALE, tSTP	61	-	61	-	nS	9b
27	140-155	ADDR0-15 to -DS-M delay valid	138	-	138	-	nS	9b

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
29	156-171	-DS-M to DATA0-15 out valid	-	66	-	66	nS	9b
31	172-187	-DS-M to DATA0-15 in required valid delay	-	277	-	277	nS	9b
30	188-203	-DS-M to DATA0-15 out valid hold time; tHLD	6	-	6	-	nS	9b
33	204-219	-DS-M to DATA0-15 in valid hold time; tHLD	73	-	73	-	nS	9b
34	220	-DS-M(write); PWL	173	-	173	-	nS	9b
35	221	-DS-M(read); PWL	263	-	263	-	nS	9b
36	222	CLK to -RBEN; tPLH	-	193	-	193	nS	9b
37	223-225	-DS-M to BYTE, N/-S, M/-IO valid hold time; tHLD	61	-	61	-	nS	9b
38	226,227	CLK to -TBEN, -RBEN; tPHL	-	93	-	93	nS	9b
37A	228	-DS-M to READ-M valid hold time; tHLD	63	-	63	-	nS	9b
39	229	CLK to -TBEN; tPLH	-	189	-	189	nS	9b
40	230,231	CLK to M/-IO, N/-S valid delay	-	94	-	94	nS	9b
40A	232	CLK to READ-M valid delay	-	167	-	167	nS	9b
41	233-235	M/-IO, BYTE, N/-S valid to ALE	60	-	60	-	nS	9b
41A	236	READ-M to ALE	38	-	38	-	nS	9b
42	237	-WAIT-M to CLK setup time; tSTP	42	-	42	-	nS	9e

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
43	238	-WAIT-M to CLK hold time; tHLD	45	-	45	-	nS	9e
44	239	-DREQ1,-DREQ2; PWL	41	-	41	-	nS	9f
45	240,241	-DREQ1,-DREQ2 valid to CLK setup time; tSTP	73	-	73	-	nS	9f
46	242,243	CLK to -DREQ1, -DREQ2 valid hold time; tHLD	45	-	45	-	nS	9f, 9g
47	244	CLK to -INT; tPHL	-	185	-	185	nS	9c
61	245-260	-INTACK to DATA0-15 output valid	4	180	4	180	nS	9h
62	261-276	-INTACK to DATA0-15 output valid hold time; tHLD	4	-	4	-	nS	9h
63	277-292	-DS SLAVE to DATA0-15 output delay	6	197	6	197	nS	9h
65	293-308	DATA0-15 IN valid to -DS SLAVE setup time; tSTP	58	-	58	-	nS	9h
66	309-324	DATA0-15 to -DS SLAVE valid hold time; tHLD	284	-	284	-	nS	9h
67	325	-DS SLAVE PWL	174	-	174	-	nS	9h
69	326	-CS SLAVE to -DS SLAVE valid hold time; tHLD	264	-	264	-	nS	9h
68	327	-INTACK PWL	164	-	164	-	nS	9h
70	328	PNTR SLAVE to -DS SLAVE valid hold time; tHLD	264	-	264	-	nS	9h
71	329	PNTR SLAVE valid to -DS SLAVE setup time (read); tSTP	29	-	29	-	nS	9h

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
71	330	PNTR SLAVE valid to -DS SLAVE setup time (write); tSTP	69	-	69	-	nS	9h
72	331	-CS SLAVE valid to -DS SLAVE setup time; tSTP	49	-	49	-	nS	9h
73	332	-DS SLAVE to -DS SLAVE recovery time (commands only)	4Tcc +24	-	4Tcc +24	-	nS	9h
74	333	-RST SLAVE PWL	3Tcc +21	-	3Tcc +21	-	nS	9h, 9i
75	334	CLK to BREQ; tPLH	-	198	-	198	nS	9c
76	335	CLK to BREQ; tPHL	-	185	-	185	nS	9c
77A	336	BREQ to READ-M; tPHL	2	167	2	167	nS	9c
79	337	BREQ to CBACK; tSTP	0	-	0	-	nS	9c
80	338	CBACK valid to CLK setup time; tSTP	73	-	73	-	nS	9c
83	339	-RST SLAVE to READ-M; tPLH	4	225	4	225	nS	9i
83	340	-RST SLAVE to -DS-M; high	4	196	4	196	nS	9i
84	341	-READ SLAVE valid to -DS SLAVE setup time (READ); tSTP	107	-	107	-	nS	9h
85	342	-DS SLAVE to -READ SLAVE valid hold time (read); tHLD	264	-	264	-	nS	9h
86	343	-READ SLAVE valid to -DS SLAVE setup time (write); tSTP	12	-	12	-	nS	9h

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
87	344	-DS SLAVE to -READ SLAVE hold time (write); tHLD	234	-	234	-	nS	9h
90	345-346	CLK to pulsed -DACK 1, -DACK2; tPHL	-	131	-	131	nS	9d
91	347-348	CLK to pulsed -DACK1, -DACK2; tPLH	-	129	-	129	nS	9d
92	349-350	-DS-M to level -DACK1, -DACK2	0	-	0	-	nS	9d
93	351-352	CLK to level -DACK1, -DACK2 valid; tPHL	-	131	-	131	nS	9d
94	353-354	CLK to level -DACK1, -DACK2 valid; tPLH	-	129	-	129	nS	9d
95	355	CLK to -EOP; tPHL	-	157	-	157	nS	9j
96	356	CLK to -EOP; tPLH	-	391	-	391	nS	9j
97	357	EOP input valid to CLK setup time during DMA operation; tSTP	24	-	24	-	nS	9j
98	358	EOP input pulse width required during DMA operation; PWH	40	-	40	-	nS	9j
99	359	EOP input valid to CLK setup time during bus hold; tSTP	24	-	24	-	nS	9j
100	360	EOP input pulse width required during bus hold; tPWH	40	-	40	-	nS	9j
103	361-376	-BACK to ADDR0-15; tPZH	5	43	5	43	nS	9c
103	377-392	-BACK to ADDR0-15; tPZL	5	55	5	55	nS	9c

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
101	393	-LR (read) to CLK setup time	-	64	-	64	nS	9b
102	394	-LR (write) to CLK setup time	-	22	-	22	nS	9b
104	395-410	-LR (slave) to -DS SLAVE setup time	-	58	-	58	nS	9h
105	411-426	-DEN SLAVE (slave) to -DS SLAVE setup time	-	110	-	110	nS	9h
6	427-434	CLK to ADDR16-23 valid	20	133	20	133	nS	9b
101A	435	-LR low to CLK hold time (READ) (Th)	85	-	85	-	nS	9b
102A	436	-LR high to CLK hold time (WRITE) (Th)	89	-	89	-	nS	9b
103A	437	-BACK to CLK set-up time (Ts)	0	-	0	-	nS	9c
106A	438	-BACK to CLK hold time (Th)	0	-	0	-	nS	9c
109	439	EOP (External) to -EOP (Output) tPLH	4	254	4	259	nS	9j
110		tPHL	4	34	4	34	nS	9j
107	440	-LR to -DS SLAVE hold time (Th)	286	-	286	-	nS	9h
104A	441	-LR high to -INTACK set-up time (Ts)	0	-	0	-	nS	9h
107A	442	-LR low to -INTACK hold time (Th)	54	-	54	-	nS	9h
108	443	-DEN SLAVE low to -DS SLAVE hold time (Th)	282	-	282	-	nS	9h
105A	444	-DEN SLAVE low to -INTACK set-up time (Ts)	0	-	0	-	nS	9h

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
108A	445	-DEN SLAVE low to -INTACK hold time (Th)	50	-	50	-	nS	9h
64	446-461	-DS SLAVE to DATA0-15 OUT hold time (READ) (Th)	8	-	8	-	nS	9h
6	462-469	CLK to ADDR16-23 valid	-	133	-	133	nS	9b
103	470-485	-BACK (L) to ADDR0-15 valid	5	66	5	66	nS	9c
29	486-501	-DS-M to DATA0-15 OUT valid	2	84	2	84	nS	9b
61	502-517	-INTACK to DATA0-15 Output valid	4	188	4	188	nS	9h
63	518-533	-DS SLAVE to DATA0-15 Output delay	6	200	6	200	nS	9h
	534	-LR to -DS SLAVE (FE) set-up time (Ts) (READ)	0	-	0	-	nS	9h
	535	-DEN SLAVE to -DS SLAVE (FE) set-up time (Ts) (READ)	0	-	0	-	nS	9h

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

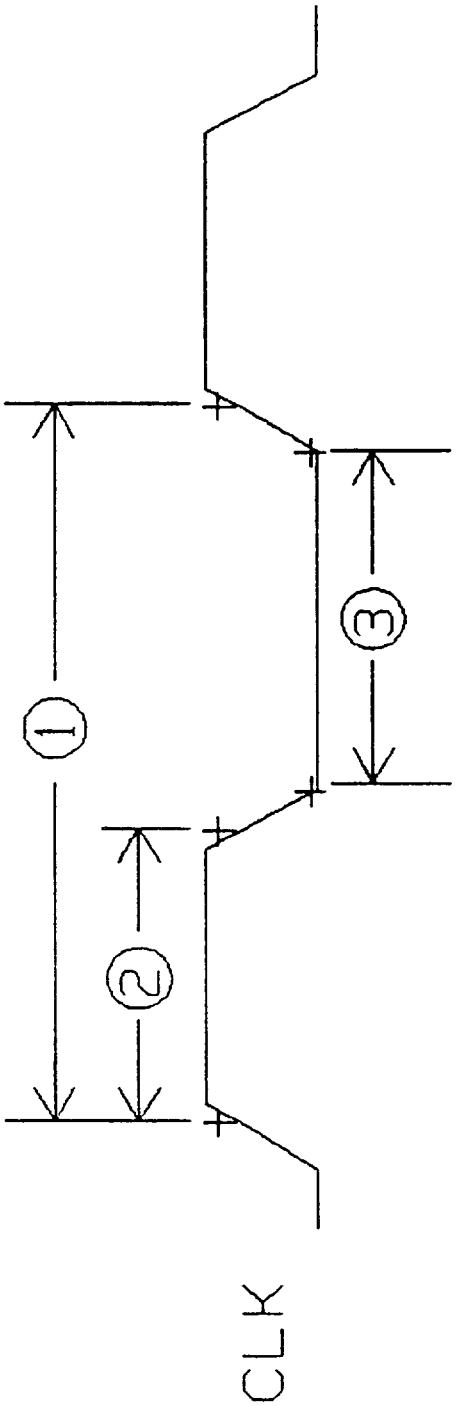


FIGURE 9a. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

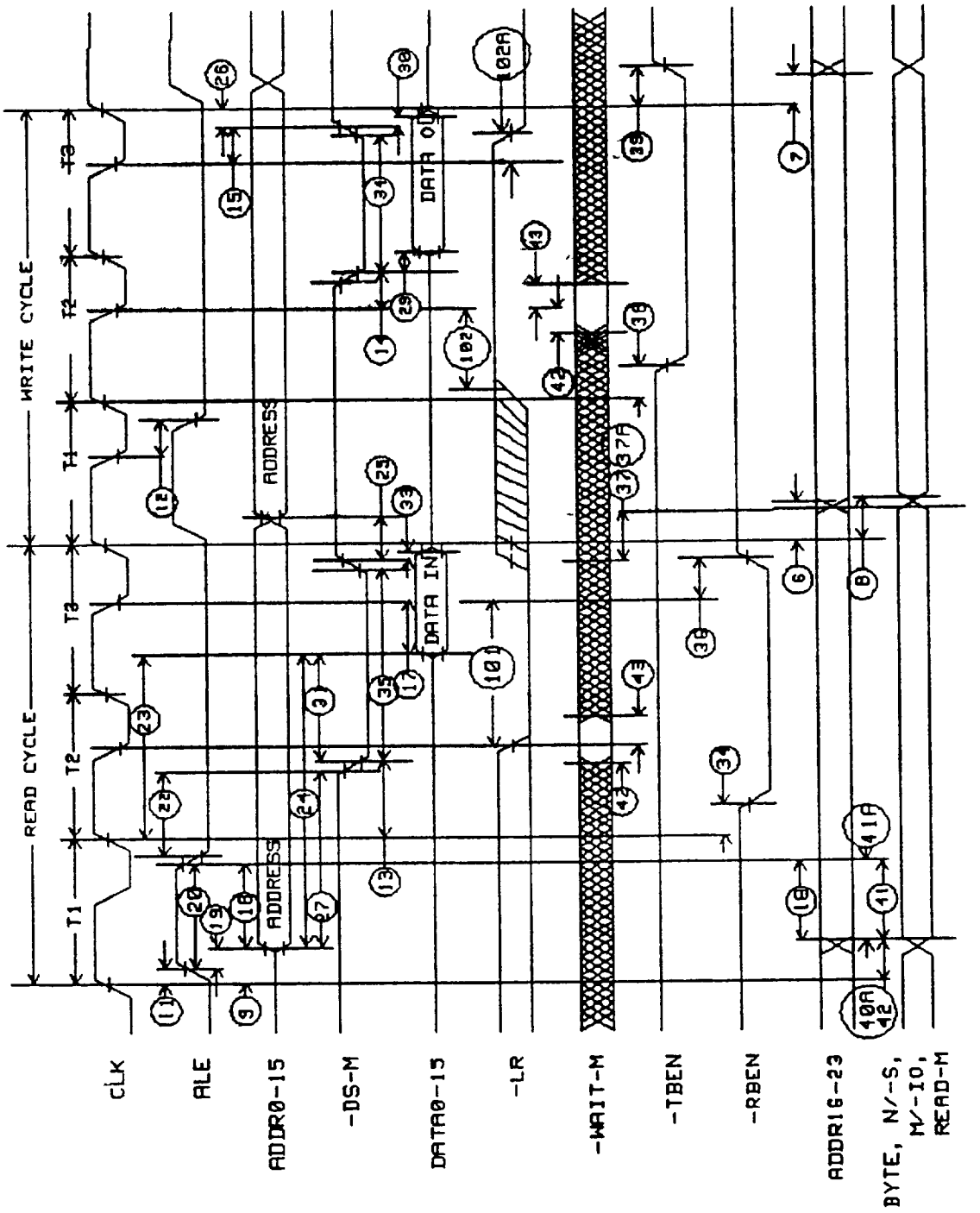
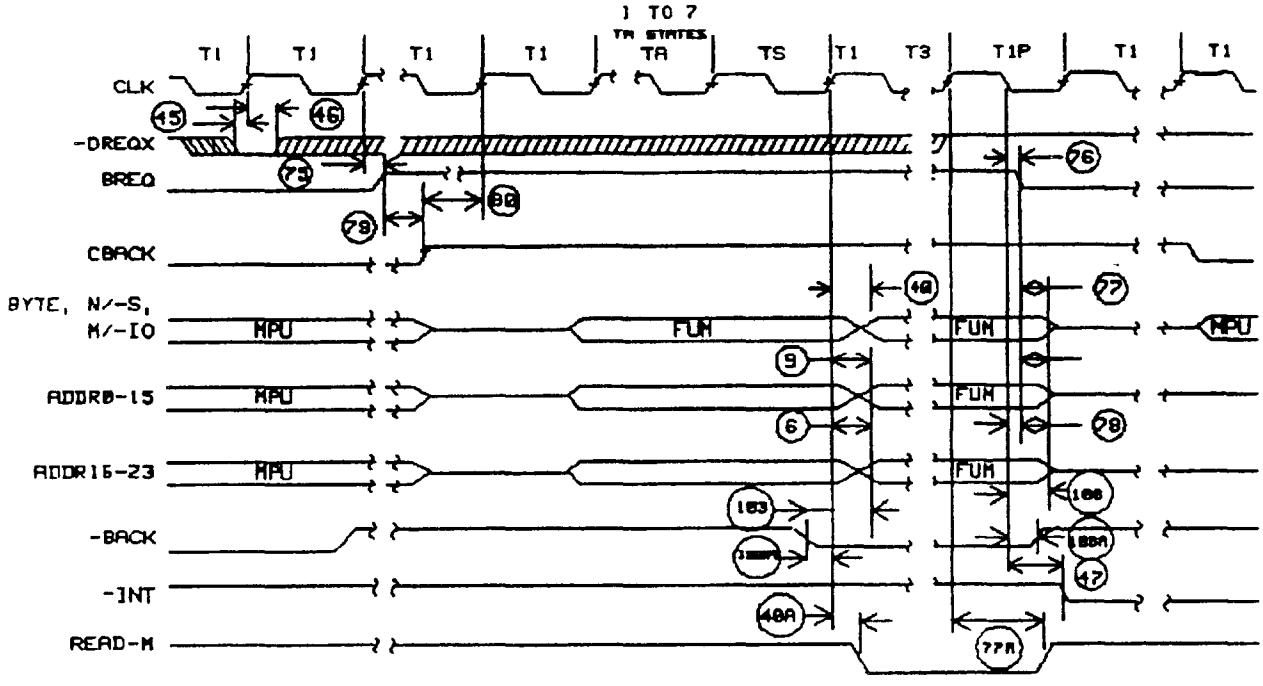


FIGURE 9b. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. Under no circumstances can CBACK be removed prior to BREQ.
2. The above waveforms are bus exchange timing waveforms.

FIGURE 9c. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

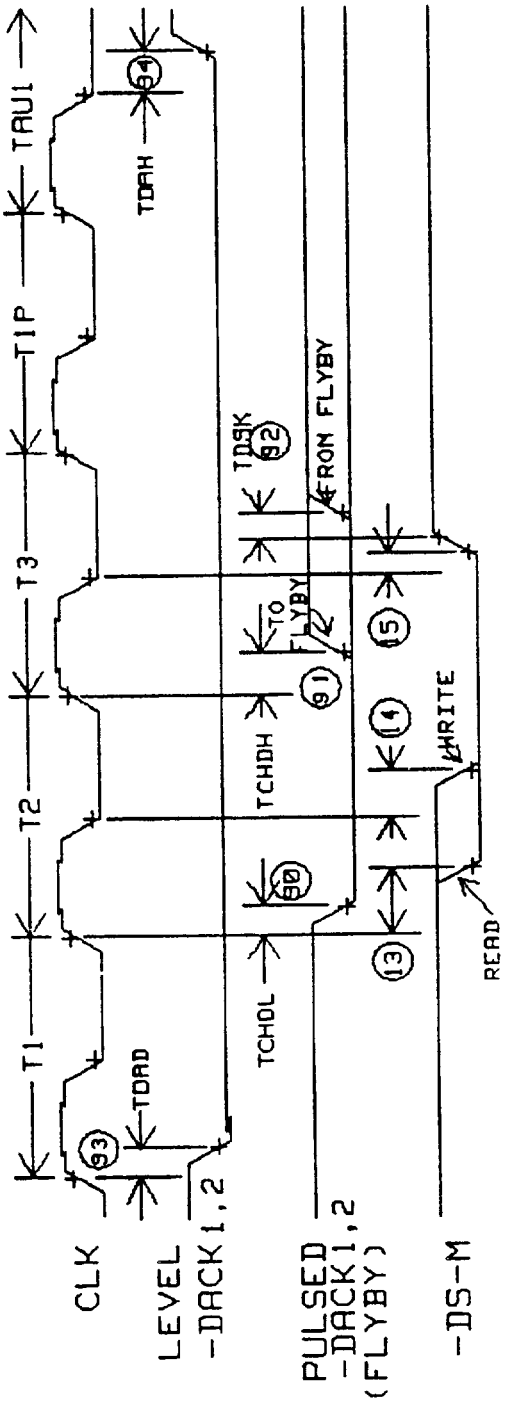


FIGURE 9d. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTE:

1. Level -DACK 1,2 low-to-high transition occurs as shown if auto-reloading is not programmed, otherwise it stays low for three additional clocks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

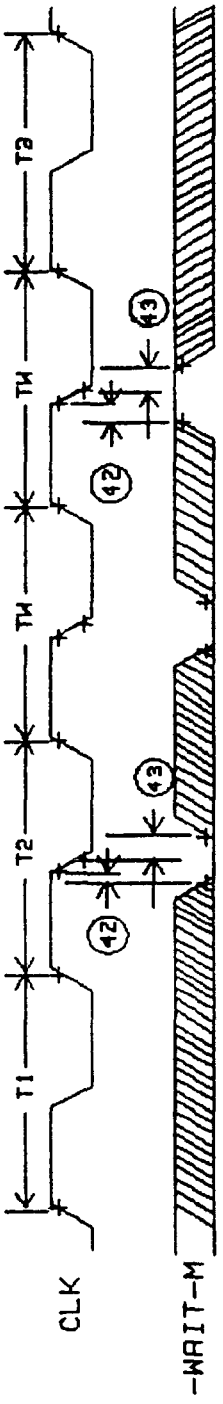
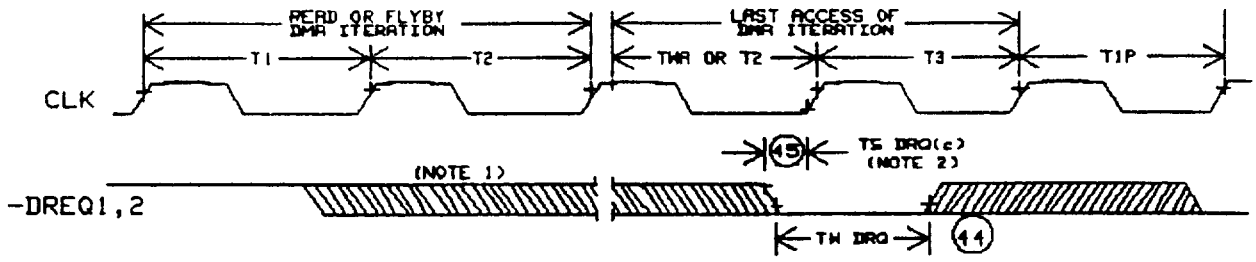


FIGURE 9e. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. High-to-low -DREQX transitions will only be recognized after the high-to-low transition of the clock during T1 of a read or flyby DMA iteration.

2. A high-to-low -DREQX transition must meet the conditions in note 1 and must occur TsDRQ(c) before state T3 of the last access of the DMA iteration if the channel is to retain bus control and immediately start the next iteration. -DREQX may go high before TsDRQ(c) if it has met the TwDRQ parameter.

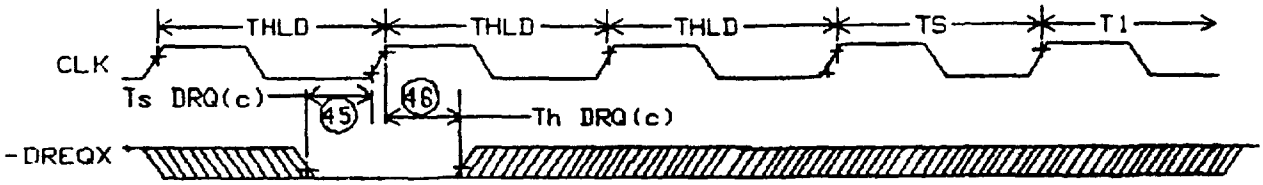
3. Flyby and search transactions have only a single access: parameter TsDRQ(c) should be referenced to the start of T3 of the access. All other operations will always have two or three accesses per iteration.

4. Above waveform is a sampling of -DREQX during single transfer DMA operations.

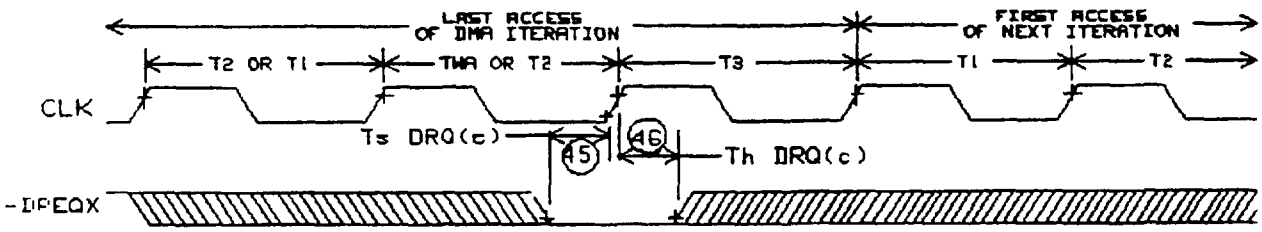
FIGURE 9f. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

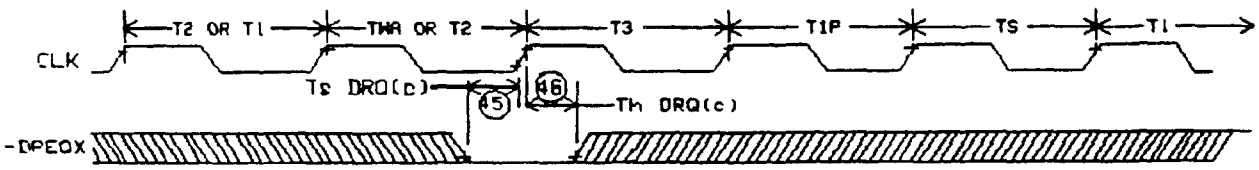
a. Sampling of -DREQX while in bus hold mode.



b. -DREQX sampling in demand mode during DMA operations.



c. Sampling of -DREQX at the end of chaining.



d. Sampling of -DREQX at the end of base-to-current reloading.

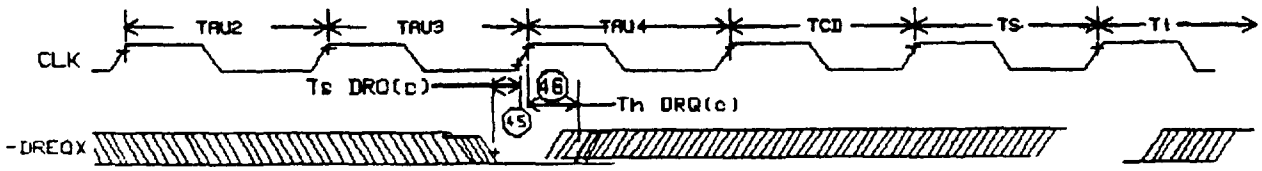


FIGURE 9g. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES:

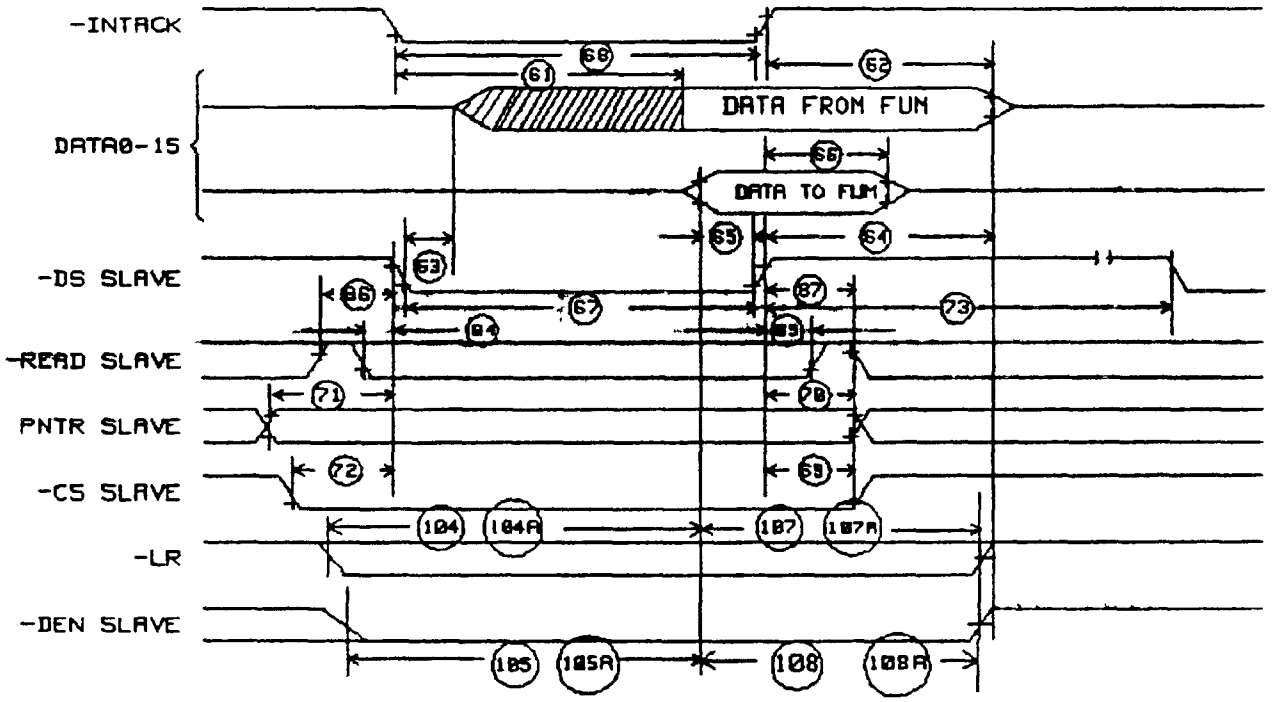
1. -DREQX must be low from the start of TsDRQ(c) to the end of ThDRQ(c) to ensure that the request is recognized.

2. Failure to meet this setup time will result in the channel releasing the bus.

3. Ts is a setup state, generated before entering DMA operation cycle.

4. TAU2, TAU3 and TAU4 are auto-reload states, followed by TCD (chain decision) state.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTE:

1. Above waveforms are ac timing waveforms when FUM module is a bus slave.

FIGURE 9h. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

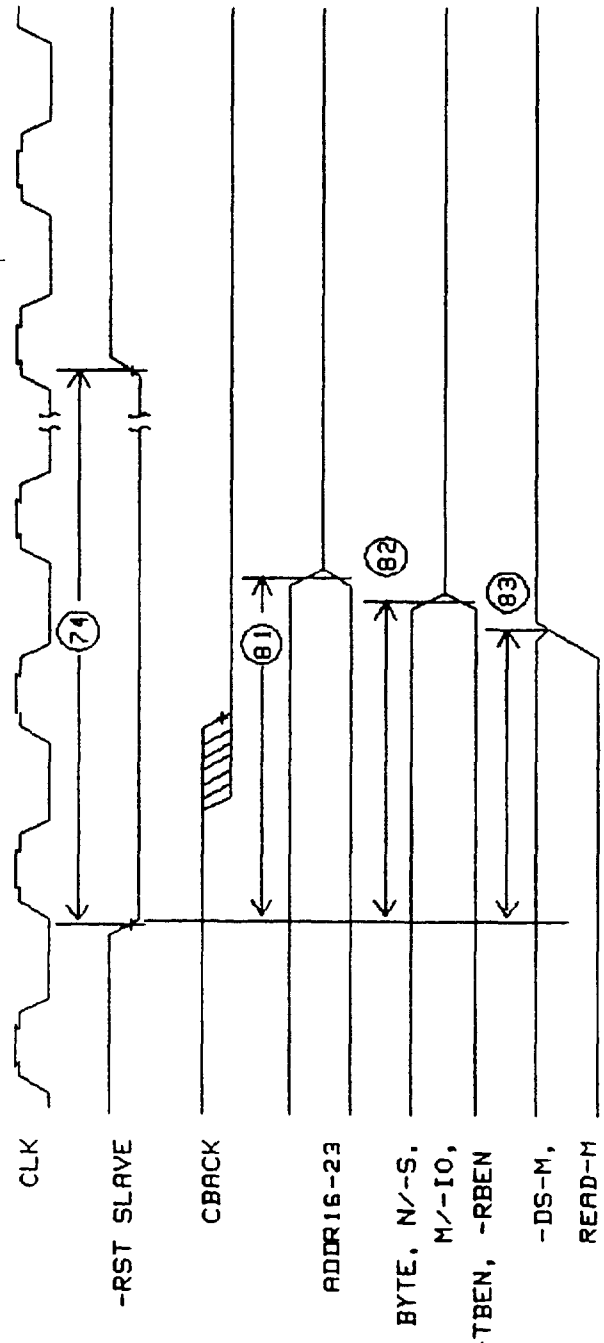
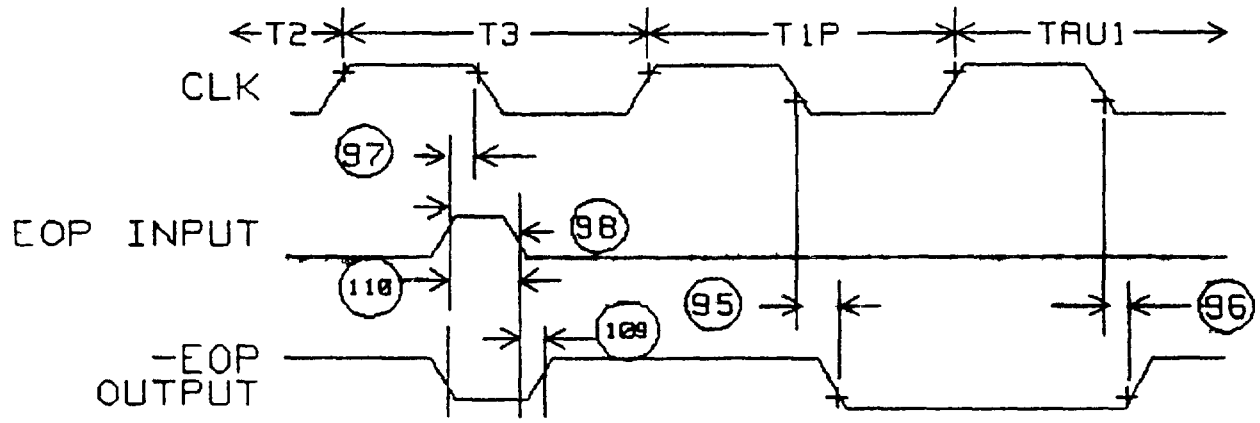


FIGURE 9i. Dynamic waveforms for table XVII.

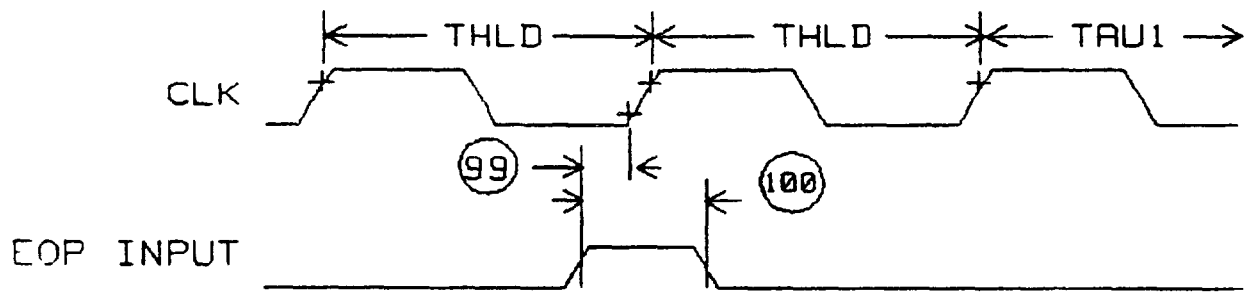
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

a. Sampling and generation of EOP during DMA operations.



b. Sampling of EOP during bus hold.

CHANNEL HOLDS BUS



NOTES:

1. The diagram lists state names for both I/O and memory accesses. Sampling of EOP will occur on the falling edge of state T3.

2. State T1P is a pseudo-T1 state, generated following termination of any DMA operation.

3. TAU1 is an auto-initialization state, generated following the TC, MC, or EOP termination.

4. Above waveforms are end-of-process timing waveforms.

FIGURE 9j. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.92	7.28	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 168 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1,31	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.23 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX. The module inputs shall not violate the minimum timing requirements referenced on figure 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 10 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 12 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

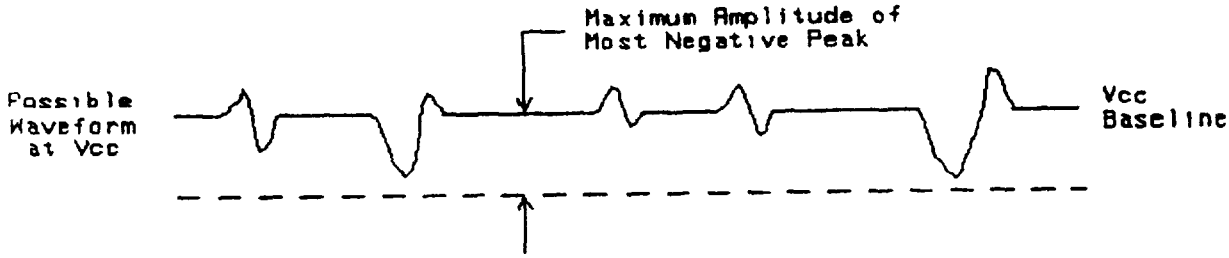


FIGURE 10. Power supply transient amplitude waveform for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
T1	ADDR0	29	0	400	mV	1-4	11
T2	ADDR1	78					
T3	ADDR2	82					
T4	ADDR3	32					
T5	ADDR4	33					
T6	ADDR5	84					
T7	ADDR6	37					
T8	ADDR7	36					
T9	ADDR8	92					
T10	ADDR9	93					
T11	ADDR10	44					
T12	ADDR11	48					
T13	ADDR12	95					
T14	ADDR13	47					
T15	ADDR14	41					
T16	ADDR15	45					
T17	DATA0	79					
T18	DATA1	80					
T19	DATA2	83					
T20	DATA3	34	0	400	mV	1-4	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
T21	DATA4	85	0	400	mV	1-4	11
T22	DATA5	86					
T23	DATA6	88					
T24	DATA7	38					
T25	DATA8	43					
T26	DATA9	94					
T27	DATA10	46					
T28	DATA11	96					
T29	DATA12	97					
T30	DATA13	98					
T31	DATA14	99					
T32	DATA15	49					
T33	READ-M	35					
T34	M/-IO	67					
T35	N/-S	19					
T36	-TBEN	16					
T37	-RBEN	17					
T38	-DACK1	9					
T39	-DACK2	60					
T40	BYTE	13	0	400	mV	1-4	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
T41	BREQ	53	0	400	mV	1-4	11
T42	-INT	51					
T43	ALE	3					
T44	ADDR16	20					
T45	ADDR17	72					
T46	ADDR18	73					
T47	ADDR19	74					
T48	ADDR20	24					
T49	ADDR21	23					
T50	ADDR22	22					
T51	ADDR23	21	0	400	mV	1-4	11
T52	-EOP	7	0	500	mV	1-4	11
T53	-DS-M	42	0	500	mV	1-4	11

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T53
Power supply current limit	1.23 A	NA	T1-T53

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T53 T1-T53
Output crosstalk measurement	 NA NA	 ± 20 mV ± 25 mV	 T1-T51 T52,T53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing requirements referenced on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 11.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

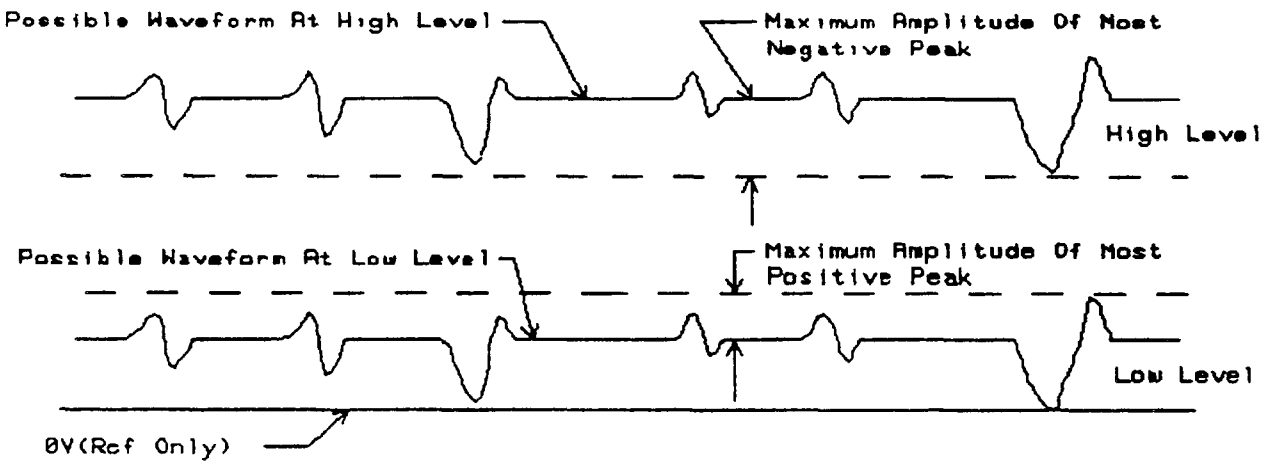


FIGURE 11. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
U1	ADDR0	29	0	400	mV	1-6	11
U2	ADDR1	78					
U3	ADDR2	82					
U4	ADDR3	32					
U5	ADDR4	33					
U6	ADDR5	84					
U7	ADDR6	37					
U8	ADDR7	36					
U9	ADDR8	92					
U10	ADDR9	93					
U11	ADDR10	44					
U12	ADDR11	48					
U13	ADDR12	95					
U14	ADDR13	47					
U15	ADDR14	41					
U16	ADDR15	45					
U17	DATA0	79					
U18	DATA1	80					
U19	DATA2	83					
U20	DATA3	34	0	400	mV	1-6	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
U21	DATA4	85	0	400	mV	1-6	11
U22	DATA5	86					
U23	DATA6	88					
U24	DATA7	38					
U25	DATA8	43					
U26	DATA9	94					
U27	DATA10	46					
U28	DATA11	96					
U29	DATA12	97					
U30	DATA13	98					
U31	DATA14	99					
U32	DATA15	49					
U33	READ-M	35	0	400	mV	1-6	11
U34	M/-IO	67	0	564	mV	1-6	11
U35	N/-S	19	0	540	mV	1-6	11
U36	-TBEN	16	0	400	mV	1-6	11
U37	-RBEN	17					
U38	-DACK1	9					
U39	-DACK2	60	0	400	mV	1-6	11
U40	BYTE	13	0	576	mV	1-6	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
U41	BREQ	53	0	564	mV	1-6	11
U42	-INT	51	0	672	mV	1-6	11
U43	ALE	3	0	400	mV	1-6	11
U44	ADDR16	20					
U45	ADDR17	72					
U46	ADDR18	73					
U47	ADDR19	74					
U48	ADDR20	24					
U49	ADDR21	23					
U50	ADDR22	22					
U51	ADDR23	21	0	400	mV	1-6	11
U52	-EOP	7	0	500	mV	1-6	11
U53	-DS-M	42	0	500	mV	1-6	11
U54	BYTE	13	0	400	mV	1-6	11

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	U1-U54

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit	1.23 A	NA	U1-U54
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U54 U1-U54
Output crosstalk measurement	NA NA NA NA NA NA NA NA NA NA	± 20.0 mV ± 28.2 mV ± 27.0 mV ± 20.0 mV ± 28.8 mV ± 28.2 mV ± 33.6 mV ± 20.0 mV ± 25.0 mV ± 20.0 mV	U1-U33 U34 U35 U36-U39 U40 U41 U42 U43-U51 U52,U53 U54

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing requirements referenced on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 6.

4. Test condition U41 to be tested when -DS SLAVE input is low.

5. Test condition U54 to be tested when -DS-M output is low.

6. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk duration, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
V1	M/-IO	67	0	50	nS	1-4	12
V2	N/-S	19					
V3	BYTE	13	0	50	nS	1-4	12
V4	BREQ	53	0	60	nS	1-4	12
V5	-INT	51	0	50	nS	1-4	12

NOTES:

1. Output crosstalk duration, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	V1-V5
Power supply current limit	1.23 A	NA	V1-V5
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V5 V1-V5
Output crosstalk measurement	NA NA NA	± 2.5 nS ± 3.0 nS ± 2.5 nS	V1-V3 V4 V5

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement for the test conditions shown in table XXII. The module inputs shall not violate the minimum input timing requirements shown on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

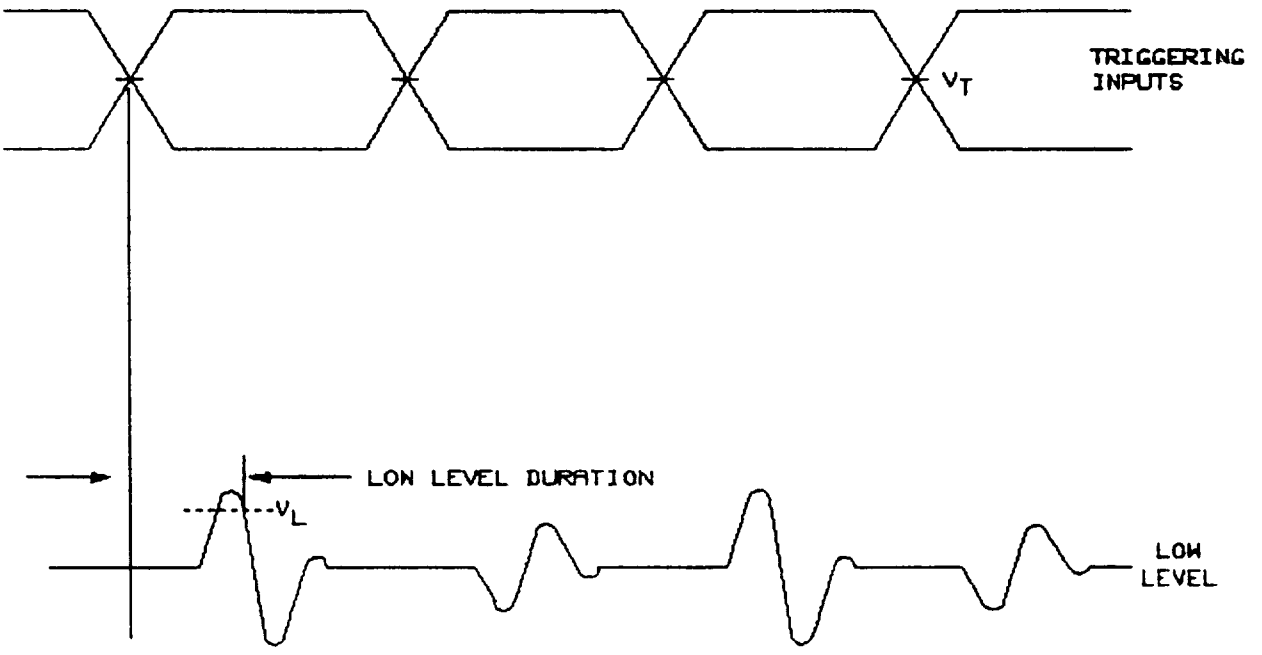
(d) The duration shall be measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage as shown on figure 12. If the crosstalk amplitude is smaller than the duration measurement voltage (VL = 400 mV), the crosstalk duration shall be recorded as 0 nS.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTES:

1. Vl = 400 mV (AC coupled)

2. The crosstalk duration is measured from the triggering edge of the input to the point where crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than VL, the crosstalk duration shall be recorded as 0 nS.

FIGURE 12. Output crosstalk, duration waveforms for table XXII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Output crosstalk pulse width, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
W1	-INT	51	0	10	nS	1-4	13
W2	BREQ	53	0	10	nS	1-4	13

NOTES:

1. Output crosstalk pulse width, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	W1,W2
Power supply current limit	1.23 A	NA	W1,W2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 W1,W2 W1,W2
Output crosstalk measurement	NA	± 0.5 nS	W1,W2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement for the test conditions shown in table XXIII. The module inputs shall not violate the minimum input timing requirements shown on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

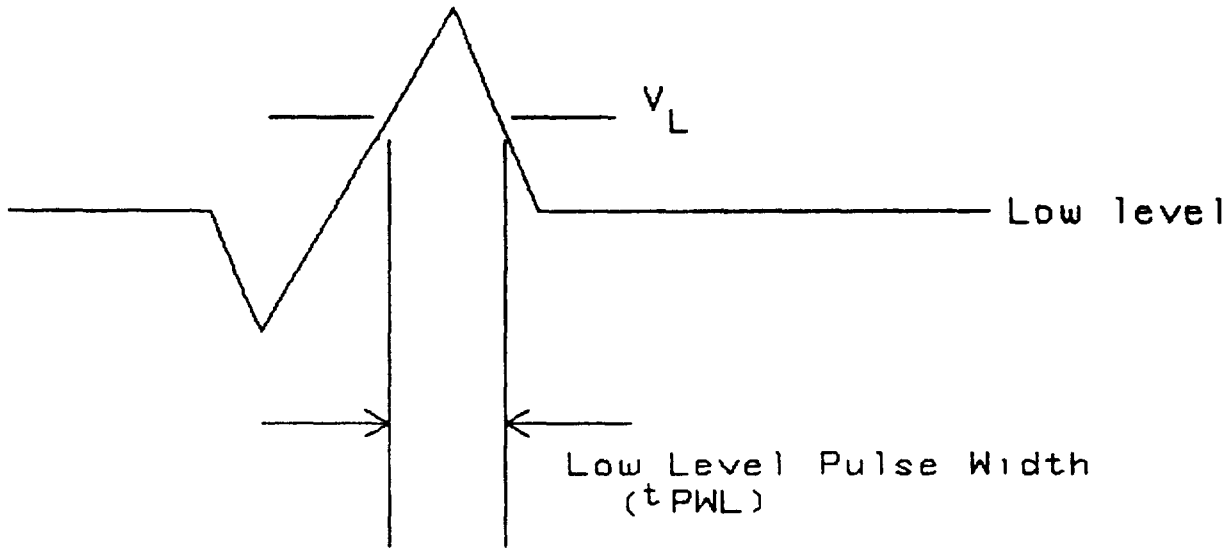
(d) The pulse width shall be measured from the rising edge to the falling edge of the output crosstalk at the specified voltage as shown on figure 13. If the crosstalk is less than the specified reference voltage, the output pulse width shall be recorded as 0 nS.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.



NOTE:

1. VL = 400 mV.

FIGURE 13. Output crosstalk pulse width, low level, test circuit for table XXIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	-CS SLAVE	26	2	30	pF	1-3
X2	PNTR SLAVE	81				
X3	-RST SLAVE	27				
X4	-DREQ2	89				
X5	-DS SLAVE	18				
X6	-DEN SLAVE	8	2	30	pF	1-3
X7	-BACK	28	2	55	pF	1-3
X8	READ-M	35	2	30	pF	1-3
X9	-WAIT-M	77	2	25	pF	1-3
X10	CBACK	30				
X11	-READ SLAVE	71				
X12	-INTACK	56				
X13	ADDR0	29				
X14	ADDR1	78				
X15	ADDR19	74				
X16	-INT	51	2	25	pF	1-3
X17	DATA0	79	2	35	pF	1-3
X18	DATA1	80				
X19	DATA2	83				
X20	DATA3	34				
X21	DATA4	85	2	35	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X22	DATA5	86	2	35	pF	1-3
X23	DATA6	88				
X24	DATA7	38				
X25	DATA8	43				
X26	DATA9	94				
X27	DATA10	46				
X28	DATA11	96				
X29	DATA12	97				
X30	DATA13	98				
X31	DATA14	99				
X32	DATA15	49				
X33	-DREQ1	39				
X34	EOP	65	2	35	pF	1-3
X35	CLK	4	2	20	pF	1-3
X36	ADDR2	82				
X37	ADDR3	32				
X38	ADDR4	33				
X39	ADDR5	84				
X40	ADDR6	37				
X41	ADDR7	36	2	20	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X42	ADDR8	92	2	20	pF	1-3
X43	ADDR10	44				
X44	ADDR11	48				
X45	ADDR12	95				
X46	ADDR13	47				
X47	ADDR14	41				
X48	ADDR15	45				
X49	ADDR16	20				
X50	ADDR17	72				
X51	ADDR18	73				
X52	ADDR20	24				
X53	ADDR21	23				
X54	ADDR22	22				
X55	ADDR23	21				
X56	M/-IO	67				
X57	N/-S	19				
X58	-TBEN	16				
X59	-RBEN	17				
X60	-DACK1	9				
X61	-DACK2	60	2	20	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X62	BYTE	13	2	20	pF	1-3
X63	BREQ	53	"	"	"	"
X64	ADDR9	93	2	20	pF	1-3
X65	ALE	3	6	75	pF	1-3
X66	-LR	91	4	45	pF	1-3
X67	-DS-M	42	4	45	pF	1-3

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X67
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X67
Capacitance measurement	NA	± 1.4 pF	X1-X8
	NA	± 1.15 pF	X9-X16
	NA	± 1.65 pF	X17-X34
	NA	± 1.9 pF	X35-X64
	NA	± 3.45 pF	X65
	NA	± 2.05 pF	X66,X67

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXV. Threshold test.

Test cond	Parameter	Notes
Y	Threshold	1-3

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Y
Power supply current limit	1.23 A	NA	Y
Pattern voltage			
high, (VH)	2.0 V	± 20 mV	Y
low, (VL)	0.7 V	± 7 mV	Y
Output voltage measurement			
high, (Voh)	NA	± 10.0 mV	Y
high, (Voh)	NA	± 105.0 mV	Y
high, (Voh)	NA	± 100.0 mV	Y
low, (Vol)	NA	± 22.5 mV	Y
low, (Vol)	NA	± 15.0 mV	Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output disable time test (Tphz and Tplz).

Test cond	Parameter	Pin no.	TA = 25 ± 5°C Initial limits Min	Max	Units	Figs	Ref.
Z1	-DEN SLAVE to DATA0 Slave Mode (Tphz)	79	4	53	nS	14,15	1
Z2	-DEN SLAVE to DATA1 Slave Mode (Tphz)	80					
Z3	-DEN SLAVE to DATA2 Slave Mode (Tphz)	83					
Z4	-DEN SLAVE to DATA3 Slave Mode (Tphz)	34					
Z5	-DEN SLAVE to DATA4 Slave Mode (Tphz)	85					
Z6	-DEN SLAVE to DATA5 Slave Mode (Tphz)	86					
Z7	-DEN SLAVE to DATA6 Slave Mode (Tphz)	88					
Z8	-DEN SLAVE to DATA7 Slave Mode (Tphz)	38					
Z9	-DEN SLAVE to DATA8 Slave Mode (Tphz)	43					
Z10	-DEN SLAVE to DATA9 Slave Mode (Tphz)	94					
Z11	-DEN SLAVE to DATA10 Slave Mode (Tphz)	46					
Z12	-DEN SLAVE to DATA11 Slave Mode (Tphz)	96					
Z13	-DEN SLAVE to DATA12 Slave Mode (Tphz)	97					
Z14	-DEN SLAVE to DATA13 Slave Mode (Tphz)	98					
Z15	-DEN SLAVE to DATA14 Slave Mode (Tphz)	99					
Z16	-DEN SLAVE to DATA15 Slave Mode (Tphz)	49					
Z17	-DEN SLAVE to DATA0 Slave Mode (Tplz)	79					
Z18	-DEN SLAVE to DATA1 Slave Mode (Tplz)	80					
Z19	-DEN SLAVE to DATA2 Slave Mode (Tplz)	83					
Z20	-DEN SLAVE to DATA3 Slave Mode (Tplz)	34					
Z21	-DEN SLAVE to DATA4 Slave Mode (Tplz)	85	4	53	nS	14,15	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output disable time test (Tphz and Tplz) - Continued.

Test cond	Parameter	Pin no.	TA = 25 ± 5°C Initial limits Min	Max	Units	Figs	Ref.
Z22	-DEN SLAVE to DATA5 Slave Mode (Tplz)	86	4	53	nS	14,15	1
Z23	-DEN SLAVE to DATA6 Slave Mode (Tplz)	88					
Z24	-DEN SLAVE to DATA7 Slave Mode (Tplz)	38					
Z25	-DEN SLAVE to DATA8 Slave Mode (Tplz)	43					
Z26	-DEN SLAVE to DATA9 Slave Mode (Tplz)	94					
Z27	-DEN SLAVE to DATA10 Slave Mode (Tplz)	46					
Z28	-DEN SLAVE to DATA11 Slave Mode (Tplz)	96					
Z29	-DEN SLAVE to DATA12 Slave Mode (Tplz)	97					
Z30	-DEN SLAVE to DATA13 Slave Mode (Tplz)	98					
Z31	-DEN SLAVE to DATA14 Slave Mode (Tplz)	99					
Z32	-DEN SLAVE to DATA15 Slave Mode (Tplz)	49	4	53	nS	14,15	1
Z33	-BACK to ADDR0 Master Mode (Tphz)	29	2	56	nS	14,16	2
Z34	-BACK to ADDR1 Master Mode (Tphz)	78					
Z35	-BACK to ADDR2 Master Mode (Tphz)	82					
Z36	-BACK to ADDR3 Master Mode (Tphz)	32					
Z37	-BACK to ADDR4 Master Mode (Tphz)	33					
Z38	-BACK to ADDR5 Master Mode (Tphz)	84					
Z39	-BACK to ADDR6 Master Mode (Tphz)	37					
Z40	-BACK to ADDR7 Master Mode (Tphz)	36					
Z41	-BACK to ADDR8 Master Mode (Tphz)	92					
Z42	-BACK to ADDR9 Master Mode (Tphz)	93	2	56	nS	14,16	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output disable time test (Tphz and Tplz) - Continued.

Test cond	Parameter	Pin no.	TA = 25 ± 5°C Initial limits Min	Max	Units	Figs	Ref.
Z43	-BACK to ADDR10 Master Mode (Tphz)	44	2	56	nS	14,16	2
Z44	-BACK to ADDR11 Master Mode (Tphz)	48					
Z45	-BACK to ADDR12 Master Mode (Tphz)	95					
Z46	-BACK to ADDR13 Master Mode (Tphz)	47					
Z47	-BACK to ADDR14 Master Mode (Tphz)	41					
Z48	-BACK to ADDR15 Master Mode (Tphz)	45	2	56	nS	14,16	2
Z49	-BACK to ADDR0 Master Mode (Tplz)	29	2	39	nS	14,16	2
Z50	-BACK to ADDR1 Master Mode (Tplz)	78					
Z51	-BACK to ADDR2 Master Mode (Tplz)	82					
Z52	-BACK to ADDR3 Master Mode (Tplz)	32					
Z53	-BACK to ADDR4 Master Mode (Tplz)	33					
Z54	-BACK to ADDR5 Master Mode (Tplz)	84					
Z55	-BACK to ADDR6 Master Mode (Tplz)	37					
Z56	-BACK to ADDR7 Master Mode (Tplz)	36					
Z57	-BACK to ADDR8 Master Mode (Tplz)	92					
Z58	-BACK to ADDR9 Master Mode (Tplz)	93					
Z59	-BACK to ADDR10 Master Mode (Tplz)	44					
Z60	-BACK to ADDR11 Master Mode (Tplz)	48					
Z61	-BACK to ADDR12 Master Mode (Tplz)	95					
Z62	-BACK to ADDR13 Master Mode (Tplz)	47					
Z63	-BACK to ADDR14 Master Mode (Tplz)	41	2	39	nS	14,16	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output disable time test (Tphz and Tplz) - Continued.

Test cond	Parameter	Pin no.	TA = 25 ± 5°C Initial limits Min	Max	Units	Figs	Ref.
Z64	-BACK to ADDR15 Master Mode (Tplz)	45	2	39	nS	14,16	2
Z65	-RST SLAVE to M/-IO Master Mode (Tphz)	67	4	286	nS	14,17	3
Z66	-RST SLAVE to N/-S Master Mode (Tphz)	19					
Z67	-RST SLAVE to BYTE Master Mode (Tphz)	13	4	286	nS	14,17	3
Z68	-RST SLAVE to M/-IO Master Mode (Tplz)	67	4	203	nS	14,17	3
Z69	-RST SLAVE to N/-S Master Mode (Tplz)	19					
Z70	-RST SLAVE to BYTE Master Mode (Tplz)	13	4	203	nS	14,17	3
Z71	-RST SLAVE to ADDR16 Master Mode (Tphz)	20	4	296	nS	14,17	4
Z72	-RST SLAVE to ADDR17 Master Mode (Tphz)	72					
Z73	-RST SLAVE to ADDR18 Master Mode (Tphz)	73					
Z74	-RST SLAVE to ADDR19 Master Mode (Tphz)	74					
Z75	-RST SLAVE to ADDR20 Master Mode (Tphz)	24					
Z76	-RST SLAVE to ADDR21 Master Mode (Tphz)	23					
Z77	-RST SLAVE to ADDR22 Master Mode (Tphz)	22					
Z78	-RST SLAVE to ADDR23 Master Mode (Tphz)	21	4	296	nS	14,17	4
Z79	-RST SLAVE to ADDR16 Master Mode (Tplz)	20	4	163	nS	14,17	4
Z80	-RST SLAVE to ADDR17 Master Mode (Tplz)	72					
Z81	-RST SLAVE to ADDR18 Master Mode (Tplz)	73					
Z82	-RST SLAVE to ADDR19 Master Mode (Tplz)	74					
Z83	-RST SLAVE to ADDR20 Master Mode (Tplz)	24					
Z84	-RST SLAVE to ADDR21 Master Mode (Tplz)	23	4	163	nS	14,17	4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Output disable time test (Tphz and Tplz) - Continued.

Test cond	Parameter	Pin no.	TA = 25 ± 5°C Initial limits Min	Max	Units	Figs	Ref.
Z85	-RST SLAVE to ADDR22 Master Mode (Tplz)	22	4	163	nS	14,17	4
Z86	-RST SLAVE to ADDR23 Master Mode (Tplz)	21	4	163	nS	14,17	4

NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Current limits	Test condition
Power supply	5.0 V	± 50 mV	1235 mA	Z1-Z86
Pattern voltage				
high (VH)	3.0 V	± 100 mV	N/A	Z1-Z86
low (VL)	0.0 V	± 100 mV	N/A	Z1-Z86
Output time measurement	N/A	± 2.45nS	N/A	Z1-Z32
	N/A	± 2.70nS	N/A	Z33-Z48
	N/A	± 1.85nS	N/A	Z49-Z64
	N/A	± 14.10nS	N/A	Z65-Z67
	N/A	± 9.95nS	N/A	Z68-Z70
	N/A	± 14.66nS	N/A	Z71-Z78
	N/A	± 7.95nS	N/A	Z79-Z86

2. The following procedure is required to perform this test:

(a) Apply power to the module. The module shall have the power supply voltage to all VCC pins. The dc return of the power supply shall be connected to all ground pins of the modules. The power supply voltage overshoot shall not exceed 2.0 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Use the pattern sets contained in the appendix for preconditioning the module under test and verify the output patterns by monitoring the outputs and comparing the results with the guaranteed output low and high levels specified in tables X and XI. Input pulses shall meet the requirements shown on figure 8.

(c) When making the test indicated on a pin, ensure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Module inputs shall meet the timing requirements shown on figures 14-17.

4. Output pins 79, 80, 83, 34, 85, 86, 88, 38, 43, 94, 46, 96, 97, 98, 99, 49, 29, 78, 82, 32, 33, 84, 37,36, 92, 93, 44, 48, 95, 47, 41, 45, 67, 19, 13, 20, 72, 73, 74, 24, 23, 22, 21 shall be loaded as specified on figure 6.

5. The output disable time shall be measured from the triggering edge of the input, as shown on figure 14.

6. N/A = not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

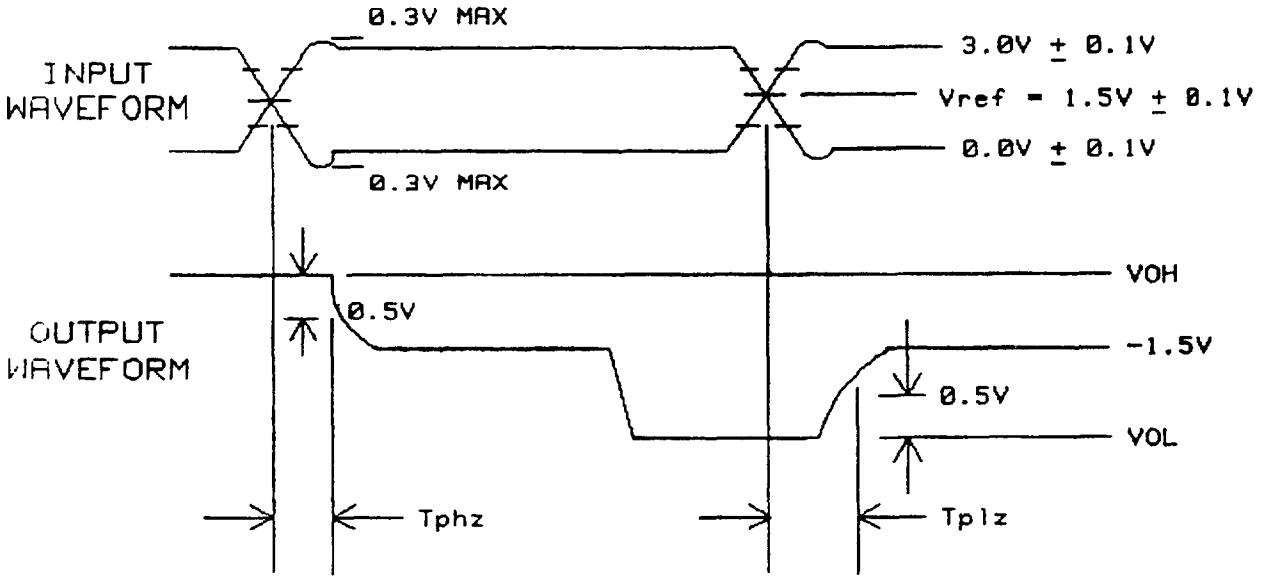


FIGURE 14. Disable time test circuit waveform for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

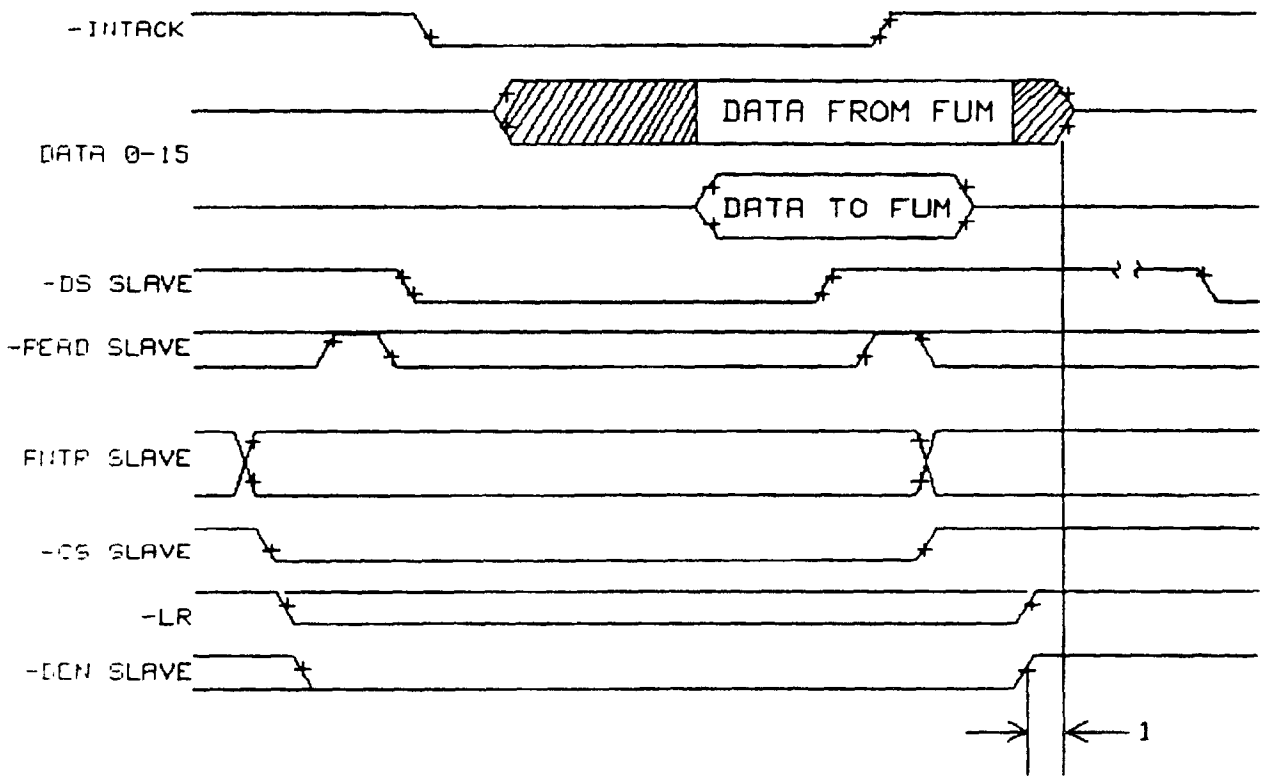


FIGURE 15. Disable time test circuit waveforms for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

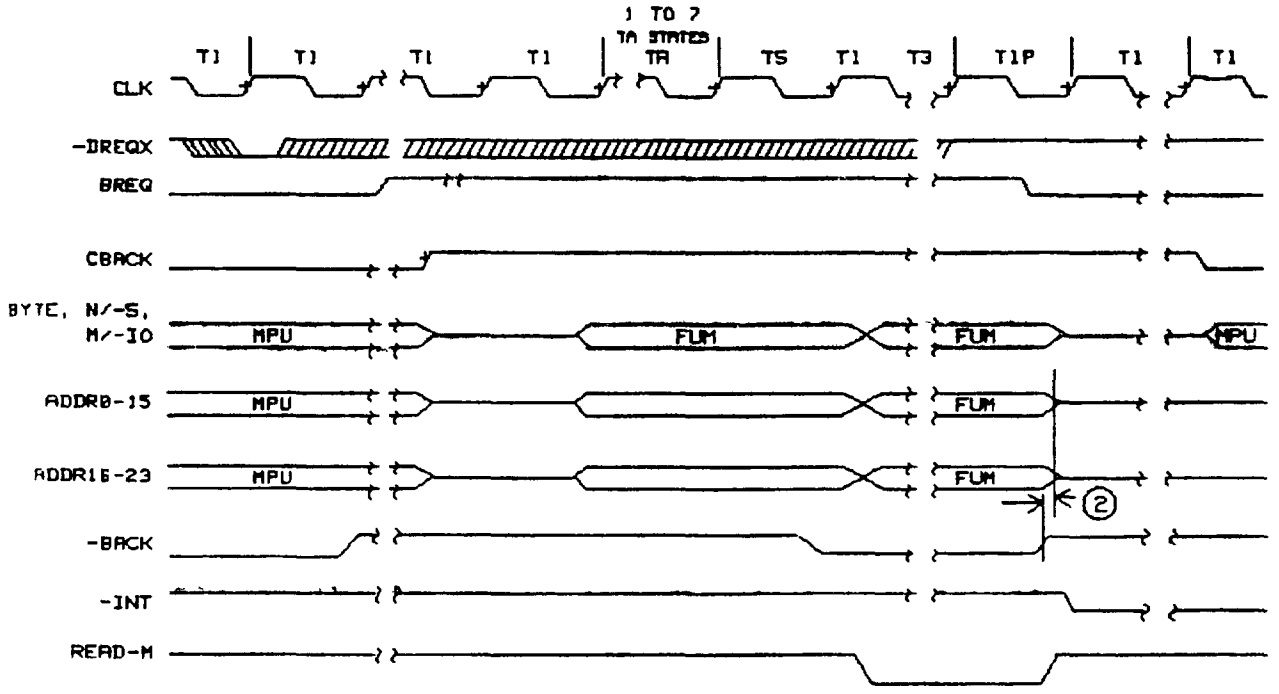


FIGURE 16. Disable time test circuit waveforms for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

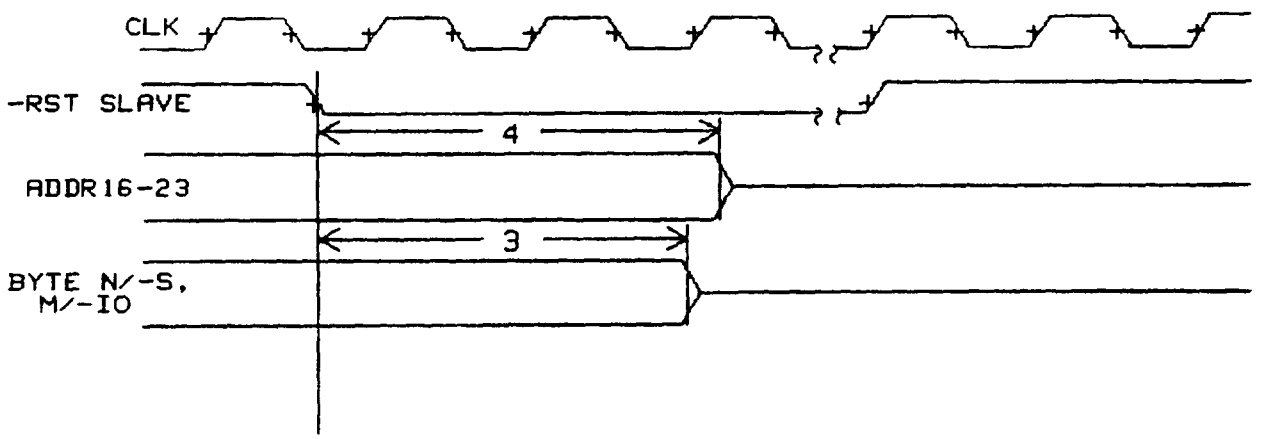


FIGURE 17. Disable time test circuit waveforms for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30. Module function.

30.1 AM9516 register description. The AM9516 block diagram (figure 1) illustrates the internal registers. Figure 2 lists each register along with its size and read/write access restrictions. Registers which can be read by the microprocessor (MPU) are either fast (F) or slow (S) readable. Fast registers may be read by a normal MPU I/O operation without additional wait states. Reading slow registers requires multiple wait states. Registers can be written to by the host MPU (W) and/or can be loaded by the DMA channel itself during chaining (C). All reads or writes must be word accesses since the AM9516 ignores the BYTE output line in slave mode. It is the responsibility of the user to supply the necessary external logic if slow readable registers are to be read. The internal registers are read or written in two steps. First, the address of the register to be accessed is written to the pointer register, when the PNTR SLAVE input is HIGH. Then, the data is read from or written into the desired register which is indicated by the pointer register, when the PNTR SLAVE input is LOW. Note that a read with PNTR SLAVE HIGH causes the contents of the pointer register to be read on the output data bus (DATA 0-15).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

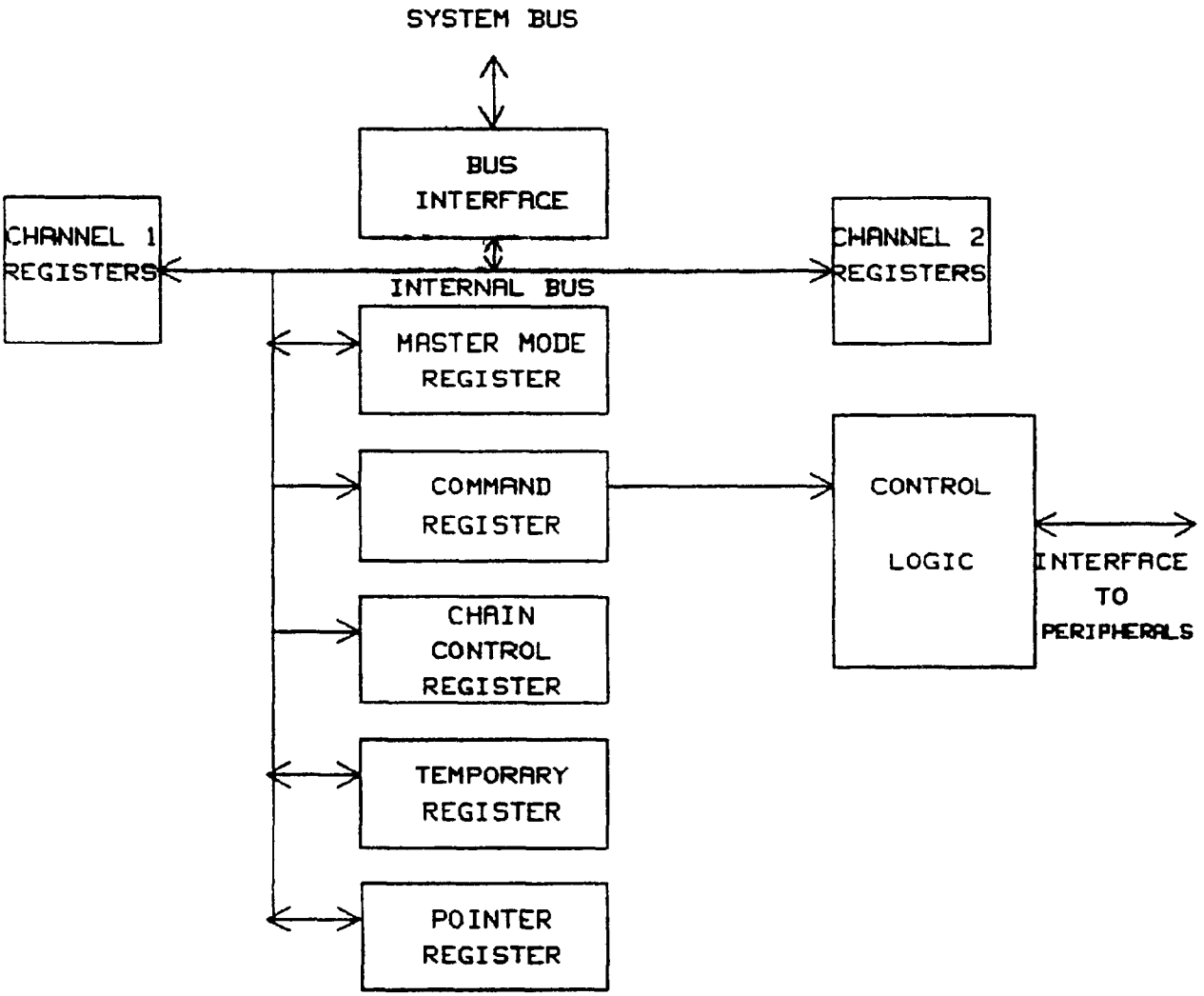


FIGURE 1. AM9516 block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

Name	Size	Number	Access	Port address
Master mode register	4 bits	1	FW	38
Pointer register	6 bits	1	FW	
Chain-control register	10 bits	1	C	
Temporary register	16 bits	1	D	
Command register	8 bits	1	W	2E/2C*
Current address register-A:				
Up-addr/tag field	14 bits	2	CFW	1A/18
Lower address field	16 bits	2	CFW	0A/08
Current address register-B:				
Up-addr/tag field	14 bits	2	CFW	12/10
Lower address field	16 bits	2	CFW	02/00
Base address register-A:				
Up-addr/tag field	14 bits	2	CFW	1E/1C
Lower address field	16 bits	2	CFW	0E/0C
Base address register-B:				
Up-addr/tag field	14 bits	2	CFW	16/14
Lower address field	16 bits	2	CFW	06/04
Current operation count	16 bits	2	CFW	32/30
Base operation count	16 bits	2	CFW	36/34

Access codes: C = Chain loadable, D = Accessible only by AM9516 during transfers, F = Fast readable, S = Slow readable, W = Writeable by MPU

NOTE:

1. The address of the register to be accessed is stored in the pointer register.

*Port addresses of command register can be used alternately for both channels except when issuing a "set of clear IP" commands.

FIGURE 2. AM9516 internal registers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

Name	Size	Number	Access	Port address
Pattern register	16 bits	2	CSW	4A/48
Mask register	16 bits	2	CSW	4E/4C
Status register	16 bits	2	F	2E/2C
Interrupt save register	16 bits	2	F	2A/28
Interrupt vector register	8 bits	2	CSW	5A/58
Channel mode register - High	5 bits	2	CS	56/54
Channel mode register - Low	16 bits	2	CSW	52/50
Chain address register:				
Up-addr/tag field	10 bits	2	CFW	26/24
Lower address field	16 bits	2	CFW	22/20

Access codes: C = Chain loadable, D = Accessible only by AM9516 during transfers, F = Fast readable, S = Slow readable, W = Writeable by MPU

NOTE:

1. The address of the register to be accessed is stored in the pointer register.

FIGURE 2. AM9516 internal registers - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.2 Master mode register. The 4-bit master mode register, shown on figure 8 controls the AM9516 chip-level interface. It can be read from and written to by the host MPU, but is not loadable by chaining. On a reset, the master mode register is cleared to all zeroes. The function of each of the master mode bits is described in the following paragraphs:

30.2.1 DMA enable. The chip enable bit CE = 1 enables the AM9516 to request the bus. When enabled, the AM9516 can perform DMA operations and reload registers. It can always issue interrupts and respond to interrupt acknowledges. When the chip enable bit is cleared, the AM9516 is inhibited from requesting control of the system bus and, therefore, inhibited from performing chaining or DMA operations.

30.2.2 Interleave enable. The MPU interleave bit enables interleaving between the MPU and the FUM.

30.2.3 Wait line enable. The wait line enable bit is used to enable sampling of the -WAIT-M line during memory and I/O transactions. Because the AM9516 provides the ability to insert software programmable wait states, many users may disable sampling of the -WAIT-M pin to eliminate the logic driving this pin.

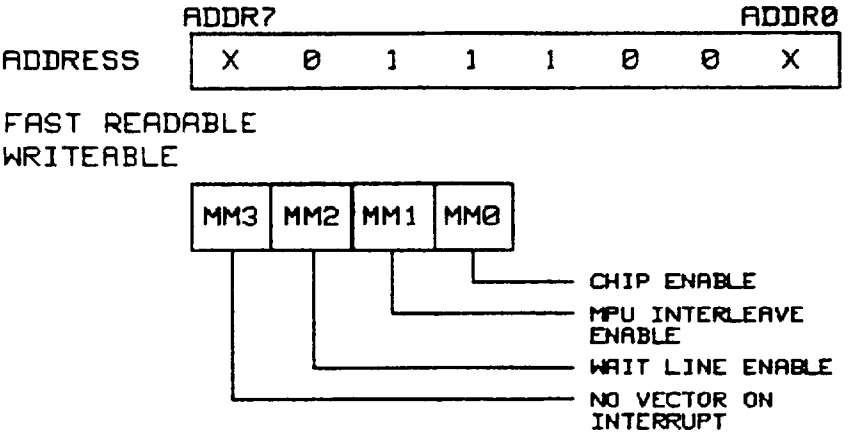


FIGURE 3. Master mode register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.2.4 No vector on interrupt. The "No Vector on Interrupt" bit selects whether the AM9516 channel or a peripheral returns a vector during interrupt acknowledge cycles. When this bit is cleared, a channel receiving an interrupt acknowledge will drive the contents of its interrupt save register onto the DATA0-15 data bus while -INTACK is low. If this bit is set, interrupts are serviced in an identical manner, but the data bus outputs are undefined throughout the acknowledge cycle.

30.3 Pointer register. The pointer register contains the address of the internal register to be accessed. It can be read or written by the MPU when the PNTR SLAVE input line is HIGH.

30.4 Chain control register. When a channel starts a chaining operation, it fetches a reload word from the memory location pointed to by the chain address register (see figure 4). This word is then stored in the chain control register (see figure 5). The chain control register cannot be written to or read from by the MPU. Once a channel starts a chain operation, the channel will not relinquish bus control until all registers specified in the reload word are reloaded unless an EOP signal is issued to the module. Issuing an EOP to a channel during chaining will prevent the chain operation from resuming and the contents of the reload word register can be discarded.

30.5 Temporary register. The temporary register is used to store data during flowthru transfers and to hold data being compared during a search of a transfer-and-search. The temporary register cannot be written to or read from by the MPU. In byte-word funnelling, data may be loaded into or from the temporary register on a byte-by-byte basis, with bytes sometimes moving between the low byte of the data bus and the high byte of the temporary register or vice versa.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

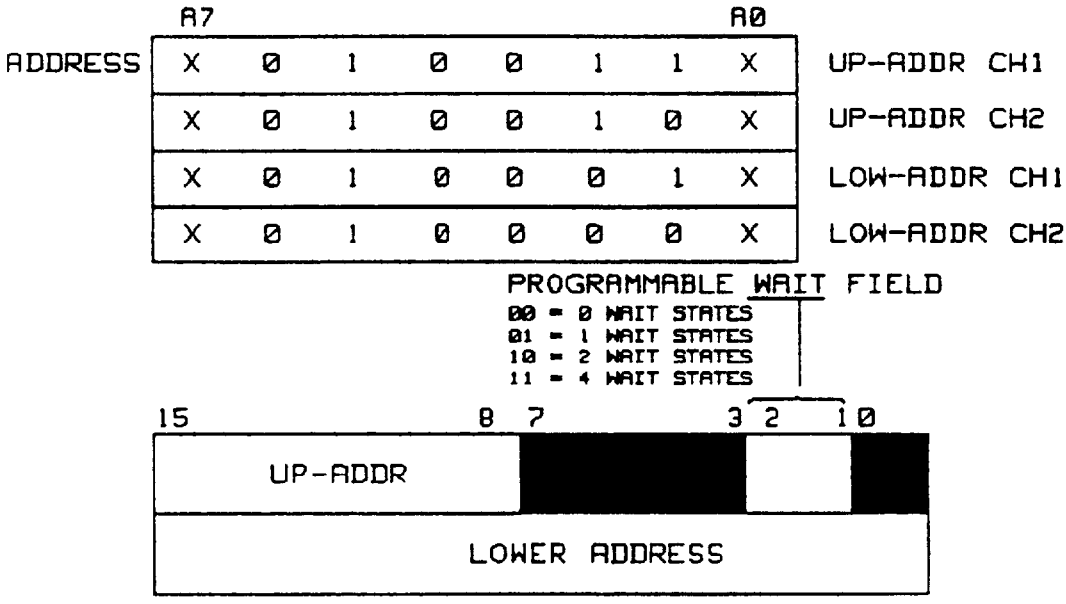


FIGURE 4. Chain address register.

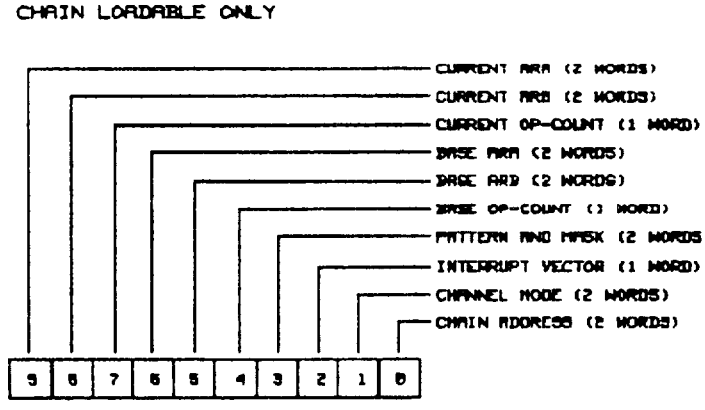


FIGURE 5. Chain control register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

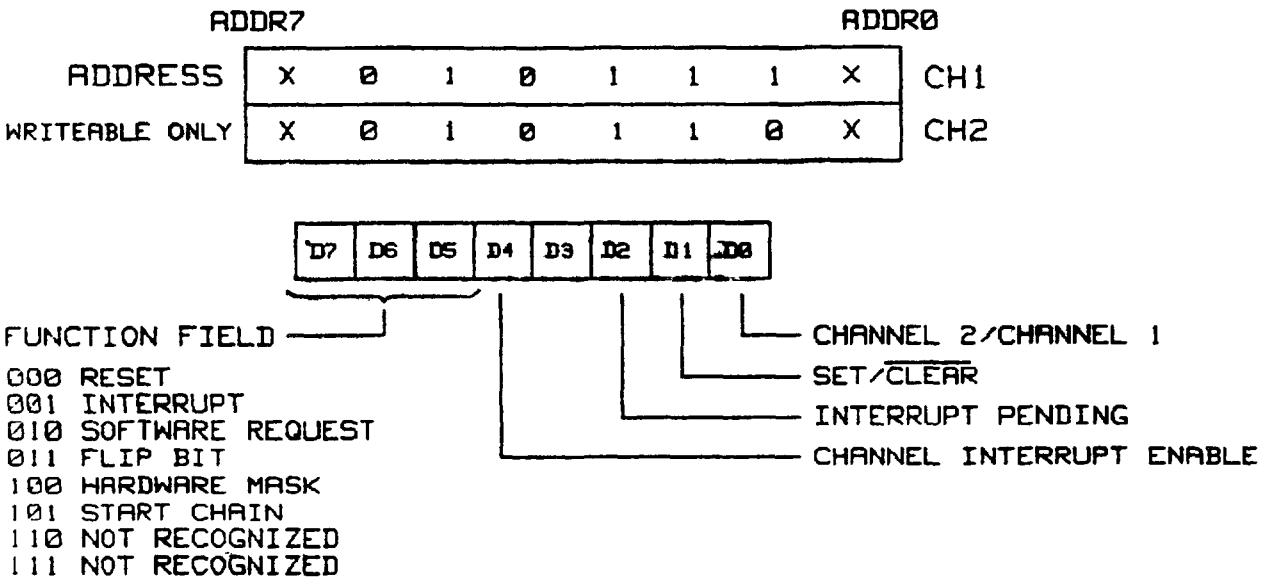


FIGURE 6. Command register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.6 Command register. The command register (see figure 6) is an 8-bit write-only register written to by the host MPU. The command register is loaded from data bits DATA0-7; data bits DATA8-15 are disregarded.

30.6.1 Command register opcode. Figure 7 shows a list of AM9516 commands. The commands are executed immediately after being written by the host MPU into the AM9516 command register. A description of each command follows.

Command	Opcode bits 7654	3210	Example Code hex
Reset	000X	XXXX	00
Start chain channel 1	101X	XXX0	A0
Start chain channel 2	101X	XXX1	A1
Set software request channel 1	010X	XX10	42
Set software request channel 2	010X	XX11	43
Clear software request channel 1	010X	XX00	40
Clear software request channel 2	010X	XX01	41
Set hardware mask channel 1	100X	XX10	82
Set hardware mask channel 2	100X	XX11	83
Clear hardware mask channel 1	100X	XX00	80
Clear hardware mask channel 2	100X	XX01	81
Set CIE, or, IP channel 1	001E	XP10	32*
Set CIE, or, IP channel 2	001E	XP11	33*
Clear CIE, or, IP channel 1	001E	XP00	30*
Clear CIE, or, IP channel 2	001E	XP01	31*
Set flip bit channel 1	011X	XX10	62
Set flip bit channel 2	011X	XX11	63
Clear flip bit channel 1	011X	XX00	60
Clear flip bit channel 2	011X	XX01	61

NOTES:

1. E = Set to 1 to perform set/clear on CIE; clear to 0 for no effect on CIE.
2. P = Set to 1 to perform set/clear on IP; clear to 0 for no effect on IP.
3. X = "don't care" bit. This bit is not decoded and may be 0 or 1.

FIGURE 7. AM9516 command summary.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.6.2 Reset (00). This command causes the AM9516 to be set to the same state generated by a hardware reset. The master mode register is set to all zeroes, the CIE, IP and SIP bits are cleared, the NAC and CA bits in each channel's status register are set and the channel activity is forbidden. The chain address must be programmed since its state may be indeterminate after a reset. The lockout preventing channel activity is cleared by issuing a start chain command.

30.6.3 Start chain channel 1/channel 2 (A0/A1). This command causes the selected channel to clear the no auto-reload or chain (NAC) bit in the channel's status register, and to start a chain reload operation of the channel's registers. These effects will take place even if the reload word fetched is all zeroes. This command will only be honored if the chain abort (CA) bit and second interrupt pending (SIP) bit in the channel's status register are clear. If either the CA or SIP bit is set, this command is disregarded. When the waiting for bus (WFB) bit of the status register is set and the "start chain" command is issued, the channel will honor the command after one DMA iteration.

30.6.4 Software request channel 1/channel 2 (Set: 42/43, Clear: 40/41). This command sets or clears the software request bit in the selected channel's mode register. If the second interrupt pending (SIP) bit and no auto-reload or chain (NAC) bit in the channel's status register are both cleared, the channel will start executing the programmed DMA operation. If either the SIP or NAC bit is set, the channel will not start executing a DMA operation until both bits are cleared. The SIP bit will clear when the channel receives an interrupt acknowledge. One way to clear the NAC bit is to issue a start chain command to the channel. If the fetched reload word is all zeroes, the channel's registers will remain unchanged and the software request bit, if set earlier by command, will cause the programmed DMA operation to start immediately. If, during chaining, new information is loaded into the channel mode register, this new information will overwirte the software request bit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.6.5 Set/clear hardware mask 1/mask 2 (Set: 82/83; clear: 80/81). This command sets or clears the hardware mask bit in the selected channel's mode register. This command always takes effect. The hardware mask bit inhibits recognition of an active signal on the channel's -DREQ input; this bit does not affect recognition of a software request. If the channel is in single transfer mode, it performs DMA operations upon receipt of a transition on -DREQ rather than in response to a -DREQ level. Transitions occurring while the hardware mask bit is set will be stored and serviced when the hardware mask is cleared, assuming the channel has not chained. The AM9516 will request the system bus 1 1/2 to 2 clocks after the receipt of any -DREQ, after which a minimum of one DMA iteration is unavoidable. -DREQ transitions are only stored for the current DMA operation. If the channel performs a chain operation in single transfer mode, any -DREQ transition stored for later service is cleared.

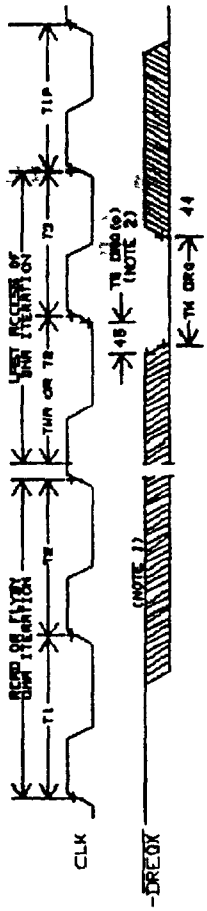
30.6.6 -DREQ timing. Figures 8 and 9 show the minimum times when a new -DREQ can be applied if it is to be serviced by the new DMA operation. Note on figure 8 the notation of first iteration and last iteration. This means, for example, -DREQ may be asserted during the write cycle T1 of a flowthru transaction, but may never be asserted during T1 of a flyby transaction since flyby is done in one iteration.

30.6.7 Set/clear CIE, and IP channel 1/channel 2. This command allows the user to either set or clear any combination of the CIE and the IP bits in the selected channel's status register. These bits control the operation of the channel's interrupt structure. Setting the IP bit causes the interrupt save register to be loaded with the current vector and status. The IP bit is cleared to facilitate an efficient conclusion to the processing of an interrupt.

30.6.8 Set/clear flip bit channel 1/channel 2 (Set: 62/63; clear: 60/61). The flip bit in the selected channel's mode register can be cleared and set by this command. This allows the user to reverse the source and destination and thereby reverse the data transfer direction without reprogramming the channel. This command will be most useful when repetitive DMA operations are being performed by the channel, using base-to-current reloading for channel reinitialization and using this command to control the direction of transfer. Chaining new information into the channel mode register will overwrite the flip bit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX



NOTES:

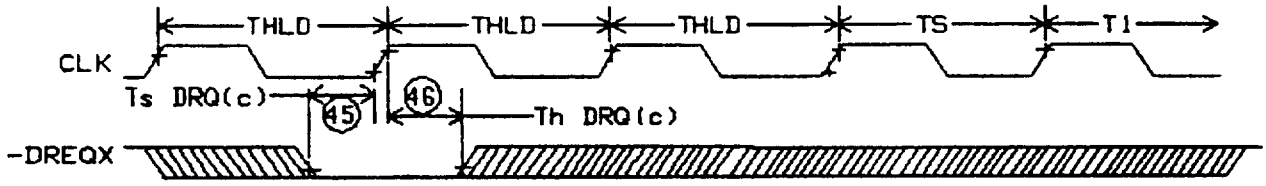
1. HIGH-to-LOW-DREQX transitions will only be recognized after the HIGH-to-LOW transition of the clock during T1 of a read or flyby.
2. A HIGH-to-LOW-DREQX transition must meet the conditions in Note 1 and must occur TsDRQ(c) before state T3 of the last access of the DMA iteration if the channel is to retain bus control and immediately start the next iteration. -DREQX may go HIGH before TsDRQ(c) if it has met the TwDRQ parameter.
3. Flyby and search transactions have only a single access; parameter TsDRQ(c) should be referenced to the start of T3 of the access. All other operations will always have two or three accesses per iteration.
4. See figure 30 for description of AM9516 internal states.

FIGURE 8. Sampling -DREQX during single transfer DMA operations.

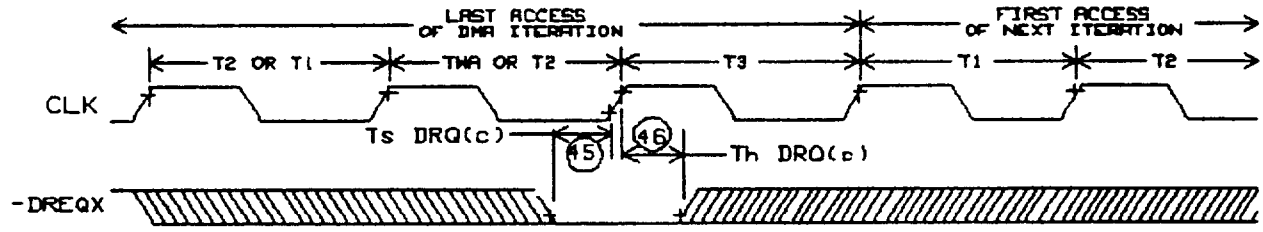
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

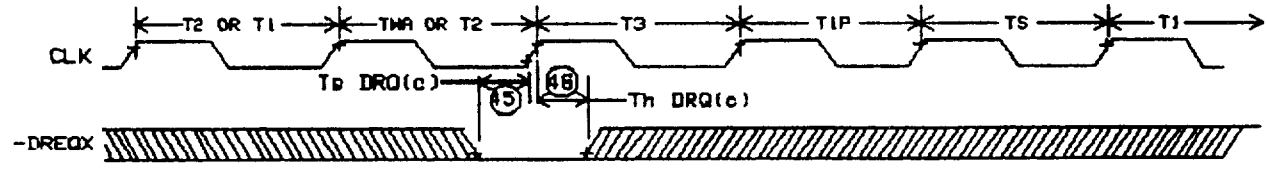
a) Sampling of -DREQX while in bus hold mode.



b) -DREQX sampling in demand mode during DMA operations.



c) Sampling -DREQX at the end of chaining.



d) Sampling -DREQX at the end of base-to-current reloading.

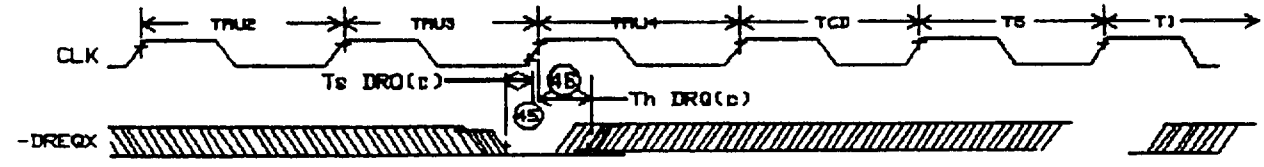


FIGURE 9. -DREQX sampling in demand mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

NOTES:

1. -DREQX must be LOW from the start of TsDRQ(c) to the end of ThDRQ(c) to ensure that the request is recognized.
2. Failure to meet this setup time will result in the channel releasing the bus.
3. Ts is a setup state, generated before entering DMA operation cycle.
4. TAU2, TAU3 and TAU4 are auto-reload states, followed by TCD (chain decision) state.
5. See figure 30 for a description of AM9516 internal states.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.7 Current and base address register A and B.

30.7.1 Address register A and B description. The current address registers (figure 10) A and B (current ARA and ARB) are used to point to the source and destination addresses for DMA operations. The contents of the base ARA and ARB registers are loaded into the current ARA and ARB registers at the end of a DMA operation if the user enables base-to-current reloading in the completion field of the channel mode register. This facilitates DMA operations without reloading of the current registers. The ARA and ARB registers can be loaded during chaining, can be written to by the host MPU without wait states and can be read by the MPU.

30.7.2 Address register A and B organization. Each of the base current ARA and ARB registers consists of two words organized as a 6-bit tag field and an 8-bit upper address in one word and a 16-bit lower address in the other. The tag field selects whether the address is to be incremented, decremented or left unchanged, and the status codes associated with the address. The tag field also allows the user to insert 0, 1, 2 or 4 wait states into memory or I/O accesses addressed by the offset and segment fields.

30.7.3 Upper address field. The address reference select field in the tag field selects whether the address pertains to memory space or I/O space. Note that the N/-S output pin may be either HIGH (indicating normal) or LOW (indicating system). At the end of each iteration of a DMA operation, the user may select to leave the address unchanged, to increment it or to decrement it. I/O addresses, if changed, are always incremented/decremented by 2. Memory addresses are changed by 1 if the address points to a byte operand (as programmed in the channel mode register's operation field) and by 2 if the address points to a word operand. Note that if an I/O or memory address is used to a word operand, the address must be even to avoid unpredictable results. An address used to point to a byte operand may be even or odd. Since memory byte operand addresses will increment/decrement by 1, they will toggle between even and odd values. Since I/O byte operand addresses will increment/decrement by 2, once programmed to an even or odd value, they will remain even or odd, allowing consecutive I/O operations to access the same half of the data bus. High bus is even address and low bus for odd.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.8 Current and base operation count registers. Both the current and base operation count registers (figure 11) may be loaded during chaining and may be written to and read from by the host MPU. The 16-bit current operation count register is used to specify the number of words or bytes to be transferred, searched or transferred-and-searched. For word-to-word operations and byte- word funnelling, the current operation count register must be programmed with the number of words to be transferred or searched.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

Address Registers

Address	ADDR7							ADDR0		
	X	0	0	1	1	0	1	X	Current ARA up-addr/tag	CH1
	X	0	0	1	1	0	0	X	Current ARA up-addr/tag	CH2
	X	0	0	0	1	0	1	X	Current ARA low-addr	CH1
	X	0	0	0	1	0	0	X	Current ARA low-addr	CH2
	X	0	0	1	0	0	1	X	Current ARB up-addr/tag	CH1
	X	0	0	1	0	0	0	X	Current ARB up-addr/tag	CH2
	X	0	0	0	0	0	1	X	Current ARB low-addr	CH1
	X	0	0	0	0	0	0	X	Current ARB low-addr	CH2
	X	0	0	1	1	1	1	X	Base ARA up-addr/tag	CH1
	X	0	0	1	1	1	0	X	Base ARA up-addr/tag	CH2
	X	0	0	0	1	1	1	X	Base ARA low-addr	CH1
	X	0	0	0	1	1	0	X	Base ARA low-addr	CH2
	X	0	0	1	0	1	1	X	Base ARB up-addr/tag	CH1
	X	0	0	1	0	1	0	X	Base ARB up-addr/tag	CH2
	X	0	0	0	0	1	1	X	Base ARB low-addr	CH1
	X	0	0	0	0	1	0	X	Base ARB low-addr	CH2

Chain loadable
Fast readable and writeable

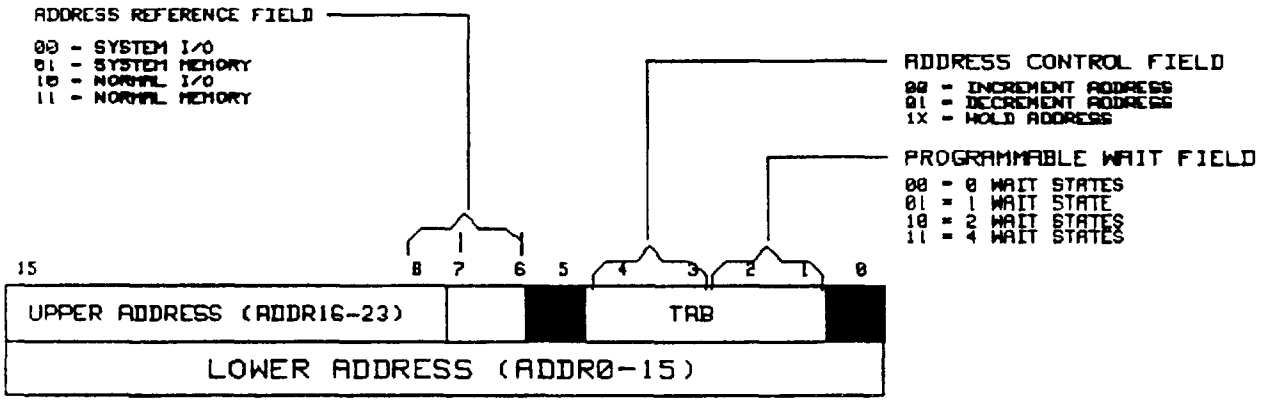


FIGURE 10. Address register A and B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

ADDRESS	ADDR7							ADDR0	
	X	0	1	1	0	0	1	X	CURRENT OPERATION COUNT CH1
	X	0	1	1	0	0	0	X	CURRENT OPERATION COUNT CH2
	X	0	1	1	0	1	1	X	BASE OPERATION COUNT CH1
	X	0	1	1	0	1	0	X	BASE OPERATION COUNT CH2
	X	1	0	0	1	0	1	X	PATTERN CH1
	X	1	0	0	1	0	0	X	PATTERN CH2
	X	1	0	0	1	1	1	X	MASK CH1
	X	1	0	0	1	1	0	X	MASK CH2

CHAIN LOADABLE
WRITEABLE
PATTERN AND MASK - SLOW READABLE
OPERATION COUNT - FAST READABLE

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

FIGURE 11. Miscellaneous registers.

Each time a datum is transferred or searched, the operation count register is decremented by 1. Once all of the data is transferred or searched, the transfer or search operation will stop, the current operation count register will contain all zeroes, and the TC bit in the status register will be '1'. If the transfer or search stops before the current operation count register reaches 0, the contents of the register will indicate the number of bytes or words remaining to be transferred or searched. This allows a channel which had been stopped prematurely to be restarted where it left off without requiring reloading of the current operation count register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

For the byte-to-byte operations, the current operation count register should specify the number of bytes to be transferred or searched. The maximum number of bytes which can be specified is 64K bytes by setting the current operation count register to 0000.

30.9 Pattern and mask registers. The 16-bit pattern and mask registers (figure 11) are used in search and transfer-and-search operations. Both the pattern and mask registers may be loaded by chaining and may be written to by the host MPU, provided wait states are inserted, since these registers are slow readable. The pattern register contains the pattern that the read data is compared to. Setting a mask register bit to '1' specifies that the bit always matches.

30.10 Status register. The two 16-bit status registers, depicted on figure 12, are read-only registers which can be read by the MPU without wait states. Each of these registers reports on the status of its associated channel.

30.10.1 Interrupt status field. The interrupt status field in the status register contains the channel interrupt enable (CIE), and interrupt pending (IP) bits.

30.10.2 AM9516 status field. The AM9516 status field contains the current channel status. The "channel initialized and waiting for request" status is not explicitly stated - it is reflected by status register bits ST12 through ST9 being all zero. The "waiting for bus" (WFB) status will cause bit ST10 to be set and indicates that the channel wants bus control to perform a DMA operation. The channel may or may not actually be asserting BREQ high, depending on the programming of the master mode chip enable bit (MMO) when the channel decided it wanted the bus. If a channel completes a DMA operation and neither base-to-current reloading nor auto-chaining were enabled, the no auto-reload or chaining (NAC) bit will be set. The NAC bit will be reset when the channel receives a "start chain command." If two interrupts are queued, the second interrupt pending bit (SIP) will be set and the channel will be inhibited from further activity until an interrupt acknowledge occurs. Finally, if the channel is issued an EOP during chaining, the chaining abort (CA), and the NAC will

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

be set. These bits are also set when a "reset" is issued to the AM9516. The CA bit holds the NAC bit in the set state. The CA bit is cleared when a new chain upper address and tag word or lower address word is loaded into the channel.

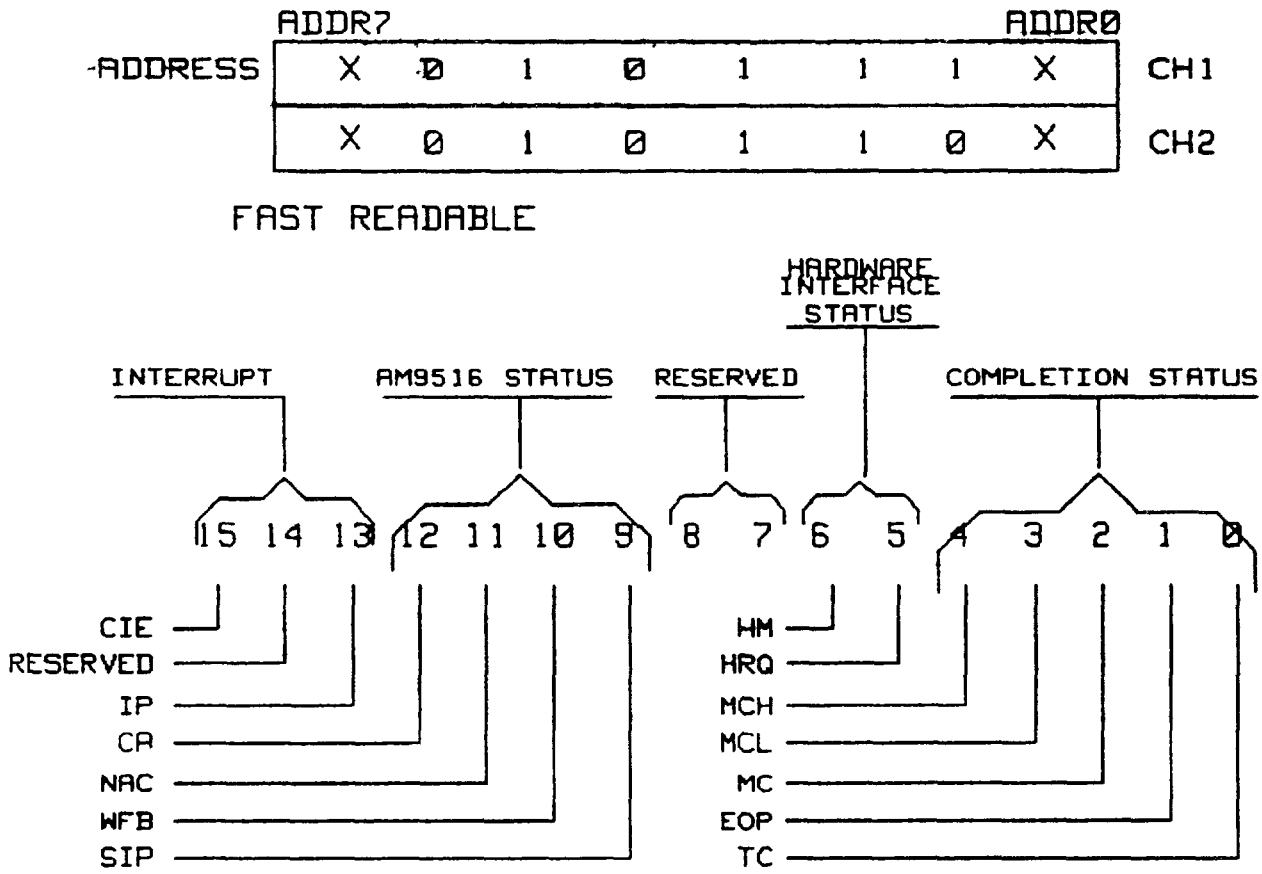


FIGURE 12. Status register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.10.3 Hardware interface status field. The hardware interface field provides a hardware request (HRQ) bit which provides a means of monitoring the channel's -DREQ input pin. When the -DREQ pin is LOW, the HRQ bit will be '1' and vice versa. The hardware mask (HM) bit, when set, prevents the AM9516 from responding to a LOW on -DREQ. Note, however, that the hardware request bit always reports the true (unmasked) status of -DREQ regardless of the setting of the HM bit.

30.10.4 Completion status field. The completion field stores data at the end of each DMA operation. This data indicates why the DMA operation ended. When the next DMA operation ends, new data is loaded into these bits overwriting, and thereby erasing the old setting. Three bits indicate whether the DMA operation is a result of a TC, MC or EOP termination. The TC bit will be '1' if the operation count reaching zero ended the DMA operation. The MC bit will be '1' if an MC termination occurred regardless of whether stop-on-match or stop-on-no-match was selected. The EOP bit is set only when an external EOP ends a DMA transfer; it is not set for EOP issued during chaining. Note that two or even all three of MC, TC and EOP may be set if multiple reasons existed for ending the DMA operation. The MCH and MCL bits report on the match state of the upper and lower comparator bytes, respectively. These bits are set when the associated comparator byte has a match and are reset otherwise, regardless of whether stop-on-match or stop-on-no-match is programmed. Regardless of the DMA operation performed, these bits will reflect the comparator status at the end of the DMA operation. These two bits are provided to help determine which byte matched or did not match when using 8-bit matches with word searches and transfer-and-searches. The three reserved bits return zeroes during reads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.11 Interrupt vector and interrupt save registers. Each channel has an interrupt vector register and interrupt save register (figure 13). The interrupt vector is 8-bit wide and is written to and read from on the data bus (DATA0-7). The interrupt save register contains the vector or identifier to be output during an interrupt acknowledge cycle. When an interrupt occurs (IP = 1) either because a DMA operation terminated or because EOP was driven LOW during chaining, the contents of the interrupt vector register and part of the channel status register are stored in the 16-bit interrupt save register. Because the vector and status are stored, a new vector can be loaded into an interrupt vector register during chaining and a new DMA operation can be performed before an interrupt acknowledge cycle occurs. If another interrupt occurs on the channel before the first is acknowledged, further channel activity is suspended.

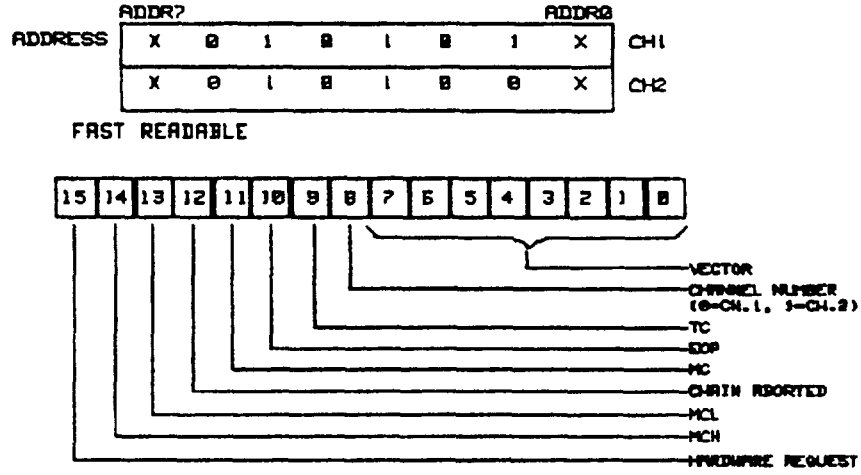


FIGURE 13. Interrupt save register.

As soon as the first clear IP command is issued, the status and vector for the second interrupt is loaded into the interrupt save register and channel operation resumes. The AM9516 can retain only two interrupts for each channel; a third operation cannot be initiated until the first interrupt has been cleared.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.12 Channel mode registers. The channel mode registers (figure 14) are two words wide. There are 21 bits defined in each channel mode register; the other 11 bits are unused. The channel mode registers may be loaded during chaining and may be read by the host MPU. MPU reads of the channel mode register are slow reads and require insertion of multiple wait states. The channel mode low word (bits 0-15) may be written to directly by the host MPU. The channel high word (bits 16-20) is chain loadable only. The channel mode register selects what type of DMA operation the channel is to perform, how the operation is to be executed, and what action, if any, is to be taken when the channel finishes.

30.12.1 Data operation and transfer fields. The data operation field and the transfer type field select the type of operation the channel is to perform. It also selects the operand size of bytes or words. The flip bit is used to select whether the current ARA register points to the source and the current ARB register points to the destination or vice versa.

30.12.2 Completion field. The completion field is used to program the action taken by the channel at the end of a DMA operation. The 2-bit match control field selects whether matches use an 8-bit or 16-bit pattern and whether the channel is to stop-on-match or stop-on-no-match. The software request bit and hardware mask bit can be set and cleared by software command in addition to being loaded in parallel with other channel mode bits.

30.12.3 -DACK control bit. The -DACK control bit is used to specify when the -DACKX pin is driven active. When this bit is cleared, the channel's -DACKX pin will be active whenever the channel is performing a DMA operation, regardless of the type of transaction. Note that the pin will not be active while the channel is chaining. If this bit is set, the -DACKX pin will be inactive during chaining, during both flowthru transfers and flowthru transfer-and-searches and during searches, but -DACKX will be pulsed active during flyby transfers and flyby transfers-and-searches at the time necessary to strobe data into or out of the flyby peripheral.

NOTE: When the FUM module is used with the DMA support module (FUN) for flowthru transfers, it is essential that the -DACK control bit be set to inhibit the generation of the -DACKX output signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

ADDR7							ADDR0	
X	1	0	1	0	1	1	X	HIGH CH1
X	1	0	1	0	1	0	X	HIGH CH2
X	1	0	1	0	0	1	X	LOW CH1
X	1	0	1	0	0	0	X	LOW CH2

CHAIN LOADABLE
WRITEABLE (LOWER 16 BITS ONLY)
SLOW READABLE

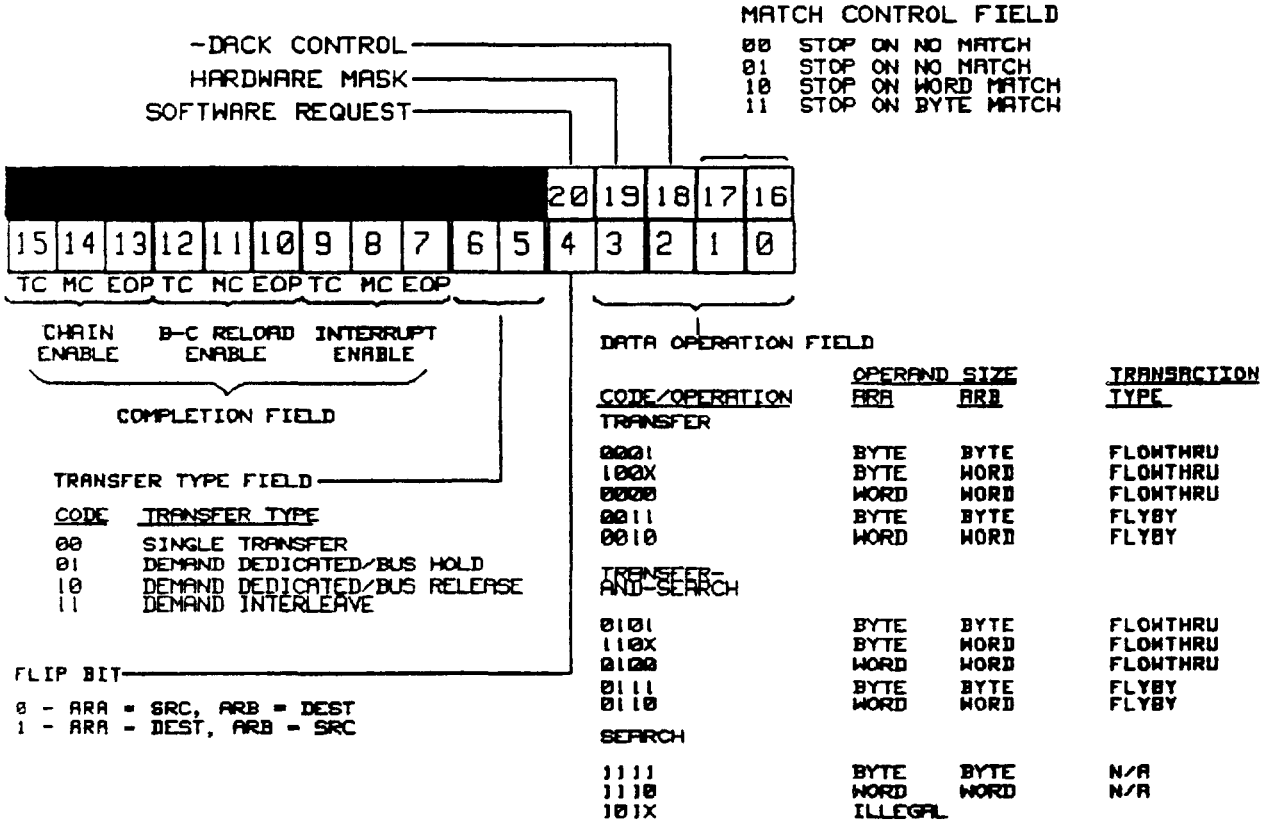


FIGURE 14. Channel mode registers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

30.13 Chain address register. Each channel has a chain address register which points to the chain control table in memory containing data to be loaded into the channel's registers. The chain address register, as shown on figure 4, is two words long. The first word consists of an upper address and tag field. The second word contains the 16-bit lower address portion of the memory address. The tag field contains 2 bits used to designate the number of wait states to be inserted during accesses to the chain control table. The chain address register may be loaded during chaining and may be read from and written to by the host MPU without wait states. If an EOP is issued to the AM9516 during chaining, the chain address register holds the old address. This is true even if the access failure occurred while new chain address data was being loaded, since the old data is restored unless both words of the new data are read successfully. Note, however, that EOPs that occur when chaining and while loading a new chain address cause the new data to be lost.

30.14 Module reset. The FUM module can be reset either by hardware or software. The software reset is accomplished by a reset command loaded into the AM9516 command register. Hardware resets are applied by pulling -RST SLAVE LOW. The AM9516 may be in control of the bus when a reset is applied. BACK is removed internally within the AM9516 causing its outputs to go tri-state. If CBACK remains high after reset, the AM9516 will not drive the bus unless BREQ is active. As soon as CBACK goes inactive, the FUM module places the ADDR16-23, N/-S, M/-IO, BYTE, -TBEN and -RBEN signals in the high impedance state. (See figure 15) Signals -DS-M and READ-M go to a HIGH output state, if the -DS SLAVE and -READ SLAVE inputs are inactive (High).

30.14.1 Register status after reset. Both software and hardware resets clear the master mode register, clear the CIE, IP and SIP bits, and set the CA and NAC bits in each channel's status register. The contents of all other AM9516 registers will be unchanged for a software reset. Since a hardware reset may have been applied partway through a DMA operation being performed by an AM9516 channel, the channel's registers should be assumed to contain indeterminate data following a hardware reset. The master mode register contains all zeroes after a reset. The AM9516 is disabled and the MPU interleave and hardware wait are inhibited. Because the CA and NAC bits in the status register are set by a reset, the channel will be prevented from starting a DMA operation until its chain address register's segment, tag and offset fields are programmed and the channel is issued a "start chain" command.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.15 Channel initialization. The upper and lower address fields of the chain address register form a 24-bit address which points to a location in system memory space. Chaining is performed by repetitively reading words from memory. Note that the chain address register should always be loaded with an even address; loading an odd address will cause unpredictable results. The 2-bit tag field facilitates interfacing to slow memory by allowing the user to select 0, 1, 2, or 4 programmable wait states. The AM9516 will automatically insert the programmed number of wait states in each memory access during chaining.

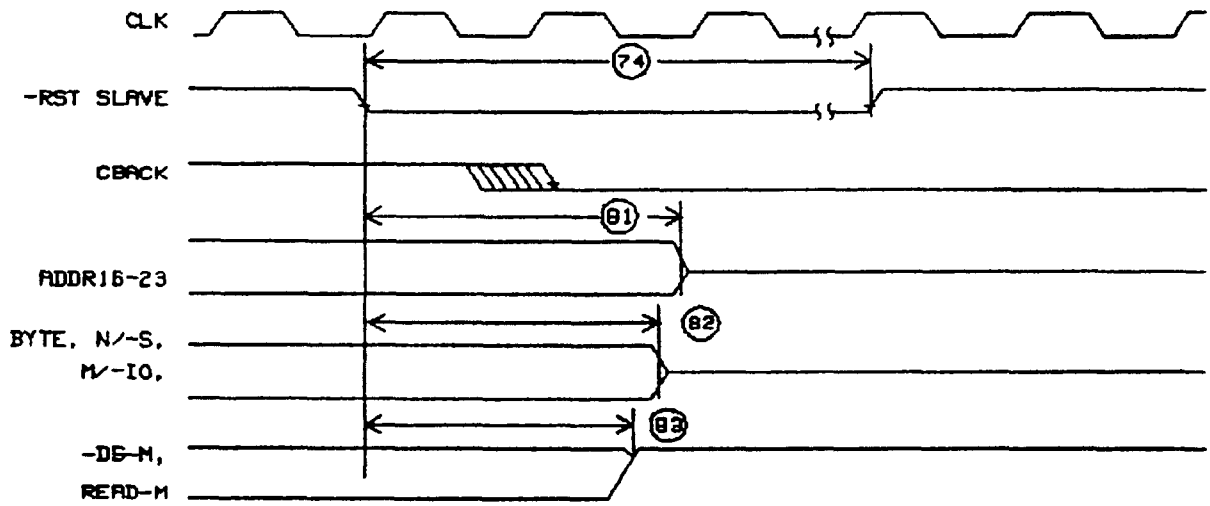


FIGURE 15. Reset timing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.15.1 Reload word operation. The chain address register points to the first word in the chain control table. This word is called the reload word (see figure 16). The purpose of the reload word is to specify which registers in the channel are to be reloaded. Reload word bits 10-15 are undefined and may be 0 or 1. Each of bits 0 through 9 in the reload word correspond to either one or two registers in the channel (see figure 17). When a reload word bit is '1', it means that the register or registers corresponding to that bit are to be reloaded. If a reload word bit is '0', the register or registers corresponding to that bit are not to be reloaded. The data to be loaded into selected register(s) follows the reload word in memory (i.e., the data are stored at successively larger memory addresses). The chain control table is a variable length table. Only the data to be loaded are in the table and the data are packed together. When the channel is to reload itself, it first uses the chain address register contents to load the reload word into the AM9516 chain control register. Next, the chain address register contents are incremented by two to point to the word in memory. The channel then scans the reload word register from bit 9 down to bit 0 to see which registers are to be reloaded. If no registers are specified (bits 9-0 are all 0), no registers will be reloaded. If at least one of bits 9-0 are set to '1', the register(s) corresponding to the set bit are reloaded, the bit is cleared and the chain address register is incremented by 2. The channel continues this operation of scanning the bits from the most significant to least significant bit position, clearing each set bit after reloading its associated register and incrementing the chain address register by 2. If all of bits 9 to 0 are set, all the registers will be reloaded in the order: current ARA, current ARB, current operation count...channel mode and chain address. Figure 18 shows two examples of chain control tables. Example 1 shows the ordering of data when all registers are to be reloaded. In example 2 only some registers are reloaded. Once the channel is reloaded, it is ready to perform a DMA operation. Note when loading address registers, the upper address and tag word are loaded first, then the lower address word. Also, the pattern register is loaded before the mask register. Figure 15 shows the FUM module timing during chaining.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.16 Initiating DMA operations. DMA operations can be initiated in one of three ways - by software request, by hardware request and by loading a set software request bit into the channel mode register during chaining.

30.16.1 Starting after chaining. If the software request bit in the channel mode register is loaded with a '1' during chaining, the channel will perform the programmed DMA operation at the end of chaining. If the channel is programmed for single operation or demand, it will perform the operation immediately. The channel will give up the bus after chaining and before the operation of the MPU interleave bit in the master mode register is set. Note that once a channel starts a chaining operation by fetching a reload word, it retains bus control at least until chaining of the last register's data is performed.

30.16.2 Software requests. The MPU can issue software request commands to start DMA operations on a channel. This will cause the channel to request the bus and perform transfers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

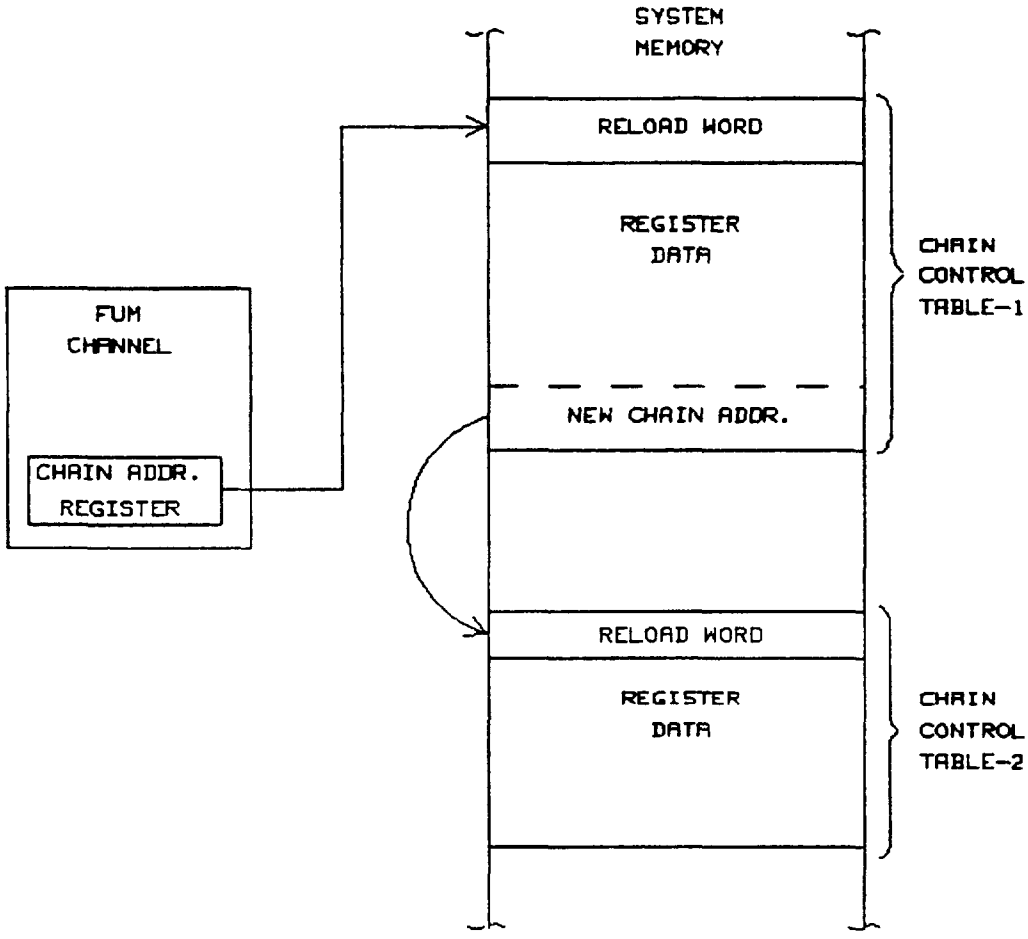


FIGURE 16. Chaining and chain control tables.

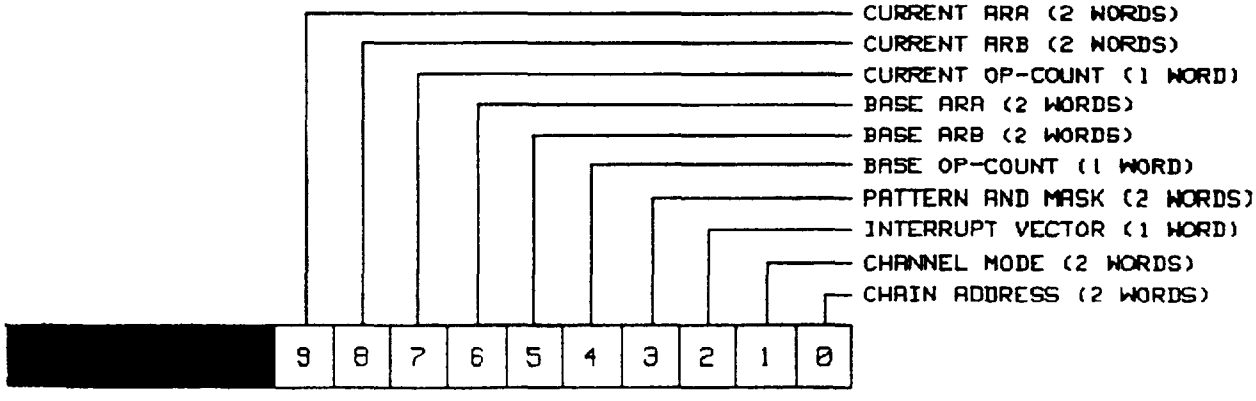


FIGURE 17. Reload word/chain control register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

30.16.3 Hardware requests. DMA operations will often be started by an external device by applying a low on the channel's -DREQ input. The channel response section describes when the low -DREQ signals are sampled and when the -DREQ requests can be applied to start the next DMA operation after chaining.

30.17 Bus request/grant. Before the AM9516 can perform a DMA operation, it must gain control of the system bus. The BREQ, CBACK and -BACK interface pins provide connections between the FUM module and the host MPU and other devices, if present, to arbitrate which device has control of the system bus. When the FUM module wants to gain bus control, it drives BREQ high. Some period of time after the FUM module drives BREQ high, the MPU will relinquish bus control. A bus arbiter function (MPU communications support module - AHH) grants the bus to the highest priority device by activating its bus acknowledge (-BACK) signal. The DMA support module (FUN) receives the bus acknowledge signal and raises the CBACK signal to the FUM module to initiate the DMA operation. When the FUM module finishes its operation, it stops driving BREQ high.

31. DMA operations. There are three types of DMA operations: transfer, search, and transfer-and-search. Transfers move data from a source location to a destination location. Two types of transfers are provided: flowthru and flyby. Searches read data from a source and compare the read data to the contents of the pattern register. A mask register allows the user to declare "don't care" bits. The user can program that the search is to stop either when the read data matches the masked pattern or when the read data fails to match the masked pattern. This capability is called stop-on-match and stop-on-no-match. Transfer-and-search combines the two functions to facilitate the transferring of variable length data blocks. Like transfer, transfer-and-search can be performed in either flowthru or flyby mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

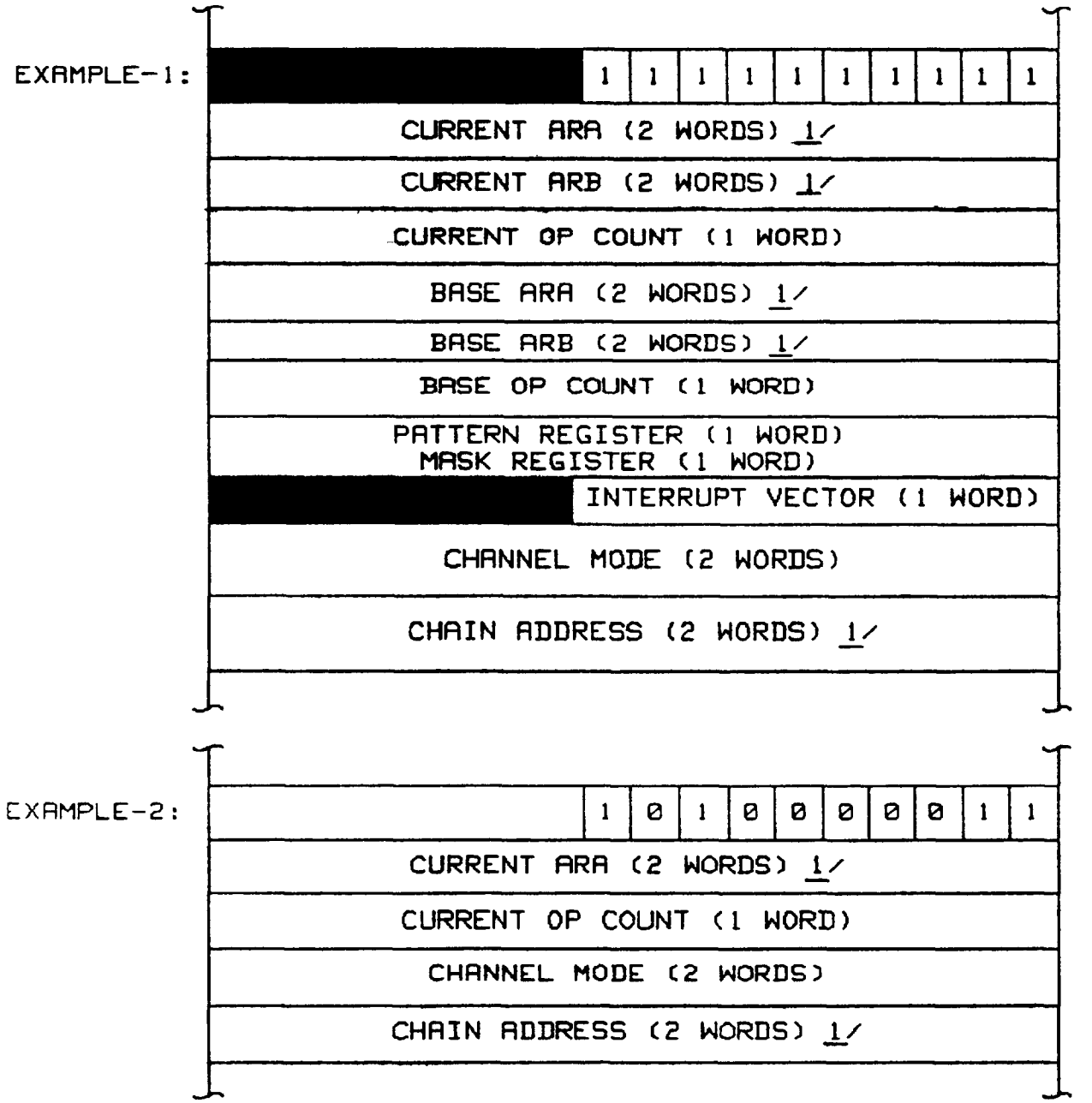
31.1 Transfers. Transfers use four of the channel registers to control the transfer operation: the current ARA and ARB registers; the current operation count register; and the channel mode register. Channel mode register bit CM4 is called the flip bit and is used to select whether ARA is to point to the source and ARB is to point to the destination or vice versa. The current operation count register specifies the number of words or bytes to be transferred.

31.1.1 Channel mode register bits CM3-CM0. Bits CM3-CM0 in the channel mode register control whether a flowthru or flyby transfer is to be performed. Flowthru transfers are performed in either two or three steps. First, the channel outputs the address of the source and reads the source data into the AM9516's temporary register. In two-step flowthru transfer, the channel will then address the destination and write the temporary register data to the destination location. The three-step flowthru operation (i.e., the byte word funnelling) is described later in this section. The source and destination for flowthru transfers can both be memory locations or both peripheral devices or one may be a memory location and the other a peripheral device. Flyby transfers provide improved transfer throughput over flowthru but are restricted to transfers between memory and peripherals or between two peripherals.

31.1.2 Channel mode register bit CM18. The -DACKX output for the transferring channel may be programmed to be inactive throughout the transfer or active during the transfer. This is controlled by bit CM18 in the channel mode register. When the FUM module is used with the DMA support module (FUN) for flow thru transfers, it is essential that the -DACK control bit be set to inhibit the generation of the -DACKX output signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

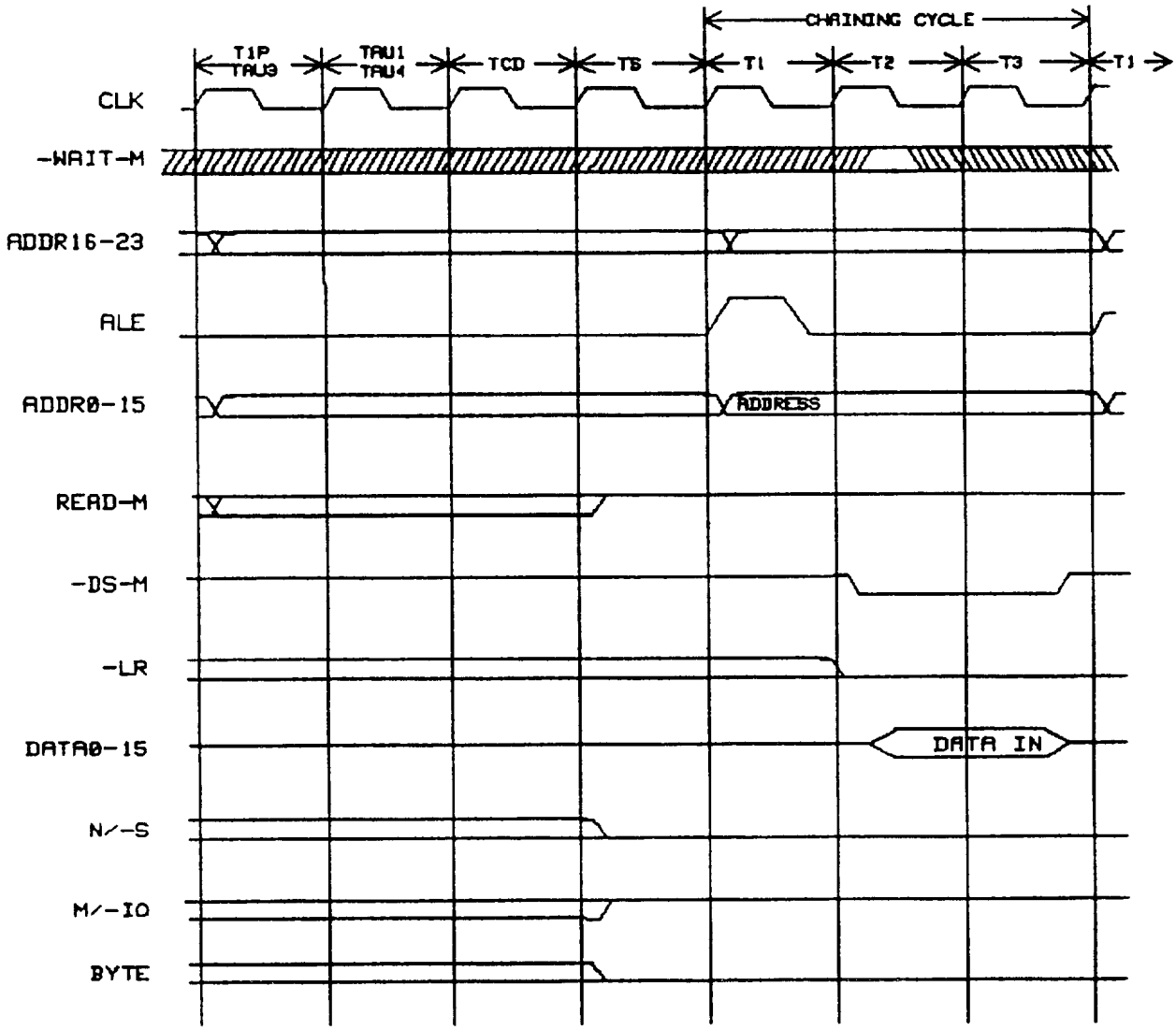


1/ Load the upper address and tag word first, then the lower address word.

FIGURE 18. Examples of chain control table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX



NOTES:

1. See figure 30 for a description of AM9516 internal states.
2. Input -BACK is assumed to be active (low).

FIGURE 19. Timing during chaining.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

31.1.3 Flowthru byte-to-byte transfers. Transfers can use both byte- and word-sized data. Flowthru byte-to-byte transfers are performed by reading a byte from the source and writing a byte to the destination. The current operation count register must be loaded with the number of bytes to be transferred. Both the current ARA and current ARB register, if programmed to increment/decrement, will change by ±1 if the register points to a memory space (TG6 = 2) and by ±2 if the register points to an I/O space (TG6 = 0).

31.1.4 Flowthru word-to-word transfers. Flowthru word-to-word transfers require that the current operation count specify the number of words to be transferred. Both the current ARA and current ARB registers, if programmed to increment/decrement, will change by ±2 regardless of whether the register points to memory or I/O space.

31.1.5 Flowthru mode, funnelling option. Byte-word funnelling packing and unpacking of byte data can be used to facilitate high speed transfers between byte and word peripherals and/or memory. This funnelling option can only be used in flowthru mode. Funnelled flowthru transfers are performed in three steps. For transfers from a byte source to a word destination, two consecutive byte reads are performed from the source address. The data read is assembled into the AM9516's temporary register. In the third step, the temporary register data is written to the destination address in a word transfer. Funnelled transfers from a word source to a byte destination are performed by first loading a word from the source into the AM9516's temporary register. The word is then written out to the destination in two byte writes. For funnel operations, the byte-oriented address must be in the current ARA register and the word-oriented address must be in the current ARB register. The flip bit (CM4) in the channel mode register is used to specify which address is the source and which is the destination. When the byte address is to be incremented or decremented, the increment/decrement operation occurs after each of the two reads or writes. The current operation occurs after each of the two reads or writes. The current operation count register must be loaded with the number of words to be transferred.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

31.1.5.1 Byte-to-word funnelling operation. In byte-to-word funnelling operations, it is necessary to specify which half of the temporary register (upper or lower byte) is loaded with the first byte of data. Similarly, for word-to-byte funnelling operations, it is necessary to define which half of the temporary register is written out first. Figure 20 summarizes these characteristics for both byte-to-word and word-to-byte funnelling operations. The criteria used to determine the packing/unpacking order is based on whether the current ARB is programmed for incrementing or decrementing of the address. Note that if the address is to remain unchanged (i.e., if bit TG4 in the tag field of the current ARB register is 1), the increment/decrement bit (bit TG3) still specifies the packing order.

Funnelling direction	Current ARB tag field		Increment/decrement and packing/unpacking rules
	TG4	TG3	
Word-to-Byte (CM4 = 1)	0	0	Increment ARB, write high byte first
	0	1	Decrement ARB, write low byte first
	1	0	Hold ARB, write high byte first
	1	1	Hold ARB, write low byte first
Byte-to Word (CM4 = 0)	0	0	Increment ARB, read high half of word first
	0	1	Decrement ARB, read low half of word first
	1	0	Hold ARB, read high half of word first
	1	1	Hold ARB, read low half of word first

FIGURE 20. Byte/word funnelling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

31.2 Search.

31.2.1 Search registers. Searches use five of the channel registers to control the operation: either the current ARA or ARB; the operation count; the pattern and mask register; and the channel mode register. Channel mode register bit CM4 is called the flip bit and is used to select either current ARA or ARB as the register specifying the source for the search. Only one of the current address registers is used for search operations since there is no destination address required. The current operation count register specifies the maximum number of words or bytes to be searched.

31.2.2 Match condition termination. Search operations involve repetitive reads from the peripheral or memory until the specified match condition is met. The search then stops. This is called a match condition or MC termination. Each time a read is performed, the source address, if so programmed, is incremented or decremented and the operation count is decremented by 1. If the match condition has not been met by the time the operation count reaches zero, the zero value will force a TC termination, ending the search. Searches can also stop due to a HIGH being applied to the EOP input pin. During a search operation, the channel's -DACKX output will be either inactive or active throughout the search. This is controlled by bit CM18 in the channel mode register.

31.2.3 Match condition options. On each read during a search operation, the AM9516 temporary register is loaded with data and compared to the pattern register. The user can select that the search is to stop when the pattern and temporary register contents match or when they don't match. This stop-on-match/stop-on-no-match feature is programmed in bit CM17 of the channel mode register. CM2 is an enable for the output of the comparator and allows the MC signal to be generated. A mask register allows the user to exclude or mask selected temporary register bits from the comparison by setting the corresponding mask register bit to '1'. The masked bits are defined to always match. Thus, in stop-on-match, successful matching of the unmasked bits, in conjunction with the always-matched masked bits, will cause the search to stop. For stop-on-no-match, the always-matched masked bits are by definition excluded from not matching and therefore excluded from stopping the search.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

31.2.4 8-bit and 16-bit search operation. For word reads, the user may select either 8-bit or 16-bit compares through channel mode register bit CM16. In an 8-bit, stop-on-match, word-read operation, successful matching of either the upper or lower byte of unmasked pattern and temporary register bits will stop the search. Both bytes do not have to match. In 16-bit stop-on-match with word reads, all unmasked pattern and temporary register bits must match to stop the search. In an 8-bit or 16-bit, stop-on-no-match, word-read search operation, failure of any bit to match will terminate the search operation.

31.2.5 Search match operation. In an 8-bit stop-on-match with byte-reads, the search will stop if either the upper or lower byte of unmasked pattern and temporary register bits match. For an 8-bit stop-on-no-match with byte reads, failure of matching or any unmasked pattern and temporary register bit will cause the search to stop. For 8-bit searches, the upper and lower bytes of the pattern and mask register should usually be programmed with the same data. Failure to set the upper and lower bytes of the pattern and mask registers to identical values will result in different comparison criteria being used for the upper and lower bytes of the temporary register. Users failing to program identical values for the upper and lower bytes can predict the results by recognizing that in 8-bit stop-on-match, the search will end if all the unmasked bits in either the upper or lower byte match, and for 8 bit stop-on-no-match, the failure of any unmasked bit to match will end the search. For accurate predictions, it is also necessary to know that for word reads the temporary register high and low bytes are located from DATA15-8 and DATA7-0, respectively. In byte reads, the read byte is duplicated in both halves of the temporary register except in funnelling.

31.3 Transfer-and-search.

31.3.1 Transfer-and-search registers. Transfer-and-search combines the operations of the transfer and the search functions. The registers used to control transfer-and-searches are the current ARA and ARB registers, the operation count register, the pattern and mask registers and the channel mode register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

31.3.2 Transfer-and-search termination. A transfer-and-search operation will end when the data transferred meets the math conditions specified in channel mode register bits CM17-CM16. The mask and pattern registers indicate those bits being compared with the temporary register contents. Like transfers and searches, transfers-and-searches will also be terminated if the operation count goes to zero or if a HIGH is applied to the EOP input pin. Regardless of whether transfer-and-search stops because of a TC, MC or EOP, it will always complete the iteration by writing to the destination address before ending (writing twice for word-to-byte funnelling).

31.3.3 Transfer-and-search mode. In flowthru mode, the transfer-and-search timing is identical to flowthru transfer. While the data is in the temporary register, it is masked by the mask register and compared to the pattern register. For word transfer and transfer-and-search, the high and low bytes of the temporary register are always writtern to and read from DATA15-8 and DATA7-0, respectively. For byte transfer and transfer-and-search, the byte read is always loaded into both halves of the temporary register and the entire register is driven directly out onto the DATA15-0 bus. Transfer-and-search can also be used with byte word funnelling. In funnelling, the match is an 8-bit match or 16-bit match as determined by the setting of bit CM16.

31.3.4 Transfer-and-search flyby operation. Flyby transfer-and-search can be used to increase throughput for transfer between two peripherals or between memory and a peripheral. Memory-to-memory flyby is not supported. Also, in flyby, the operand sizes of the source and destination must be the same; i.e., funnelling is not supported. During a flyby transfer-and-search, data is loaded into the temporary register to facilitate the comparison operation and at the same time data is transferred from the source to the destination. When byte operands are used, data is loaded into both bytes of the temporary register, from the DATA15-8 bus if the current ARA register is even and from DATA7-0 line if the current ARA register is odd. This will alternate for memory bytes so the user must drive both halves of the bus to use the search. When word operands are used, data is loaded directly from DATA15-8 and DATA7-0 into the temporary register's high and low bytes, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

32. Channel response.

32.1 Channel response types and subtypes. Channel mode register bits CM16-CM5 select the channel's response to the request to start a DMA operation. The response falls into either of two types: single operation or demand. There are three subtypes for demand operations: demand dedicated with bus hold, demand dedicated with bus release, and demand interleave. To make discussions clear, it is necessary to define the term "single iteration of a DMA operation". For search operations, one iteration consists of a single read operation and a comparison of the read data to the unmasked pattern register bits. The operation count will be decremented by 1 and the current address register used incremented or decremented if so programmed. For transfer and transfer-and-search operations, a single iteration comprises reading a datum from the source, writing it to the destination, comparing the read datum to the unmasked pattern register bits (transfer-and-search only), decrementing the operation count by 1 and incrementing/decrementing the current ARA and ARB registers if so programmed. In byte-word funnelling, a single iteration consists of two reads followed by a write (byte-to-word funnelling) or one read followed by two writes (word-to-byte funnelling). In all transfer and transfer-and-search cases, the iteration will not stop until the data in the temporary register is written to the destination. See figure 21 for flowchart.

32.2 Single Operation. The single operation response (figure 21) is intended for use with peripherals which transfer single bytes or words at irregular intervals. Each application of a software request command will cause the channel to perform a single iteration of the DMA operation (See figure 21). Similarly, if the software request bit is set by chaining, at the end of chaining the channel will perform a single iteration of the DMA operation. Each application of a HIGH-to-LOW transition on the -DREQX input will also cause a single iteration of the DMA operation. If the hardware mask bit is set when the transition is made, the iteration will be performed when the mask is cleared, providing the DMA operation has not terminated. See the set/clear hardware mask bit command for details. Each time a single operation ends, the channel will give up control of the bus unless a new transition has occurred on -DREQX. The new transition can occur anytime after the high-to-low ALE transition of a read or flyby memory or I/O access of the DMA iteration. Figure 9 shows the times after which a new transition can be applied and recognized to avoid giving up the bus at the end of the current iteration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

32.3 Demand operations.

32.3.1 Demand dedicated with bus hold. In demand dedicated with bus hold (abbreviated bus hold), the application of a software request command or the setting of the software request bit during chaining or applying a LOW level on the -DREQX input will cause the channel to acquire bus control. See figure 21.

32.3.2 -DACKX register. If -DACKX is programmed as a level output (CM18 = 0), -DACKX will be active from when the channel acquires bus control to when it relinquishes control. A software request will cause the channel to request the bus and perform the DMA operations until either TC, MC or EOP occur.

32.3.3 Operation. Once the channel gains bus control due to a low -DREQX level, it samples -DREQX as shown in figure 19. If -DREQX is low, an iteration of the DMA operation is performed. If -DREQX is high, the channel retains bus control and continues to drive all bus control signals active or inactive, but performs no DMA operation. Thus the user can start or stop execution of DMA operations by modulating -DREQX. Once TC, MC or EOP occurs, the channel will either release the bus or, if chaining or base-to-current reloading is to occur, perform the desired operation. After chaining or base-to-current reloading, if the channel is still in bus hold mode and does not have a set software request bit (set either by chaining or command), the channel will relinquish bus control unless a low -DREQX level occurs within the time limits.

32.4 Demand dedicated with bus release. In demand dedicated with bus release (abbreviated bus release), the application of a software request will cause the channel to request the bus and perform the programmed DMA operation until TC, MC or EOP occurs. See figure 21. If the channel was programmed for bus release and the software request bit was set during chaining, the channel will start the DMA operation as soon as chaining ends, without releasing the bus, and will continue performing the operation until TC, MC or EOP occurs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

32.4.1 -DREQX low operation. When an active low -DREQX is applied to a channel programmed for bus release, the channel will acquire the bus and perform DMA operations (a) until TC, MC or EOP or (b) until -DREQX goes inactive. Figure 9 shows when -DREQX is sampled to determine if the channel should perform another cycle or release the bus. Note that this sampling also occurs on the last cycle of a chaining operation. If a channel has an active -DREQX at the end of chaining, it will begin performing DMA operations immediately, without releasing the bus. When a TC, MC or EOP occurs, terminating a bus release mode operation, the channel, if enabled for chaining and/or base-to-current reloading, will perform reloading and/or chaining (assuming the status register's SIP bit is clear) without releasing the bus.

32.4.2 SIP bit register. If the SIP bit of the channel mode register is set when a DMA termination occurs, the channel will relinquish the bus control until an interrupt acknowledge has been serviced; the channel will perform the base-to-current reloading and/or chaining, if enabled for the termination.

32.4.3 Operation with no request applied. If an active request is not applied and the channel is in demand dedicated with bus hold, the channel will go into state THLD (see figure 19). If an active request is not applied and the channel is in demand dedicated with bus release or demand interleave mode, it will release the bus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

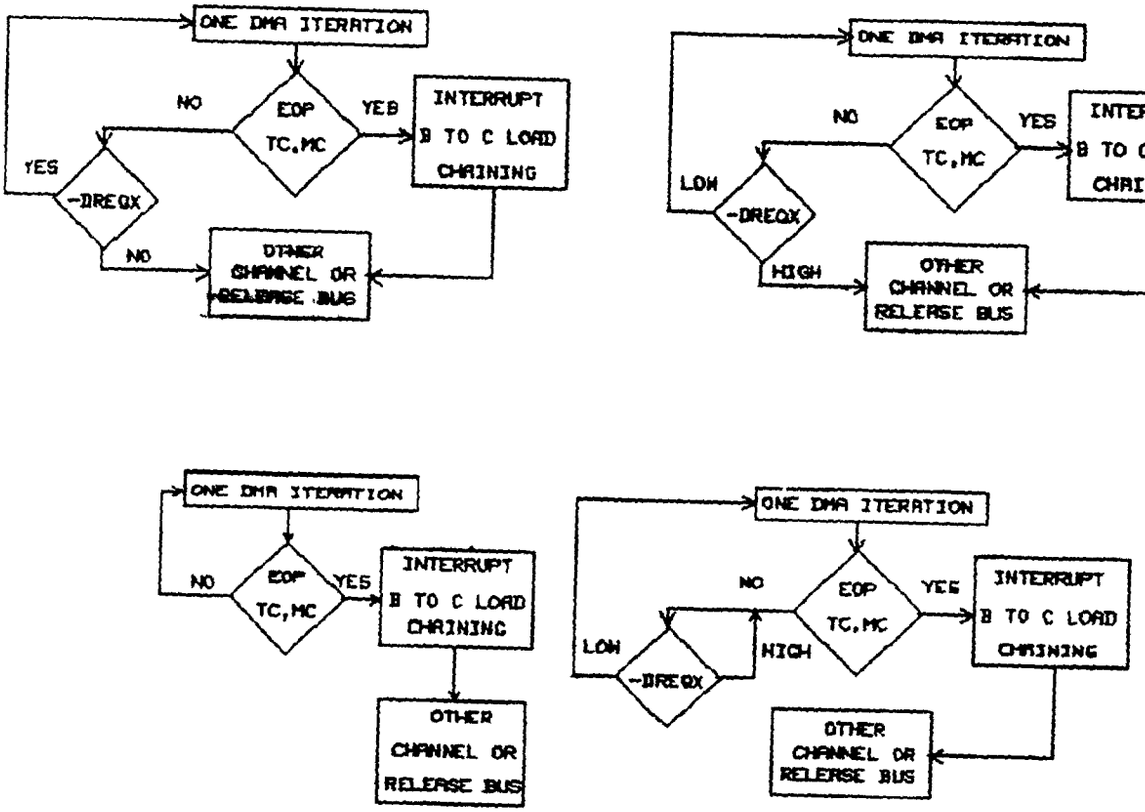


FIGURE 21. Basic DMA operations of AM9516.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

32.5 Demand interleave.

32.5.1 Demand interleave note. Note that even if an active request is applied in demand interleave, the channel may still release the bus. The request for demand interleave should continue to be applied to ensure that the channel eventually responds to the request by acquiring the bus (i.e., the request is not latched by the channel).

32.5.2 Demand interleave operation. Demand interleave behaves in different ways depending on the setting of master mode register bit MM2. If MM2 is set, the FUM module will always relinquish bus control and then re-request it after each DMA iteration. This permits the MPU and other devices to gain bus control. If MM2 is clear, control can pass from one FUM channel to the other without requiring the FUM module to release bus control. If both channels have active requests, control will pass to the channel which did not just have control. For instance, if MM2 is clear and both channels have active requests and are in demand interleave mode, control will toggle between the channels after each DMA operation iteration and the FUM module will retain bus control until both channels are finished with the bus. If MM2 is set and both channels have active requests and are in demand interleave mode, each channel will relinquish control to the MPU after each iteration resulting in the following control sequence: channel 1, MPU, channel 2, MPU, etc. Note that if there are other devices on the bus, they may gain control during the part of the sequence labelled MPU. See figure 22 for flowchart.

32.5.3 Demand interleave termination. A software or hardware request will cause a channel programmed for demand interleave to perform interleaved DMA operations until TC, MC or EOP occur. If the software requested bit is set during chaining, the channel will retain the bus after chaining and will immediately start performing a DMA iteration and will interleave all DMA iterations after the first. If -DREQX is low on the last cycle during chaining, the channel will perform a single iteration immediately after chaining and interleave thereafter until (a) TC, MC or EOP or (b) -DREQX goes high. If (b) occurs, the channel will relinquish the bus until -DREQX goes low again and the channel again starts performing interleaved operations. If (a) occurs, the channel will not interleave before first performing chaining and/or base-to-current reloading (assuming SIP is cleared).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

32.5.4 -DACKX register. The waveform of the -DACKX output is programmed in channel mode register (CM18). The pulsed -DACKX is for flyby transaction only. See figure 23. Note: This figure shows a single search or flyby iteration. State TWA is optionally inserted if programmed. For more than one iteration, the level -DACKX output would stay active during the time the channel had bus control. When CM18 is set, the -DACKX output will be inactive for all non-flyby modes.

32.6 Wait states.

32.6.1 Hardware wait states. The number of wait states to be added to the memory or I/O transfer can be programmed by the user as 0, 1, 2, or 4 and can be separately programmed for the current address registers A and B and for the chain address register. This allows different speed memories and peripherals to be associated with each of these addresses. The base address registers A and B also have a tag field which is loaded into the current ARA and ARB registers during base-to-current reloading. Because many users utilizing the software programmable wait states will not need the ability to generate hardware wait states through the -WAIT-M input pin, the wait function can be disabled by clearing the wait line enable bit (MM2) in the master mode register.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

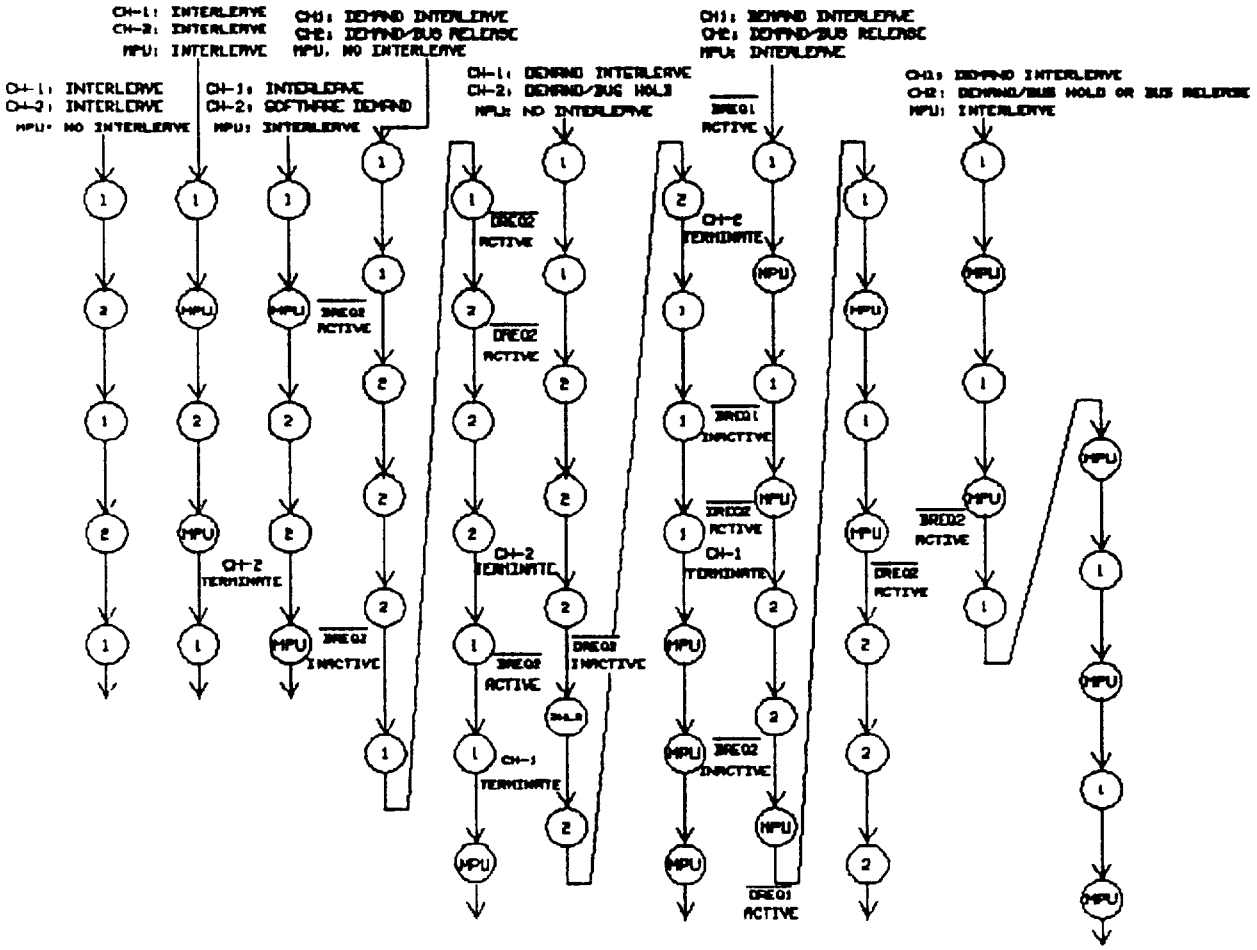


FIGURE 22. Demand interleave operations of AM9516.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

32.6.2 -WAIT-M input sampling. During DMA transactions, the -WAIT-M input is sampled in the middle of the T2 state. If -WAIT-M is high, and if no programmable wait states are selected, the AM9516 will proceed to state T3. Otherwise, at least one wait state will be inserted. The -WAIT-M line is then sampled in the middle of state TWA. If -WAIT-M is HIGH, the AM9516 will proceed to state T3. Otherwise, additional wait states will be inserted (see figure 24).

32.6.3 Operation with hardware and software wait states. Consider what happens in a transaction when both hardware and software wait states are inserted. Each time the -WAIT-M line is sampled, if it is low, a hardware wait state will be inserted in the next cycle. The software wait state insertion will be suspended until -WAIT-M is sampled and is high. The hardware wait states may be inserted anytime during the software wait state sequence. It is important to note that hardware wait states are served consecutively rather than concurrently with software wait states. For example, assume for a flowthru I/O transaction that a user has programmed 4 software wait states. Driving a low on the -WAIT-M input during T2 for 2 cycles would insert 2 hardware wait states. Driving -WAIT-M high for 3 cycles would allow insertion of three of the four software wait states. Driving -WAIT-M low for 2 more cycles would insert 2 more hardware wait states. Finally, driving -WAIT-M high would allow the final software wait state to be inserted. During this last software wait state, the -WAIT-M pin would be sampled for the last time. If it is high, the channel will proceed to state T3. If the pin is low, the channel will insert hardware wait states until the pin goes high and the channel would then enter state T3 to complete the I/O transaction.

33. DMA transactions. There are three types of DMA transactions performed by the FUM module: flowthru, flyby and search. Figures 25 and 26 show the configurations of flowthru and flyby transactions.

33.1 Flowthru transactions. A flowthru transaction consists of read and write cycles. Each cycle consists of three states: T1, T2, and T3 as shown on figure 24. The user may elect to insert software states through the tag fields of the current ARA and ARB registers. In addition, if master mode register bit MM2 = 1, hardware wait states may be inserted by driving a low signal on the -WAIT-M input pin.

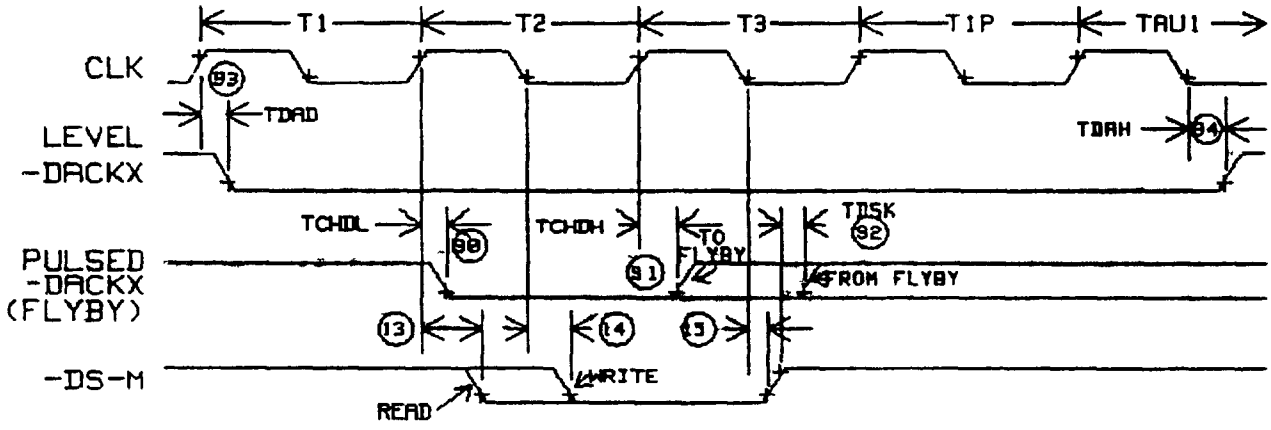
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

33.1.1 Flowthru transaction timing. The M/-IO and N/-S outputs will reflect the appropriate level for the current cycle early in T1. The TG6 and TG7 bits of the current ARA and ARB registers should be programmed properly. The ALE output will be pulsed high to mark the beginning of the cycle. The offset portion of the address for the peripheral being accessed will appear on ADDR0-15 during T1 and remain until after T3. The READ-M and BYTE lines will select a read or write operation for bytes or words. The READ-M, N/-S, M/-IO and BYTE lines will become stable during T1 and will remain stable until after T3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

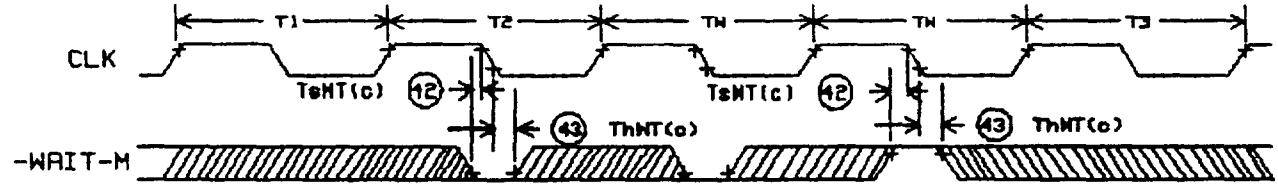


NOTES:

1. Level -DACKX low-to-high transition occurs as shown if auto-reloading is not programmed; otherwise, it stays LOW for three additional clocks.

2. See figure 30 for description of AM9516 internal states.

FIGURE 23. -DACKX timing.



NOTE:

1. See figure 30 for description of AM9516 internal states.

FIGURE 24. -WAIT-M timing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

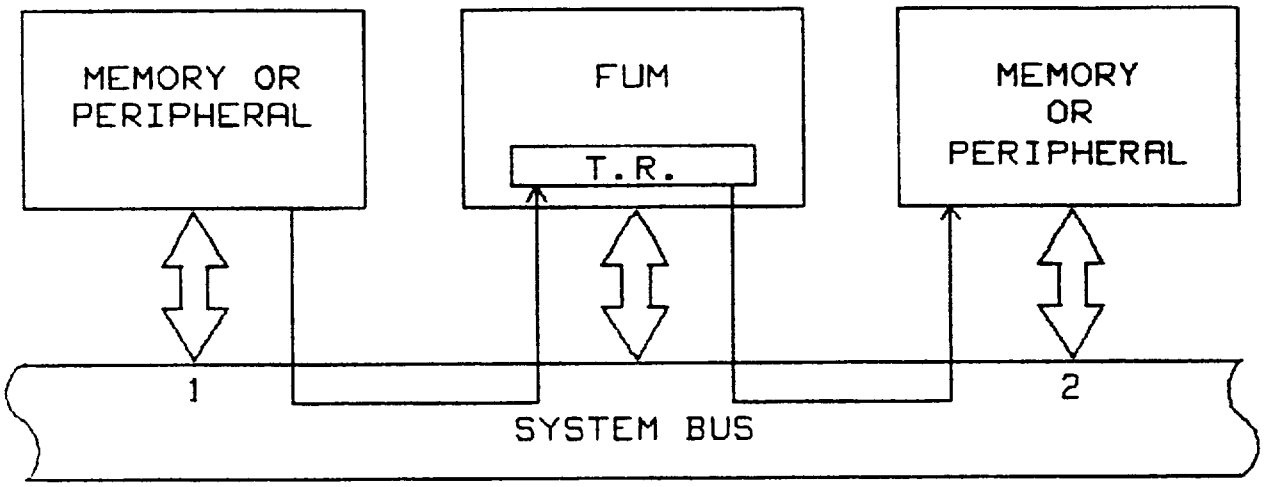


FIGURE 25. Configuration of flowthru transaction.

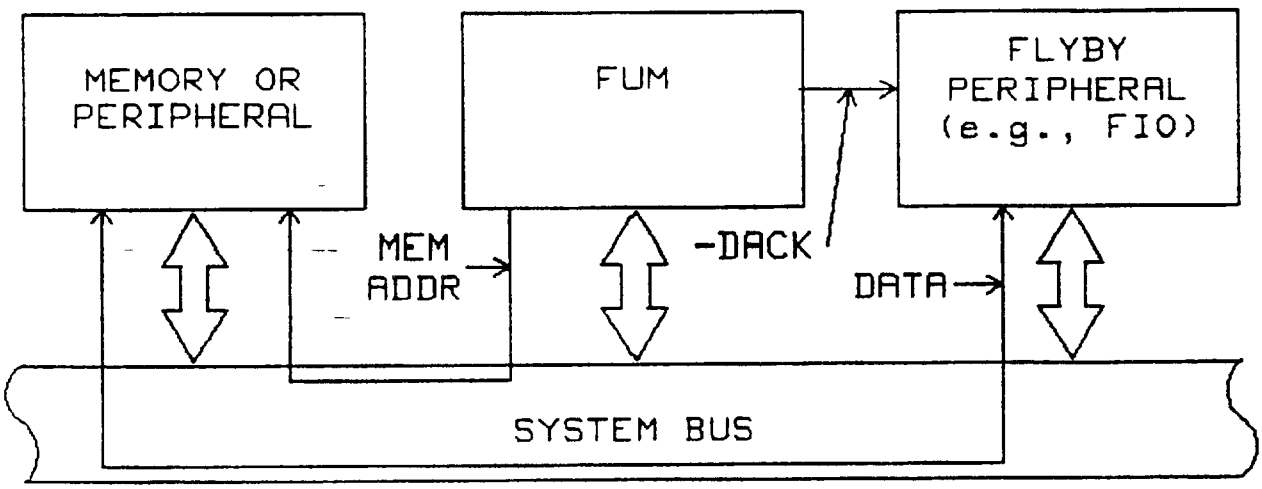
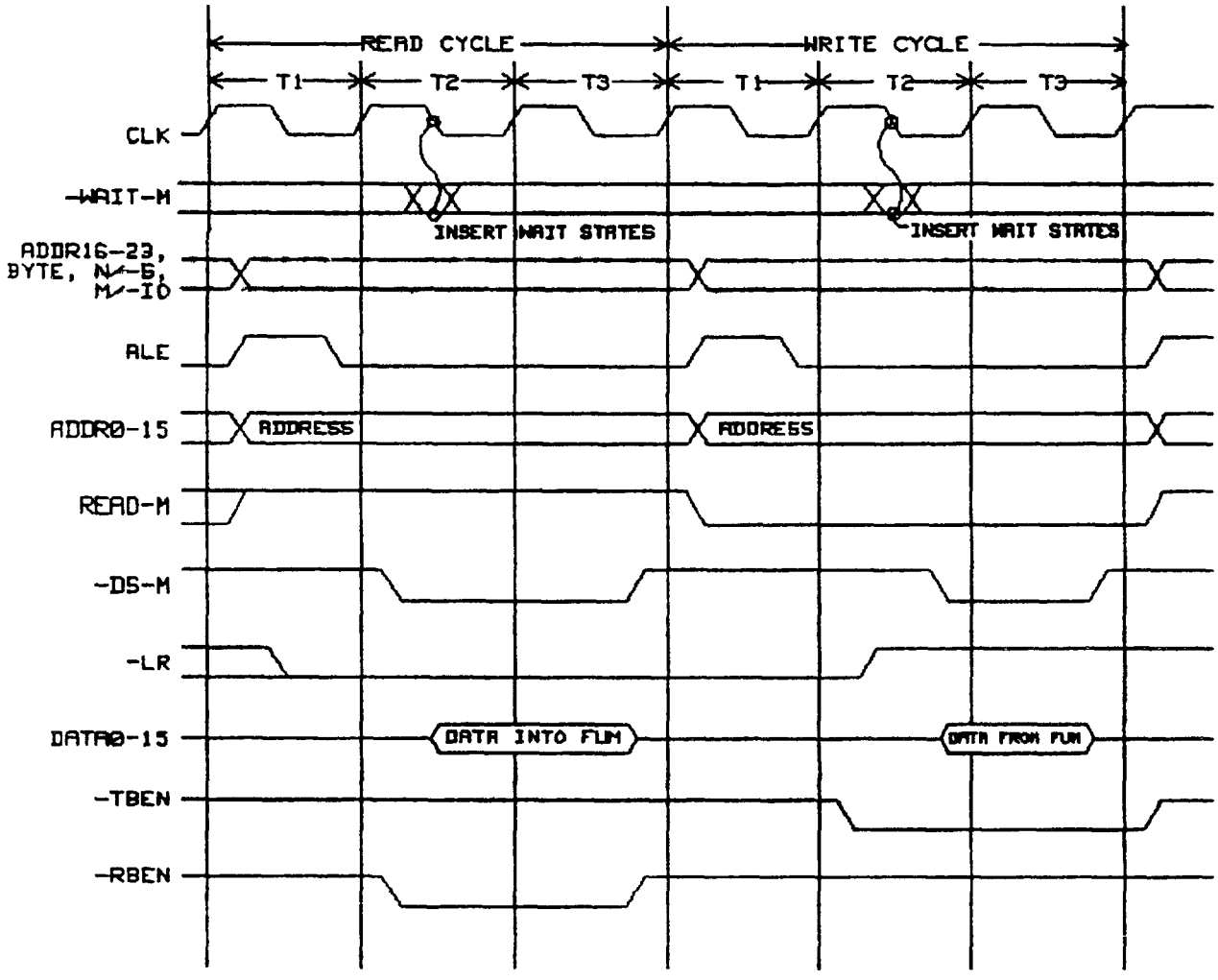


FIGURE 26. Configuration of flyby transaction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX



NOTES:

1. See figure 30 for description of AM9516 internal states.
2. Input -BACK is assumed to be active (low).

FIGURE 27. Flowthru transactions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

33.1.2 BYTE output. I/O address space is byte-addressed, but both 8-bit and 16-bit data sizes are supported. During I/O transactions, the BYTE output signal will be high for byte transactions and low for word transactions. For I/O transactions, both even and odd addresses can be output, hence the address bit output on ADDR0 may be 0 or 1.

33.1.3 I/O read and write operations. The channel can perform both I/O read and I/O write operations by keeping the M/-IO line low. During an I/O read, the FUM module will drive the -DS-M output low to signal the peripheral that data can be gated onto the bus. The FUM will strobe the data into the AM9516's temporary register during T3. -DS-M will be driven high to signal the end of the I/O transaction. During I/O write, the FUM module will enable the contents of the temporary register onto the DATA0-15 bus with the -DS-M output low until T3. Peripherals may strobe the data on DATA0-15 bus into their internal registers on the rising edge of -DS-M.

33.1.4 Byte I/O read and write. For byte I/O writes, the channel will drive the same data bus lines DATA0-7 and DATA8-15. During byte I/O reads, when the address bit on ADDR0 is 0, the FUM module will strobe data in from data line DATA8-15. During byte I/O reads when the address bit on ADDR0 is 1, the FUM module will strobe data in from data line DATA0-7. Thus, when an 8-bit peripheral is connected to the bus, its internal registers will typically be mapped at all even or all odd addresses. To simplify accesses to 8-bit peripherals, byte oriented I/O addresses are incremented/decremented by 2.

33.1.5 Flowthru timing. The channel can perform the I/O read and memory write operation, the memory read and I/O write operation, and the memory read and memory write operation, also. The timing for all flowthru transactions is the same.

33.1.6 Flowthru chaining operation. During chaining operations, the FUM module reads words from an address in system memory pointed by the active channel's chain address register. Those chaining operations are performed identically to the flowthru memory read transactions, except that the data is loaded into an internal AM9516 channel register rather than the temporary register. Note that chaining never causes a write or a byte read; thus all memory writes or all byte accesses are due to DMA operations. A typical memory operation consists of three states: T1, T2, and T3, as shown on figure 27.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

33.1.7 Flowthru wait state programming. The user may elect to insert 1, 2 or 4 software wait states after state T2 and before state T3 by programming the tag field of the current address register or the chain address register. If the wait line enable bit in the master mode register is set, the user may also insert hardware wait states after state T2 and before state T3 by driving a LOW on the -WAIT-M line. The operation of flowthru memory transactions is performed identical to the flowthru I/O transactions.

33.2 Flyby transactions. Flyby transfer and flyby transfer-and-search operations are performed in a single cycle, providing a transfer rate significantly faster than that available from flowthrus. In flyby, operations can only be performed between memory and peripheral or between peripheral and peripheral. Memory-to-memory operations cannot be performed in flyby mode; these must be done using flowthru.

33.2.1 Peripheral requirements for flyby transactions. The flyby transaction can only be used with peripherals having a special flyby signal or with external logic. This flyby input is connected to one of the FUM module's -DACKX outputs. For memory-peripheral flyby, the address of the source memory location must be programmed in the current ARA register. The current ARB register must be programmed with the destination memory location for peripheral-memory flyby. For flyby peripheral-to-peripheral transaction, if both peripherals have a flyby input, only one (called "flyby peripheral") should be connected to -DACKX; the other peripheral's flyby input should be held high during the flyby operation. The address of the peripheral (called "non-flyby peripheral") not connected to the channel's -DACKX output should be programmed in the current ARB register when it is a destination. When the non-flyby peripheral is a source, its address should be programmed in the current ARA register. Note that a set flip bit (CM4 = 1) is for flyby peripheral to non-flyby peripheral or memory write transaction (defined as "from flyby transaction") and a clear flip bit (CM4 = 0) is for the memory or non-flyby peripheral read to flyby peripheral transaction (defined as "to flyby transaction").

Transaction	CM4	READ-M	Address of memory or non-flyby peripheral
To Flyby	0	HIGH	ARA
From Flyby	1	LOW	ARB

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

33.2.2 Flyby operation state timing. A flyby operation is performed using three states: T1, T2, and T3. During T1, the channel pulses ALE and outputs the address information. See figure 28. The READ-M line is high for "to flyby" transaction and low for "from flyby" transaction.

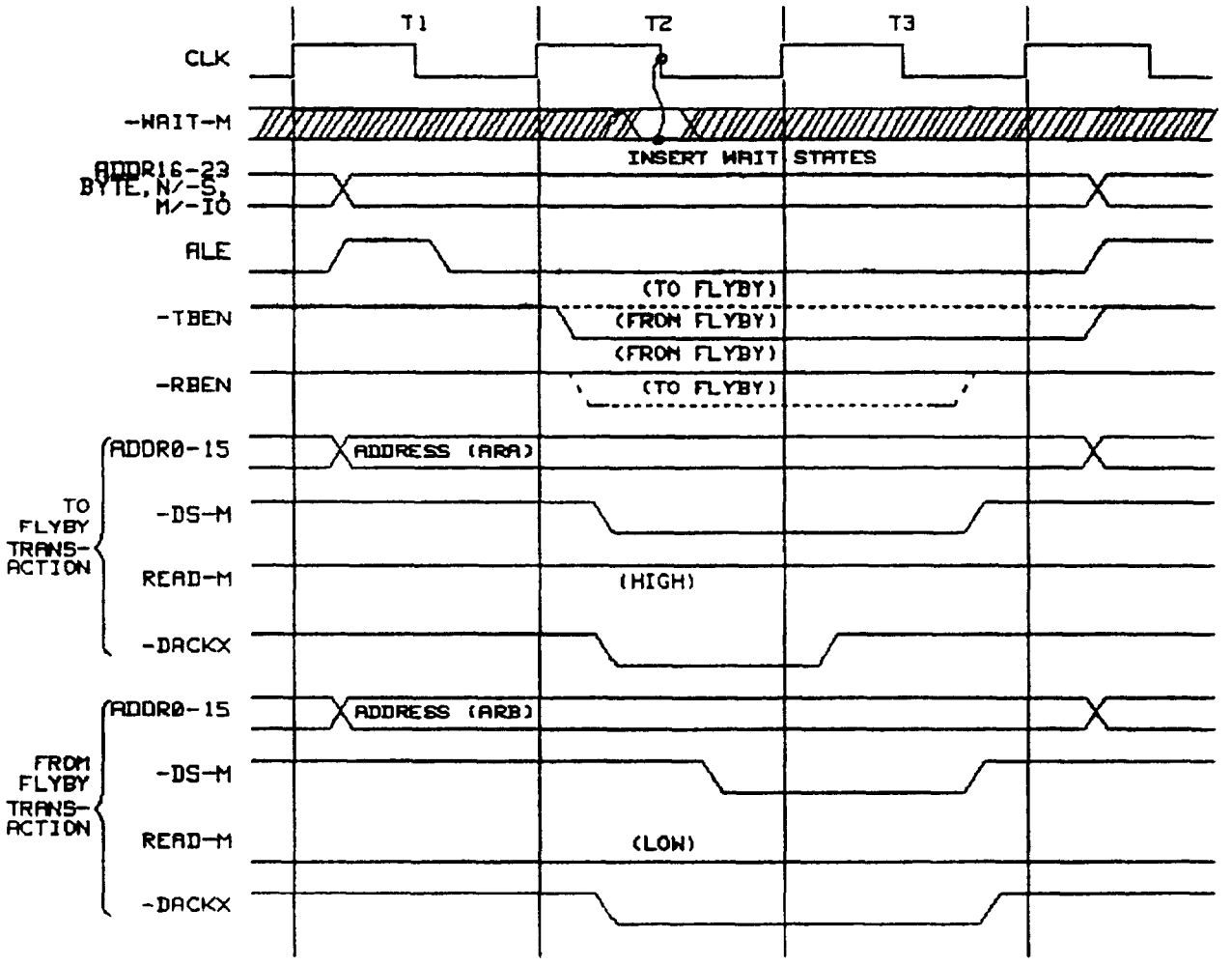
33.2.3 States T1 and T2. The FUM module's M/-IO and N/-S lines are coded as specified by the current ARA or ARB tag field. The BYTE output indicates the operand size programmed in the channel mode register operation field. During state T1, the FUM module drives the READ-M line to indicate the transaction direction, and during state T2, the FUM drives both -DS-M and -DACKX active.

33.2.4 State T3. The pulsed -DACKX input to the peripheral serves two purposes: to select the peripheral for the read/write operation, and to provide timing information on when to drive data onto or input data from the DATA0-15 bus. Note that because the "flyby peripheral" never gets explicitly addressed by the FUM address bus, it must know which internal register is to be loaded from or driven onto the DATA0-15 bus. On state T3, the -DS-M and -DACKX lines are driven inactive to conclude the transfer. In transfer-and-search mode, data is loaded into the FUM module AM9516's temporary register on the low-to-high transition of the -DS-M output in order to perform the search function.

33.2.5 Wait state. To provide adequate data setup time, the rising edge of -DS-M or -DACKX should be the edge used to perform the write to the transfer destination. To extend the active time of -DS-M and -DACKX, wait states can be inserted between T2 and T3. Software wait states can be inserted by programming the appropriate code in the tag field of the current ARA or ARB registers. Hardware wait states can be inserted by pulling the -WAIT-M input low if the wait line enable bit in the master mode register is set. The -WAIT-M line is sampled in the middle of the T2 or TWA state.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX



NOTES:

1. The FUM module data bus (DATA 0-15) transceivers must be configured in the DMA read direction (-LR a low) during the flyby transaction; i.e., the FUM module does not drive the data bus.

2. Input -BACK is assumed to be active (low).

3. See figure 30 for a description of AM9516 internal states.

FIGURE 28. Flyby transactions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APENDIX

34. DMA operation termination. There are three ways a transfer-and-search or search operation can end and two ways a transfer operation can end. When a channel's current operation count goes to 0, the DMA operation being performed will end. This is called a TC or terminal count termination. A DMA operation can also be stopped by driving the EOP input high with external logic. This is called an EOP termination. Search and transfer-and-search operations have a third method of terminating called match condition or MC termination. An MC termination occurs when the data being transferred-and-searched or searched meets the match condition programmed in channel mode register bits CM17-CM16. These bits allow the user to stop when a match occurs between the unmasked pattern register bits and the data read from the source, or when a no-match occurs. Both byte and word matches are supported. MC terminations do not apply to transfer operations, since the pattern matching logic is disabled in transfer mode.

34.1 End-of-process. Whenever a TC, MC or EOP termination occurs, the FUM module will drive the -EOP output pin low. During DMA operations, the EOP input pin is sampled by the FUM module to determine if EOP is being driven high by external logic. Figure 29 shows when the internal FUM AM9516 EOPs are generated marking termination of all transfers. These figures also show the point during the DMA iteration when the EOP input pin is sampled. The generation of internal EOPs and sampling of external EOPs for transfer-and-searches follows the same timing used for transfers. Since there is a single EOP input pin for both channels, EOP should only be driven by a channel while that channel is being serviced.

34.1.1 Response to EOP while reloading chain address register. If an EOP is detected while the channel is trying to reload the chain address register, the new chain address offset and segment are discarded and the old address + 2 is preserved to allow inspection of the erroneous address.

34.2 Programming completion options. When a channel ends a DMA operation, the reason for ending is stored in the completion status field of the channel's status register. This information

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

is retained until the next DMA operation ends at which time the status register is updated to reflect the reason(s) for the latest termination. Note that it is conceivable that more than one bit in the completion field could be set. As an extreme example, if a channel decremented its current operation count to zero, causing a TC termination, input data from the source generated a match causing an MC termination, and there was a high on the EOP input pin resulting in an EOP termination, all three of the channel's status register completion bits would be set.

34.2.1 When a DMA operation ends, the channel can:

(a) Issue an interrupt request (i.e., setting the IP or SIP bit of the channel's status register);
(b) Perform base-to-current reloading;
(c) Chain reload the next DMA operation;
(d) Perform any combination of the above; or
(e) None of the above.

34.2.2 Completion option field. The user selects the action to be performed by the channel in the completion option field of the channel mode register. For each type of termination (TC, MC or EOP), the user can choose which action or actions are to be taken. If no reloading is selected for the type of termination that occurred, the NAC bit in the status register will be set. More than one action can occur when a DMA operation ends. This may arise because more than one action was programmed for the applicable termination. The priorities of those actions are interrupt request first, base-to-current reloading second, and then chaining. The interrupt cannot be serviced unless the FUM module has relinquished the bus.

34.3 Interrupts. In order to allow the FUM to start executing a new DMA operation after issuing an interrupt, but before an interrupt acknowledge is received, a two-deep interrupt queue is implemented in each channel. The following discussion will describe the standard interrupt structure and then elaborate on the additional interrupt queuing capability of the FUM.

34.3.1 Interrupt cycle. A complete interrupt cycle consists of an interrupt request followed by an interrupt-acknowledge transaction. The request, which consists of -INT being pulled low, notifies the MPU that an interrupt is pending. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

interrupt-acknowledge transaction, which is initiated by the MPU as a result of the request, performs two functions: it selects the peripheral whose interrupt is to be acknowledged, and it obtains a vector that identifies the selected device and operation; cause of interrupt.

34.3.2 Peripheral interrupt identification. A peripheral can have one or more sources of interrupt. Each interrupt source has two bits that control how it generates interrupts. These bits are a channel interrupt enable bit (CIE), and an interrupt pending bit (IP). In the FUM module, each channel is an interrupt source. The two interrupt control bits are located in bits CM15 and CM13 of each channel's status register.

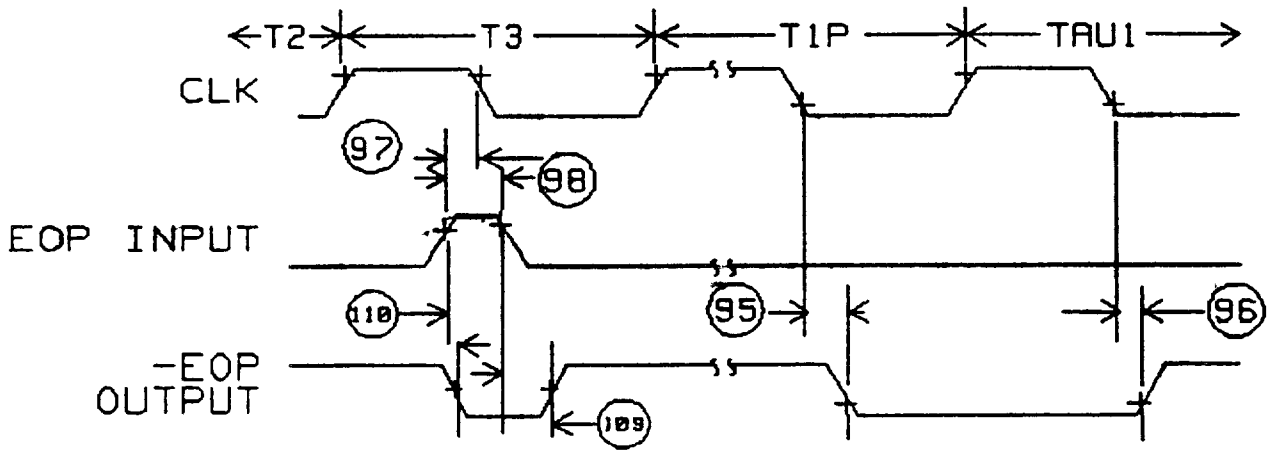
34.3.2.1 Interrupt vector register. Each channel has its own vector register for identifying the source of the interrupt during an interrupt acknowledge transaction. There is one bit (MM3) in the master mode register used for controlling interrupt behavior for the whole module.

34.3.3 Interrupt queuing operation. Once a channel issues an interrupt, it is desirable to allow the channel to proceed with the next DMA operation before the interrupt is acknowledged. This could lead to problems if the FUM channel attempted to chain reload the vector register contents. In such a situation, it may not be clear whether the old or new vector would be returned during the acknowledge. This dilemma is resolved in the FUM by providing each channel with an interrupt save register. When the channel sets IP as part of the procedure followed to issue an interrupt, the contents of the vector register and some of the status register bits are saved in an interrupt save register. See figure 13. When an interrupt acknowledge cycle is performed, the contents of the interrupt save register are driven onto the bus. Although the use of an interrupt save register allows the channel to proceed with a new task, problems can still potentially arise if a second interrupt is to be issued by the channnel before the first interrupt is acknowledged. To avoid conflicts between the first and second interrupt, each channel has a second interrupt pending (SIP) bit in its status register. When a second interrupt is to be issued before the first interrupt is acknowledged, the SIP bit is set and the channel relinquishes the bus until an acknowledge occurs. For compatibility with polled interrupt schemes, the interrupt save register can be read by the host MPU without wait states. As an aid to debugging a system's interrupt logic, whenever IP is set, the interrupt save register is loaded from the vector and status registers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

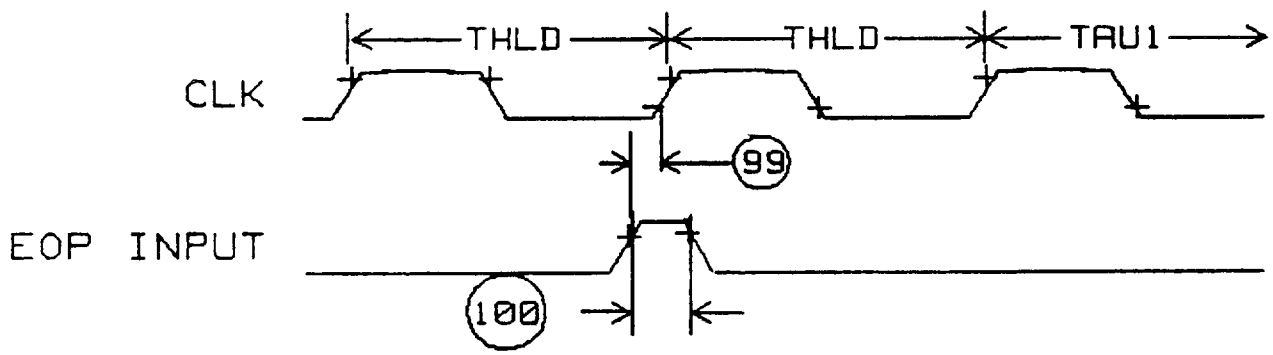
APPENDIX

a) EOP sampling and generation during DMA operations



b) Sampling of EOP during bus hold

CHANNEL HOLDS BUS



NOTES:

1. The diagram lists state names for both I/O and memory accesses. Sampling of EOP will occur on the falling edge of state T3.

2. State T1P is a pseudo-T1 state, generated following termination of any DMA operation.

3. TAU1 is an auto-initialization state, generated following the TC, MC or EOP termination.

4. See figure 30 for a description of AM9516 internal states.

FIGURE 29. End-of-process timing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

34.3.3.1 IP bit. Note that the SIP bit is transferred to the IP bit when IP is cleared by the host MPU. Whenever CIE is set, INT will go low as soon as IP is set.

34.4 Base-to-current reloading. When a channel finishes a DMA operation, the user may elect to perform a base-to-current reload (base-to-current reloading is also referred to as auto-reloading). In this type of reload, the current address registers A and B are loaded with the data in the base address registers A and B, respectively, and the current operation count register is loaded with the data in the base operation count. The base-to-current reload operation facilitates repetitive DMA operations without the multiple memory accesses required by chaining. Although the channel must have bus control to perform base-to-current reloading, the complete reloading operation occurs in four clock cycles (i.e., TAU1 through TAU4). Note that if the channel had to relinquish the bus because two unacknowledged interrupts were queued, it would have to regain bus control to perform any base-to-current reloading (or chaining, for that matter). In this case, it acquires the system bus once an interrupt acknowledge is received, even if it immediately afterward will relinquish the bus because no hardware or software request is present.

34.5 Chaining. If the channel is programmed to chain at the end of a DMA operation, it will use the chain address register to point to a chain control table in memory. The first word in the table is a reload word, specifying the register(s) to be loaded. Following the reload word are the data values to be transferred into the register(s). Because chaining occurs after base-to-current reloading, it is possible to reset the current address registers A and B and the current operation count register to the values used for previous DMA operation only. If the base values are not reloaded during chaining, the channel can revert back to the base values at a later cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

34.5.1 Zero reload word operation. If an all zero reload word is fetched during chaining, the chain operation will not reload any registers, but in all other respects it will perform like any other operation. Thus, the chain address will be incremented by 2 to point to the next word in memory and at the end of the all zero-reload word chain operation, the channel will be ready to perform a DMA operation. All zero reload words are useful as "stubs" to start or terminate linked lists of DMA operations traversed by chaining. On the other hand, care must be taken in their use, since the channel may perform an erroneous operation if it is unintentionally started after the chaining operation.

34.6 AM9516 internal operation routines. Figure 30 shows the AM9516 state diagram which indicates the internal operations of the AM9516.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

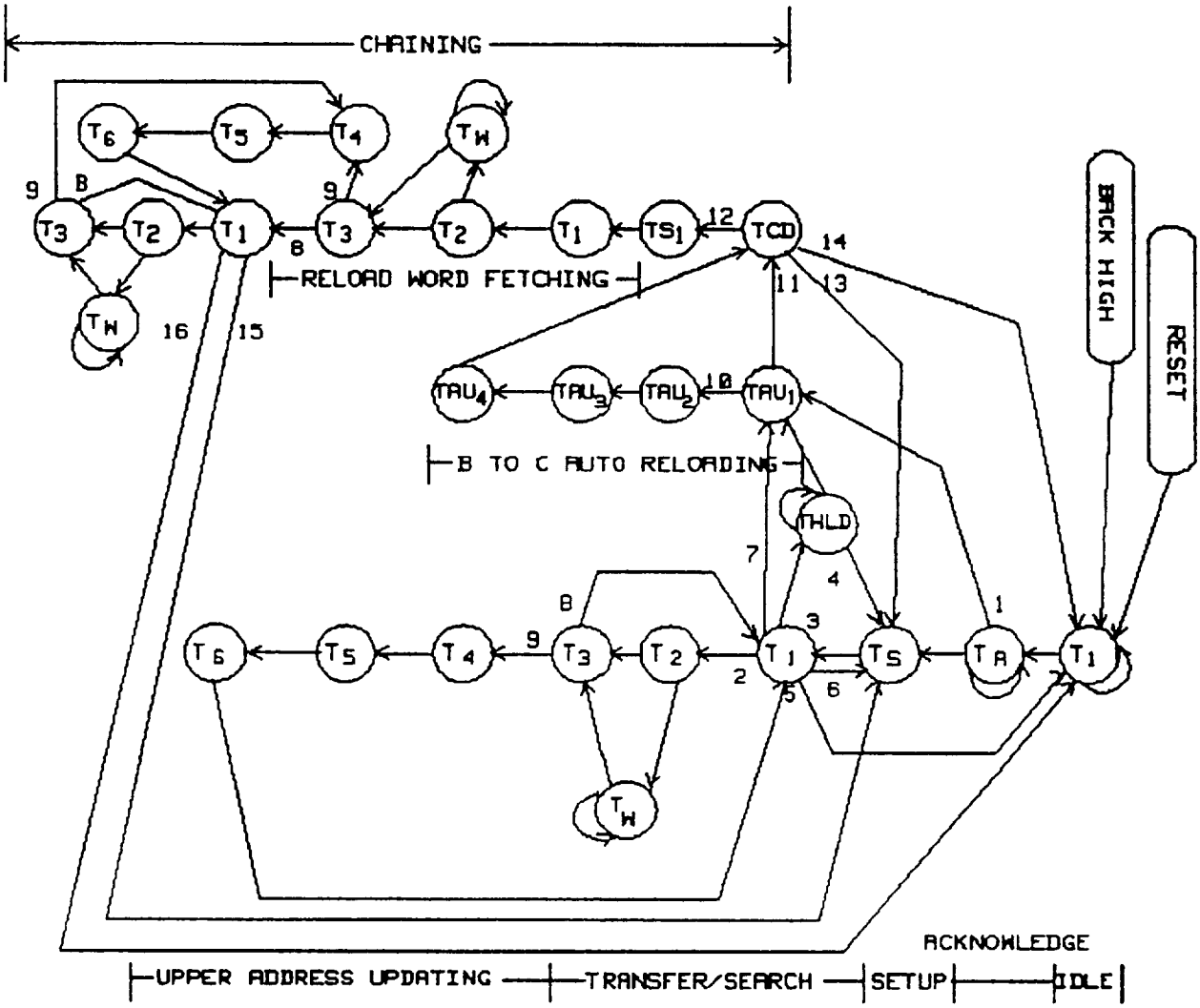


FIGURE 30. AM9516 state diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

APPENDIX

Figure 30 AM9516 state diagram - Continued.

1. "Start Chain" command issued or start updating routine* after an interrupt has been served.**
2. Normal DMA operation.
3. Demand with bus hold while -DREQX is inactive.
4. -DREQX is active while bus held.
5. Single transfer, MPU interleave enabled, or demand with bus release while current -DREQX is inactive and no DMA request is pending.
6. Single transfer or demand/bus release while current -DREQX is inactive, but the other DMA request is pending.
7. TC, MC, or EOP termination occurs.
8. One DMA or chain transaction is done and the upper address is not changed.
9. One DMA or chain transaction is done and the upper address is changed.
10. Base-to-current auto-reloading is enabled.
11. Base-to-current auto-reloading is disabled.
12. Chaining is enabled.
13. Chaining is disabled and another DMA request is pending.
14. Chaining is disabled and no DMA request is pending.
15. Chaining ends and another DMA request is pending.
16. Chaining ends and no DMA request is pending.
17. EOP termination of bus hold.

* Updating routine includes base-to-current auto-reloading and chaining.

** When a second interrupt is to be issued before the first interrupt is acknowledged, the SIP bit of a status register is set and the channel relinquishes the bus until the first interrupt has been served. If the channel was to perform the updating routine, once the SIP bit is cleared, the AM9516 DTC will re-acquire the bus and perform the appropriate operation (i.e., 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) ASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	(1) Commercial TPOC: 812-854-1528	(2) AUTOVON 88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — *Previous editions are obsolete* — 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:03Z
Check the source to verify that this is the current version before use.